



Hardware and Software
Engineered to Work Together

Oracle Fusion Middleware 12c: Multitenancy Administration

Activity Guide
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Table of Contents

Practices for Lesson 1: Course Overview	1-1
Practices for Lesson 1.....	1-2
Practices for Lesson 2: Multitenancy Concepts.....	2-1
Practices for Lesson 2.....	2-2
Practices for Lesson 3 - Domain Partitions	3-1
Practices for Lesson 3: Overview.....	3-2
Practice 3-1: Creating a Single Application Domain Partition.....	3-3
Practice Solution: Creating a Single Application Domain Partition.....	3-9
Practice 3-2: Exporting and Importing a Single Application Domain Partition	3-10
Practice Solution: Exporting and Importing a Single Application Domain Partition	3-13
Practice 3-3: Creating a Domain to Support Multitenancy Using Fusion Middleware Control	3-14
Practice Solution: Creating a Domain to Support Multitenancy using Fusion Middleware Control.....	3-23
Practice 3-4: Creating Domain Partitions Using Fusion Middleware Control.....	3-24
Practice Solution: Creating Domain Partitions Using Fusion Middleware Control	3-36
Practice 3-5: Creating a Full JRF Domain to Support Multitenancy using Fusion Middleware Control	3-37
Practices for Lesson 4 - Resource Groups.....	4-1
Practices for Lesson 4: Overview.....	4-2
Practice 4-1: Creating a Resource Group Template	4-3
Practice Solution: Creating a Resource Group Template.....	4-15
Practice 4-2: Creating a Domain Partition from a Resource Group Template.....	4-16
Practice Solution: Creating a Domain Partition from a Resource Group Template	4-27
Practice 4-3: Creating a Resource Group from a Resource Group Template	4-28
Practice Solution: Creating a Resource Group from a Resource Group Template	4-32
Practices for Lesson 5 - Oracle Traffic Director	5-1
Practices for Lesson 5: Overview.....	5-2
Practice 5-1: Creating an Oracle Traffic Director Configuration	5-3
Practice Solution: Creating an Oracle Traffic Director Configuration.....	5-10
Practice 5-2: Creating Domain Partitions That Use Oracle Traffic Director.....	5-11
Practice 5-3: Configuring Oracle Traffic Director Logging Level	5-17
Practices for Lesson 6 - Database Multitenancy.....	6-1
Practices for Lesson 6: Overview.....	6-2
Practice 6-1: Creating a Container Database.....	6-3
Practice 6-2: Creating Pluggable Databases	6-6
Practice 6-3: Using Pluggable Databases in a Java Application	6-12
Practices for Lesson 7 - Coherence with Multitenancy	7-1
Practices for Lesson 7: Overview.....	7-2
Practice 7-1: Creating Coherence Clusters Managed by WebLogic Server.....	7-3
Practice Solution: Creating Coherence Clusters Managed by WebLogic Server	7-14
Practice 7-2: Creating Coherence Resource Group Templates	7-15
Practice Solution: Creating Coherence Resource Group Templates	7-21
Practice 7-3: Creating Domain Partitions Using Coherence Resources.....	7-22

Course Practice Environment: Security Credentials

For OS usernames and passwords, see the following:

- If you are attending a classroom-based or live virtual class, ask your instructor or LVC producer for OS credential information.
- If you are using a self-study format, refer to the communication that you received from Oracle University for this course.

For product-specific credentials used in this course, see the following table:

Product-Specific Credentials		
Product/Application	Username	Password
Oracle WebLogic Server	weblogic	Welcome1
Oracle WebLogic Server – bayland Security Realm	administrator	bayland123
Oracle WebLogic Server – valley Security Ream	administrator	valley123
Oracle WebLogic Server	wlstadmin	Welcome1
Oracle Database	system	welcome1
Oracle Database	oracle	welcome1
Oracle Database	sys	welcome1
Oracle Database	MEDREC	MEDREC
Oracle Database	sys as sysdba	welcome1
Oracle Database	FULL_STB	welcome1
Oracle Database	FULL	welcome1
Oracle Database	Administrative User	welcome1
Oracle Database	trade	trade

Practices for Lesson 1: Course Overview

Chapter 1

Practices for Lesson 1

There are no practices for this lesson.

Practices for Lesson 2: Multitenancy Concepts

Chapter 2

Practices for Lesson 2

There are no practices for this lesson.

Practices for Lesson 3 - Domain Partitions

Chapter 3

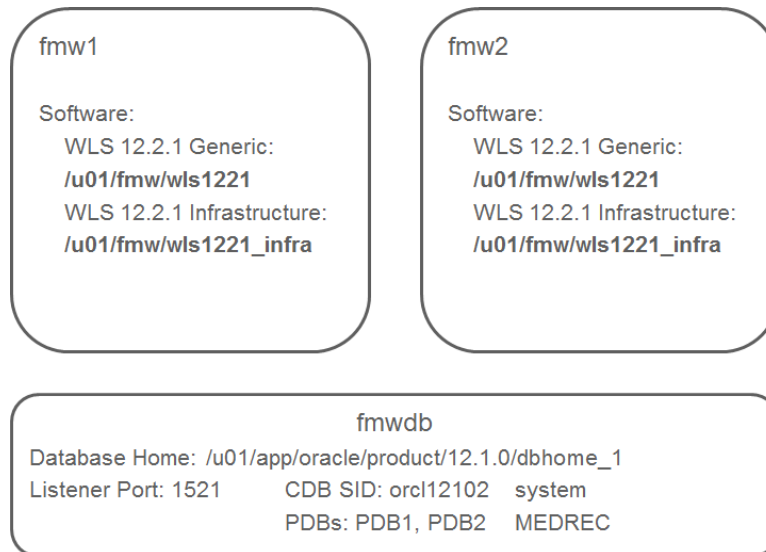
Practices for Lesson 3: Overview

Overview

In these practices, you create WebLogic Server domain partitions for two scenarios:

- Single Application Domain Partition using the WebLogic Server Administration Console
 - A scenario where an application is deployed in a domain partition and uses a separate security realm from the WebLogic Server domain
 - Also, exporting and importing the domain partition with a single application is useful in development-to-test and test-to-production deployment scenarios
- Multiple Applications in a domain partition using Fusion Middleware Control
 - This scenario deploys the same application (the MedRec sample) using different partitions and resources, such as JDBC Data Sources, to a WebLogic Server domain

The practice environment is configured with two fusion middleware hosts and one database host:



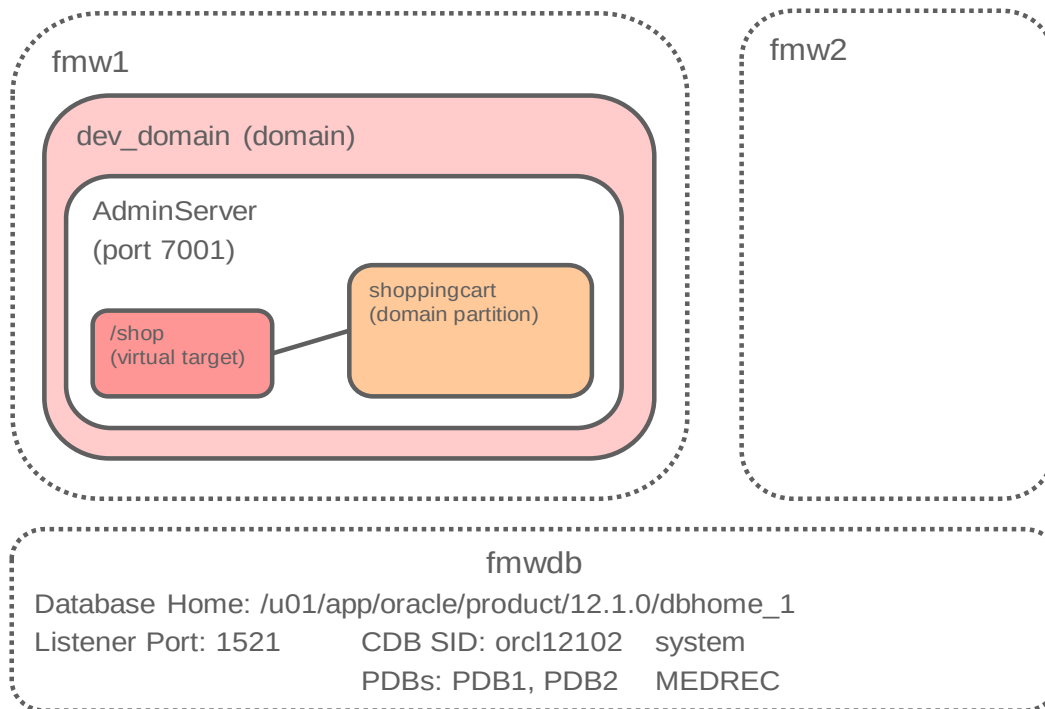
Practice 3-1: Creating a Single Application Domain Partition

Overview

In this practice, using the WebLogic Server Administration Console, you create a new WebLogic Server domain, virtual target, and domain partition for the MedRec sample application. You also create a new security realm for the domain partition and deploy the sample application. Finally, you test the sample application to ensure it functions as expected.

Because many WebLogic Server customers have requirements for this type of configuration—a Java application deployed to a single partition in WebLogic Server—this is a useful practice to execute the steps to create and use this configuration.

The diagram below illustrates the configuration:



Assumptions

- WebLogic Server 12.2.1 generic is installed in /u01/fmw/wls1221.

Tasks

- Connect to fmw1 as the oracle user.
 - Refer to the instructions provided by your instructor to connect to your machine.
 - Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
- Create the WebLogic Server Domain.
 - Open a Terminal window as the oracle user, and navigate to /u01/fmw/wls1221/.
Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221/
```
 - Execute the Fusion Middleware Configuration Wizard.

```
$ oracle_common/common/bin/config.sh
```

3. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

Step	Window/Page Description	Choices or Values
a.	Page 1 of 8 - Create Domain	Click Create a new domain . For Domain Location , enter <code>/u01/fmw/wls1221/user_projects/domains/dev_domain</code> and click Next .
b.	Page 2 of 8 - Templates	The Basic Template is preselected: Basic WebLogic Server Domain – 12.2.1 [wlserver] Click Next .
c.	Page 3 of 8 – Administrator Account	For Name , enter <code>weblogic</code> . For Password , enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. For Confirm Password , enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click Next .
d.	Page 4 of 8 – Domain Mode and JDK	For Domain Mode select Development , and click Next .
e.	Page 5 of 8 – Advanced Configuration	Click Next .
f.	Page 6 of 8 – Configuration Summary	Click Create .
g.	Page 7 of 8 – Configuration Progress	Click Next .
h.	Page 8 of 8 – End of Configuration	Click Finish .

4. Log in to the WebLogic Server domain using the WebLogic Server Administration Console.
- In the Terminal window, navigate to
`/u01/fmw/wls1221/user_projects/domains/dev_domain`
`$ cd /u01/fmw/wls1221/user_projects/domains/dev_domain/`
 - Execute `startWebLogic.sh` to start the WebLogic Administration Server.
`$ ./startWebLogic.sh`
 - Open a Browser window and navigate to **`http://fmw1:7001/console`**.

- d. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the `weblogic` user for the Oracle WebLogic Server product. Click **Login**.

The WebLogic Server Administration Console home page for **dev_domain** is displayed.

5. Create a Virtual Target for shoppingcart.

- a. From the **Domain Structure** panel menu, click **Environment** and then click **Virtual Targets**.
- b. Click **New**.
- c. For **Name**, enter `shoppingcart` and for **URI Prefix** enter `/shop`.

Create a New Virtual Target

OK Cancel

Virtual Target Properties

The following properties will be used to identify your new virtual target.

What would you like to name your new virtual target?

Name:

Select one managed server or cluster on which you would like to deploy this virtual target

Target:

Enter one or more hostnames, each on separate lines within the text box.

Host Names:

Enter an URI prefix starting with a / (forward slash).

URI Prefix:

Partition specific channels may be configured for a virtual target by specifying the name of a channel (which must exist on the selected targets and be configured using the same protocol) and either a port offset or explicit port to be applied to that channel for this partition.

Partition Channel:

Explicit Port:

Port Offset:

OK Cancel

- d. For **Target**, select AdminServer and Click **OK**.
The Virtual Target **shoppingcart** is created successfully.
6. Create a domain partition for shoppingcart.
 - a. In the Browser window, from the **Domain Structure** panel, click **Domain Partitions**.
 - b. Click **New**.
 - c. For **Name**, enter shoppingcart and click **Next**.

The screenshot shows the 'Create a New Domain Partition' dialog box. At the top, there are navigation buttons: 'Back', 'Next', 'Finish', and 'Cancel'. Below this is the 'Domain Partition Properties' section. It contains the text: 'The following properties will be used to identify your new domain partition.' and '* Indicates required fields'. A question asks 'What would you like to name your new domain partition?'. The '* Name:' field is filled with 'shoppingcart'. Below this, it asks 'Do you want to create a default resource group ? You can also specify the resource group template that will be referenced here.' The 'Default Resource Group' checkbox is checked. There is a 'Resource Group' dropdown menu and a 'Template:' field. At the bottom, there are navigation buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

- d. Select the check box for the **Virtual Target shoppingcart**. Click **Next**.

The screenshot shows the 'Create a New Domain Partition' dialog box, Step 2: Available Targets. It contains the text: 'Specify one or more targets to be available for this domain partition. The target(s) for the resource group of this domain partition must be in this list.' Below this is a table with the following content:

Virtual Targets
☒ shoppingcart

At the bottom, there are navigation buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

- e. In the **Available Targets** page, select the **shoppingcart** target from **Available** to **Chosen**.
- f. From the **Security Realm** drop-down menu, select **myrealm**. Click **Finish**.

Create a New Domain Partition

Back Next Finish Cancel

Domain Partition Configuration

The following properties will be used to configure your new domain partition.

Default Targets:

Available:

Chosen:

shoppingcart

Security Realm: myrealm

Partition File System

File System: /u01/fmw/wls1221/user_projects/domains/dev_domain/partitions/shoppingcart/system

Root:

☒ Create On Demand

☒ Preserved

Notes:

Back Next Finish Cancel

The Domain Partition **shoppingcart** is created successfully.

7. Deploy applications to the shoppingcart domain partition.
 - a. From the **Domain Structure** pane, click **Deployments**.
 - b. Click **Install**.
 - c. In the **Locate deployment to install and prepare for deployment** step, navigate to the `/u01/practices/part1/practice03-01/` directory, select the `shoppingcart.war` file and click **Next**.

Install Application Assistant

Back Next Finish Cancel

Locate deployment to install and prepare for deployment

Select the file path that represents the application root directory, archive file, exploded archive directory, or application module descriptor that you want to install. You can also enter the path of the application directory or file in the Path field.

Note: Only valid file paths are displayed below. If you cannot find your deployment files, [Upload your file\(s\)](#) and/or confirm that your application contains the required deployment descriptors.

Path:

/u01/practices/part1/practice03-01/shoppingcart.war

Recently Used (none)

Paths:

Current fmw1 / u01 / practices / part1 / practice03-01

Location:

shoppingcart.war

Back Next Finish Cancel

- d. In the **Choose installation type and scope** step, click **Install this deployment as an application**, select **default in shoppingcart** for **Scope** and click **Next**.

Install Application Assistant

Back Next Finish Cancel

Choose installation type and scope

Select if the deployment should be installed as an application or library. Also decide the scope of this deployment.

The application and its components will be targeted to the same locations. This is the most common usage.

☒ **Install this deployment as an application**

Application libraries are deployments that are available for other deployments to share. Libraries should be available on all of the targets running their referencing applications.

☐ **Install this deployment as a library**

Select a scope in which you want to install the deployment.

Scope: default in shoppingcart

Back Next Finish Cancel

- e. Accept the defaults for Optional Settings and click **Next**.
- f. Review the deployment configuration and click **Finish**.
8. Start the shoppingcart domain partition.
- From the **Domain Structure** pane, click **Domain Partitions**.
 - Click the **Control** tab.
 - Select the check box for the shoppingcart domain partition and click **Start**.
 - Click the refresh icon until the domain partition state changes to **RUNNING**.
9. Test the shoppingcart Virtual Target.
- Open a new Browser tab, navigate to <http://fmw1:7001/shop/shoppingcart>.
 - Click Browse the Store, Go Shopping or View Shopping Cart.

The shopping cart application has been deployed to a domain partition. The next practice will export this domain partition and perform a domain partition import.

Practice Solution: Creating a Single Application Domain Partition

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating the `dev_domain`, this script will not work. To run the script in that case: delete the `/u01/fmw/wls1221/user_projects/domains/dev_domain` directory.

Assumptions

- WebLogic Server 12.2.1 generic is installed in `/u01/fmw/wls1221`.

Solution Tasks

1. Connect to `fmw1` as the `oracle` user.
2. Run the solution script.
 - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice03-01
```
 - b. Run the solution script.

```
$ ./solution.sh
```
 - c. Enter any passwords as you are prompted.

Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain.py`, which creates the domain, partition and starts the partition for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>>Practice 03-01 solution applied
```

```
Exiting WebLogic Scripting Tool.
```

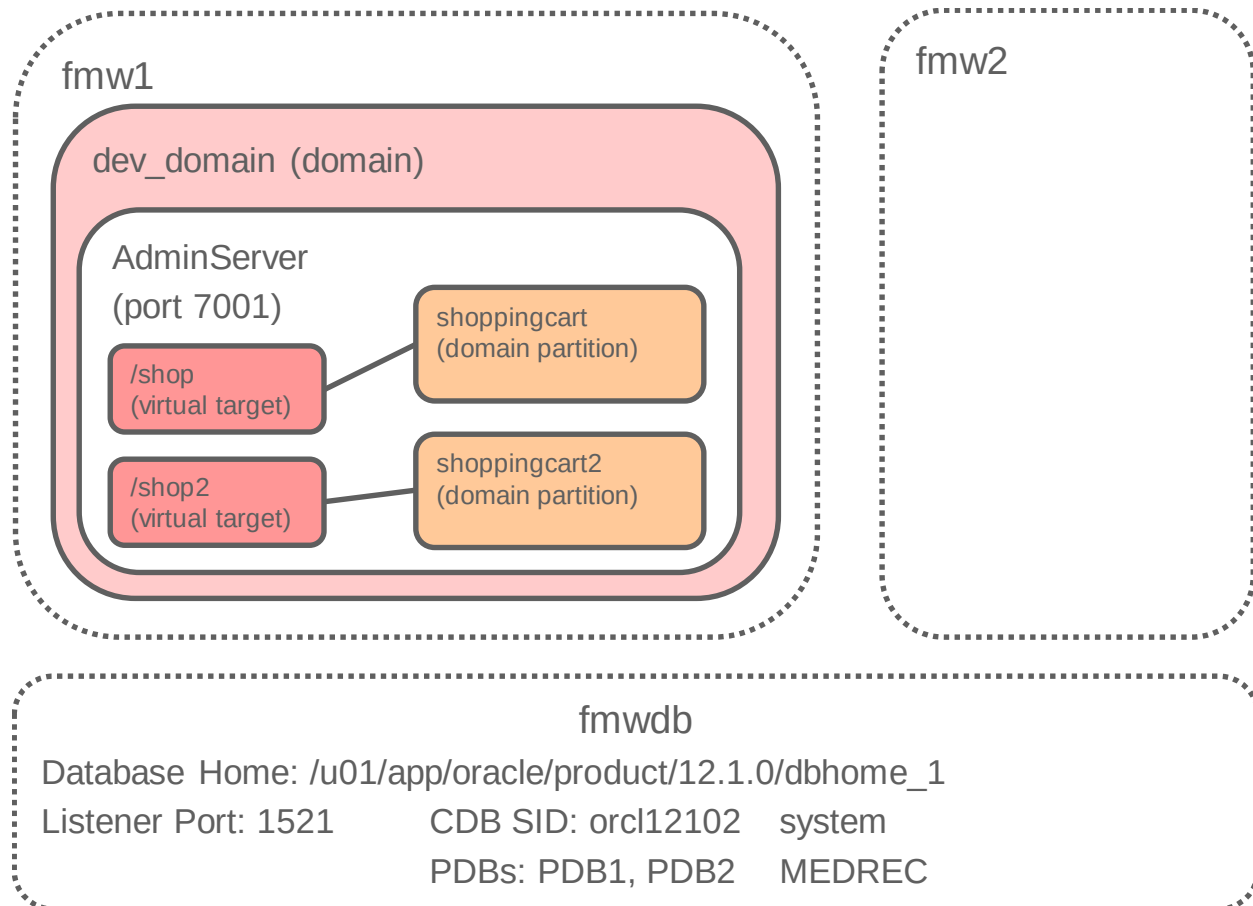
3. Close the Terminal window.

Practice 3-2: Exporting and Importing a Single Application Domain Partition

Overview

In this practice, you export the domain partition created in the previous practice, modify the domain partition archive attributes and then import the domain partition archive into a new domain partition.

The following diagram illustrates the configuration:



Assumptions

- WebLogic Server 12.2.1 generic is installed in `/u01/fmw/wls1221`.
- The `dev_domain` administration server is running.
- The `shoppingcart` domain partition has been created, started, and tested successfully.

Tasks

1. Log in to the WebLogic Server Domain using the WebLogic Server Administration Console.
 - a. In the Browser window, navigate to **`http://fmw1:7001/console`**.
 - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the `weblogic` user for the Oracle WebLogic Server product. Click **Login**.

The WebLogic Server Administration Console home page for `dev_domain` is displayed.

2. Export the shoppingcart domain partition.
 - a. In the **Domain Structure** pane, click **Domain Partitions**.
 - b. Select the check box next to the **shoppingcart** domain partition and click **Export**.
 - c. Select **Include Application Bits**.
 - d. For **Path**, enter `/u01/fmw/wls1221/user_projects` and click **OK**.
3. Create a virtual target named shoppingcart2 for the new partition.
 - a. In the **Domain Structure** pane, click **Environment** and then click **Virtual Targets**.
 - b. Click **New**.
 - c. For **Name**, enter `shoppingcart2` and for **URI Prefix** enter `/shop2`.
 - d. For Target, select **AdminServer** and Click **OK**.

The Virtual Target **shoppingcart2** is created successfully.
4. Edit the export file to use the new Virtual Target avitekclinic2.
 - a. Open a Terminal window on host `fmw1` as the `oracle` user, and navigate to `/u01/fmw/wls1221/user_projects`.
`$ cd /u01/fmw/wls1221/user_projects`
 - b. Edit the `shoppingcart-attributes.json` file.
`$ vi shoppingcart-attributes.json`
 - c. Set the default-target to shoppingcart2:
`"default-target": [{"virtual-target": {"name": "shoppingcart2"}}],`
 - d. Set the available-target to shoppingcart2:
`"available-target": [{"virtual-target": {"name": "shoppingcart2"}}]`
 - e. Save and close the `shoppingcart -attributes.json` file:
`:wq`
5. Import the domain partition.
 - a. In the **Domain Structure** pane, click **Domain Partitions**.
 - b. Click **Import**.
 - c. In the **Import a Domain Partition** page, select **Create New**, enter `shoppingcart2` for **Domain Partition Name**, enter `/u01/fmw/wls1221/user_projects` for **Path**, and select the **shoppingcart.zip** file. Click **OK**.
6. Start the imported shoppingcart2 Domain Partition.
 - a. From the **Domain Structure** pane, click **Domain Partitions**.
 - b. Click the **Control** tab.
 - c. Select the check box for the shoppingcart2 domain partition and click **Start**.
 - d. Click the refresh icon until the domain partition state changes to **RUNNING**.
7. Test the shoppingcart2 Virtual Target.
 - a. Open a new Browser tab, navigate to <http://fmw1:7001/shop2/shoppingcart>.
 - b. Click **Browse the Store**, **Go Shopping** or **View Shopping Cart**.

8. Stop the AdminServer for dev_domain.
 - a. In Browser tab, navigate to <http://fmw1:7001/console>.
 - b. For **User Name**, enter weblogic and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.

The WebLogic Server Administration Console home page for the **dev_domain** is displayed.
 - c. In the WebLogic Server Administration Console, from the **WebLogic Domain Structure** panel, click **Environment** and click **Servers**.
 - d. Click the **Control** tab.
 - e. Select the **AdminServer**.
 - f. Click **Shutdown** and select **Force Shutdown Now**.

Practice Solution: Exporting and Importing a Single Application Domain Partition

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating the shoppingcart2 domain partition or virtual target this script will not work. To run the script in that case: delete the shoppingcart2 domain partition and virtual target.

Assumptions

- WebLogic Server 12c (12.2.1) generic is installed in `/u01/fmw/wls1221`.
- The `dev_domain` from the previous practice has been created.
- The AdminServer from `dev_domain` is running.
- The `avitekclinic` domain partition has been created, started, and tested successfully.

Solution Tasks

1. Connect to `fmw1` as the `oracle` user.
2. Run the solution script.
 - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice03-02
```
 - b. Run the solution script.

```
$ ./solution.sh
```
 - c. Enter any passwords as you are prompted.

Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `export_import.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>>Practice 03-02 solution applied
```

```
Exiting WebLogic Scripting Tool.
```

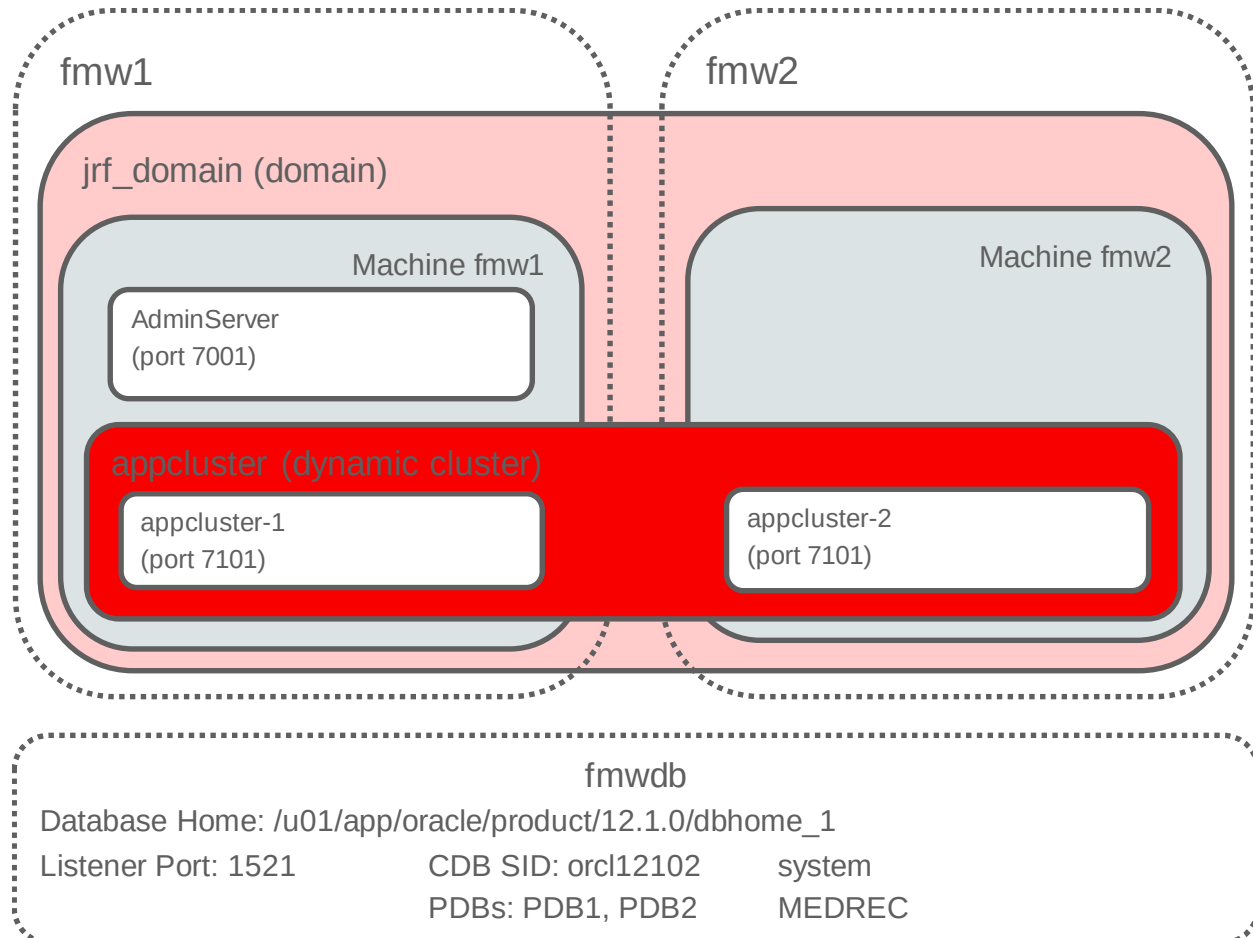
3. Run the solution script.

Practice 3-3: Creating a Domain to Support Multitenancy Using Fusion Middleware Control

Overview

In this practice, you create a WebLogic Server domain using the Fusion Middleware Configuration Wizard – this practice creates a domain that uses Fusion Middleware Control for administration, Oracle Traffic Director and Oracle Coherence.

The following diagram illustrates the configuration:



Assumptions

- WebLogic Server 12c (12.2.1) infrastructure is installed in /u01/fmw/wls1221_infra.
- The AdminServer from dev_domain has been shut down.

Tasks

1. Connect to fmw1 as the `oracle` user.
 - a. Refer to the instructions provided by your instructor to connect to your machine.
 - b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Create the WebLogic Server Domain.

- a. Open a Terminal window as the `oracle` user, and navigate to
`/u01/fmw/wls1221_infra/`.

Tip: There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/
```

- b. Execute the Fusion Middleware Configuration Wizard.

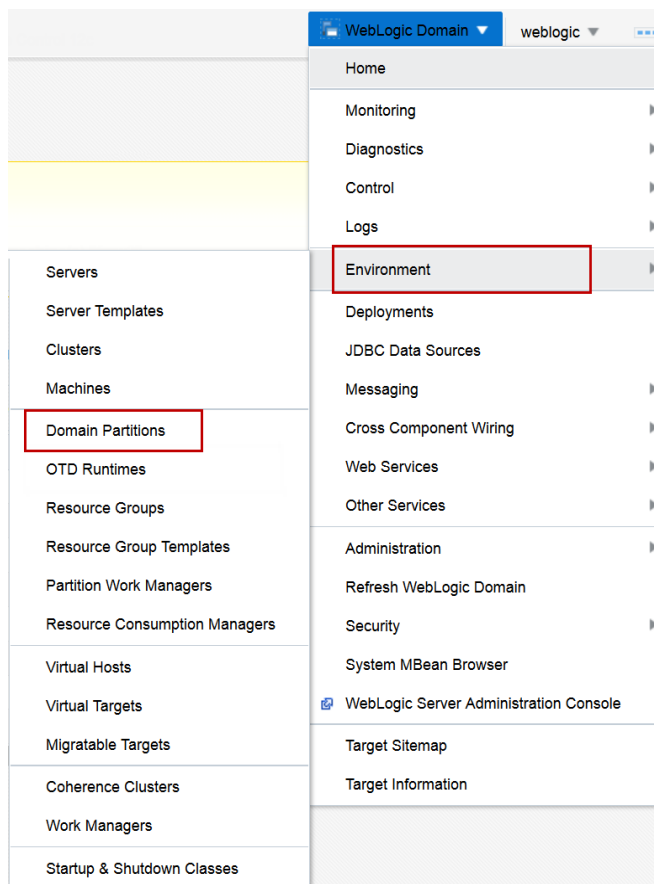
```
$ oracle_common/common/bin/config.sh
```

3. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

Step	Window/Page Description	Choices or Values
a.	Page 1 of 8 - Create Domain	Click Create a new domain . For Domain Location , enter <code>/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain</code> and click Next .
b.	Page 2 of 8 - Templates	Select these Available Templates : • Oracle Traffic Director – Restricted JRF – 12.2.1 [otd] • Oracle Enterprise Manager – Restricted JRF – 12.2.1 [em] • Oracle Restricted JRF – 12.2.1 [oracle_common] • WebLogic Coherence Cluster Extension – 12.2.1 [wlserver] Click Next .
c.	Page 3 of 9 – Application Location	For Application Location , enter <code>/u01/fmw/wls1221_infra/user_projects/applications/jrf_domain</code> and click Next .
d.	Page 4 of 9 – Administrator Account	For Name , enter <code>weblogic</code> . For Password , enter the password from the Course Practice Environment: Security Credentials document for the <code>weblogic</code> user for the Oracle WebLogic Server product. For Confirm Password , enter the password from the Course Practice Environment: Security Credentials document for the <code>weblogic</code> user for the Oracle WebLogic Server product. Click Next .
e.	Page 5 of 9 – Domain Mode and JDK	For Domain Mode select Development , and click Next .
f.	Page 6 of 9 – Advanced Configuration	Click Next .
g.	Page 7 of 9 – Configuration Summary	Click Create .
h.	Page 8 of 9 – Node Manager	Click Next .
i.	Page 9 of 9 –	Click Finish .

Step	Window/Page Description	Choices or Values
	End of Configuration	

4. Log in to the WebLogic Server domain using Fusion Middleware Control.
 - a. In the Terminal window, navigate to
`/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain.`
`$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain`
 - b. Set the maximum memory for WebLogic Server to 512m.
`$ sed -i 's/1024/512/g' bin/setStartupEnv.sh`
 - c. Execute `startWebLogic.sh` to start the WebLogic Administration Server.
`$ ./startWebLogic.sh`
 - d. Open a Browser window and navigate to **http://fmw1:7001/em.**
 - e. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the `weblogic` user for the Oracle WebLogic Server product. Click **Login**.
 The home page for the **jrf_domain** is displayed.
5. Enable the Lifecycle Manager.
 - a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Domain Partitions**.



- b. Click **Enable Lifecycle Manager**.

Domain Partitions

Domain partitions are the building blocks of WebLogic server multi-tenancy (MT). Multi-tenancy permits multiple client organizations to share a domain, improving efficiency and reducing operation costs. Before creating a domain partition, you must first create one or more virtual targets. Look at the Getting Started topics for more information.

▶ **Getting Started with Multi-Tenancy**

▲ Hide Pie Chart

Information

Lifecycle Manager services are not currently available, therefore the create and delete partition operations have been disabled.

Enable Lifecycle Manager

View ▼ Create ✕ Delete Control ▼ Import Export

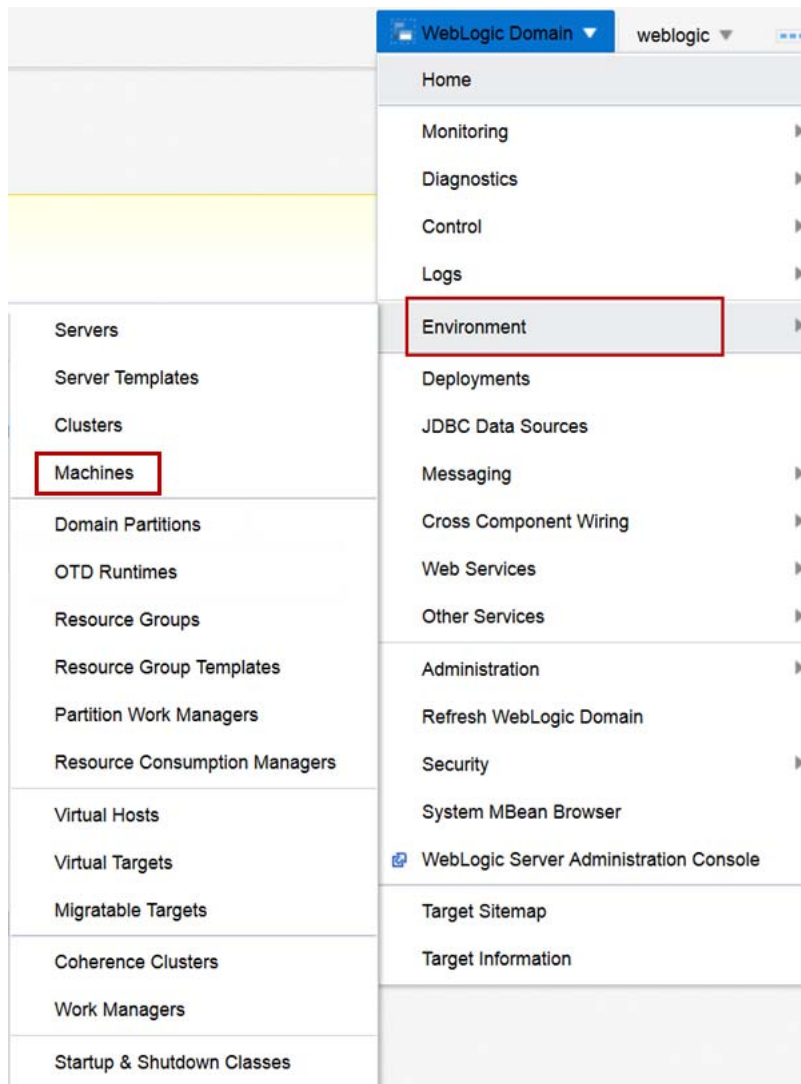
Name	Status	State	OTD Partition	Realm	D
No domain partitions found.					

Domain Partitions 0 of 0

- c. In the terminal window where `startWebLogic.sh`, is running, enter `control-c` to stop the WebLogic Server.
- d. After the WebLogic Server has stopped, execute `startWebLogic.sh` to start the WebLogic Administration Server.
- \$ `./startWebLogic.sh`
- e. In the Browser window navigate to <http://fmw1:7001/em>.
- f. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.

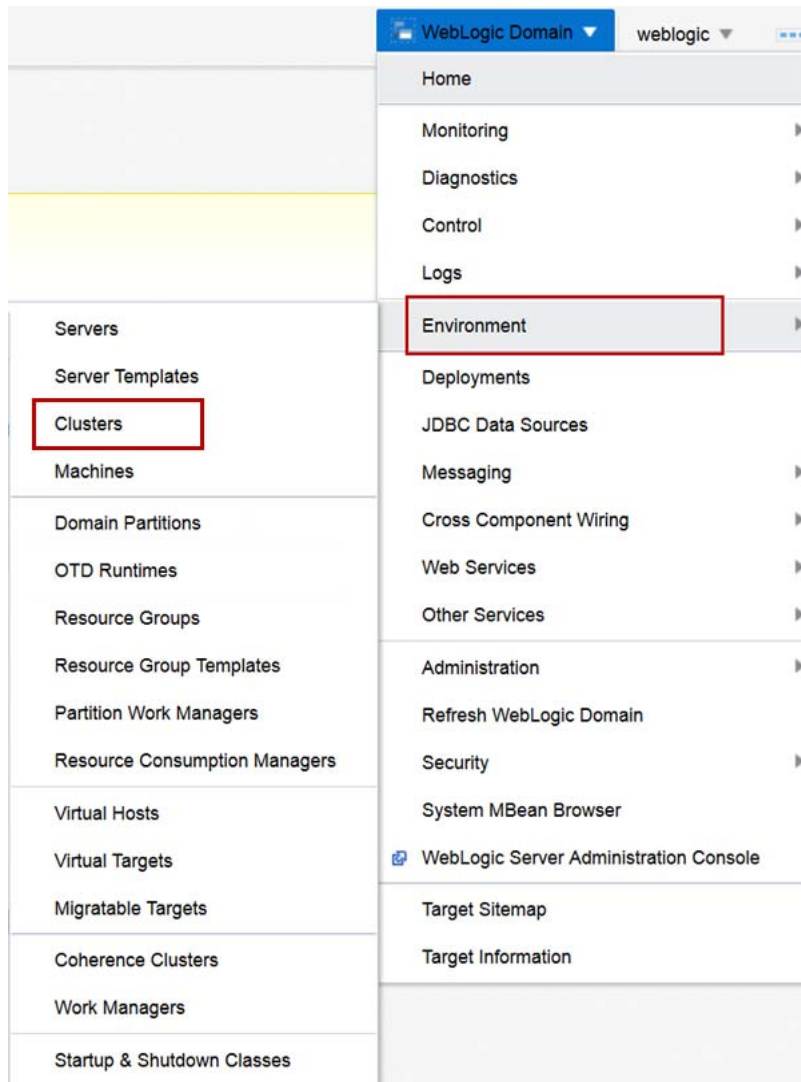
The home page for the `jrf_domain` displays and the Lifecycle Manager has been enabled.

6. Configure the WebLogic Server domain for machines for hosts fmw1 and fmw2.
 - a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Machines**.



- b. Click **Create**.
- c. For **Name**, enter `fmw1` and for **Machine OS** select **Other**. Click **Next**.
- d. For **Listen Address**, enter `fmw1` and for **Listen Port** enter `5556`. Click **Next**.
- e. Review the settings and click **Create**.
The machine **fmw1** is created successfully.
- f. Click **Create**.
- g. For **Name**, enter `fmw2` and for **Machine OS** select **Other**. Click **Next**.
- h. For **Listen Address**, enter `fmw2` and for **Listen Port** enter `5556`. Click **Next**.
- i. Review the settings and click **Create**.
The Machine **fmw2** is created successfully.

7. Configure the WebLogic Server domain with a Dynamic Cluster named appcluster.
 - a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Clusters**.



- b. Click **Create** and select **Dynamic Cluster**.
 - c. For **Name**, enter `appcluster`. Click **Next**.
 - d. For **Dynamic Cluster Size**, enter 2 and for **Max Dynamic Cluster Size** enter 4 . Click **Next**.
 - e. Select **Use any machine configured in this domain** and click Next.
 - f. Select **Assign each dynamic server unique listen ports**, and for **Base Listen Port** enter 7100 , and for **SSL Base Listen Port** enter 8100 . Click **Next**.
 - g. Review the configuration and click Create.
The cluster **appcluster** creates successfully.
8. Pack and unpack the jrf_domain to deploy it to the fmw2 host.
 - a. In a new Terminal window navigate to `/u01/fmw/wls1221_infra/`.
`$ cd /u01/fmw/wls1221_infra/`
 - b. Execute `pack.sh` to pack the jrf_domain.

```
$ oracle_common/common/bin/pack.sh -domain
/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/ -
template /u01/fmw/wls1221_infra/user_projects/jrf_domain.jar -
template_name jrf_domain -managed true
```

- c. Create the `user_projects` directory on the `fmw2` machine.

```
$ ssh fmw2 "mkdir -p
/u01/fmw/wls1221_infra/user_projects/domains"
```

- d. Enter the `oracle` user password when prompted.

- e. Copy the `jrf_domain.jar` to `fmw2` machine.

```
$ scp user_projects/jrf_domain.jar
fmw2:/u01/fmw/wls1221_infra/user_projects/jrf_domain.jar
```

- f. Enter the `oracle` user password when prompted.

- g. Connect to the `fmw2` machine as the `oracle` user.

- h. Open a new Terminal window on `fmw2` as the `oracle` user and navigate to

```
/u01/fmw/wls1221_infra/.
```

```
$ cd /u01/fmw/wls1221_infra/
```

- i. Execute `unpack.sh` to unpack the `jrf_domain`.

```
$ /u01/fmw/wls1221_infra/oracle_common/common/bin/unpack.sh -
domain /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain -
template /u01/fmw/wls1221_infra/user_projects/jrf_domain.jar
```

The `jrf_domain` is now unpacked and ready to use NodeManager and start managed servers on host `fmw2`.

9. Start NodeManager on host `fmw1`.

- a. Connect to the `fmw1` machine as the `oracle` user.

- b. Open a new Terminal window on `fmw1` as the `oracle` user and navigate to

```
/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/.
```

```
$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
```

- c. Copy the `nodemanager` directory to the `nodemanager-fmw1` directory.

```
$ cp -r nodemanager nodemanager-fmw1
```

- d. Copy the `startNodeManager.sh` script to the `nodemanager-fmw1` directory.

```
$ cp bin/startNodeManager.sh nodemanager-fmw1
```

- e. Edit the `nodemanager.properties` file.

```
$ vi nodemanager-fmw1/nodemanager.properties
```

- f. Set the `NodeManagerHome` property:

```
NodeManagerHome=/u01/fmw/wls1221_infra/user_projects/domains/jr
f_domain/nodemanager-fmw1
```

- g. Set the `ListenAddress` property:

```
ListenAddress=fmw1
```

- h. Set the `NodeManagerHome` property:

```
LogFile=/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
nodemanager-fmw1/nodemanager.log
```

- i. Save and close the `nodemanager.properties` file:

```
:wq
```

- j. Edit the `startNodeManager.sh` file.

- ```
$ vi nodemanager-fmw1/startNodeManager.sh
```
- k. Set the NODEMGR\_HOME property:  
`NODEMGR_HOME="/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/nodemanager-fmw1"`
  - l. Save and close the nodemanager.properties file:  
`:wq`
  - m. Navigate to the nodemanager-fmw1 directory.  
`$ cd nodemanager-fmw1`
  - n. Execute the startNodeManager.sh script.  
`$ ./startNodeManager.sh`  
 The NodeManager process starts successfully – leave the Terminal window open.
10. Start NodeManager on host fmw2.
- a. Connect to the fmw2 machine as the oracle user.
  - b. In the Terminal window on fmw2 as the oracle user navigate to  
`/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/`  
`$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/`
  - c. Copy the nodemanager directory to the nodemanager-fmw2 directory.  
`$ cp -r nodemanager nodemanager-fmw2`
  - d. Copy the startNodeManager.sh script to the nodemanager-fmw2 directory.  
`$ cp bin/startNodeManager.sh nodemanager-fmw2`
  - e. Edit the nodemanager.properties file.  
`$ vi nodemanager-fmw2/nodemanager.properties`
  - f. Set the NodeManagerHome property:  
`NodeManagerHome=/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/nodemanager-fmw2`
  - g. Set the ListenAddress property:  
`ListenAddress=fmw2`
  - h. Set the NodeManagerHome property:  
`LogFile=/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/nodemanager-fmw2/nodemanager.log`
  - i. Save and close the nodemanager.properties file:  
`:wq`
  - j. Edit the startNodeManager.sh file.  
`$ vi nodemanager-fmw2/startNodeManager.sh`
  - k. Set the NODEMGR\_HOME property:  
`NODEMGR_HOME="/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/nodemanager-fmw2"`
  - l. Save and close the nodemanager.properties file:  
`:wq`
  - m. Navigate to the nodemanager-fmw2 directory.  
`$ cd nodemanager-fmw2`

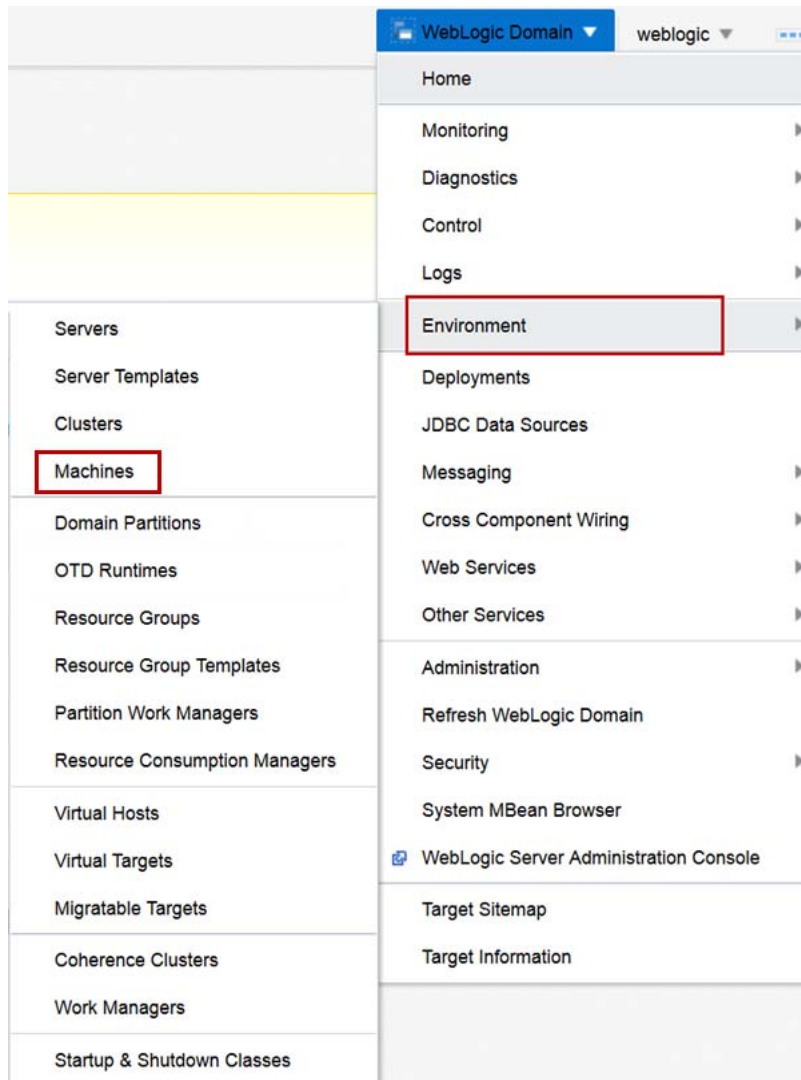
- n. Execute the `startNodeManager.sh` script.

```
$./startNodeManager.sh
```

The NodeManager process starts successfully – leave the Terminal window open.

11. Validate that both machines are reachable from the WebLogic Administration Server.

- a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Machines**.



- b. Click **fmw1**.  
c. Click the **Monitoring** tab.  
d. The **Node Manager Status** reports **Reachable**.  
e. Click **fmw2**.  
f. Click the **Monitoring** tab.  
g. The **Node Manager Status** reports **Reachable**.

The `jrf_domain` is created with a dynamic cluster and NodeManager running on both hosts to support a multitenancy deployment.



## Practice Solution: Creating a Domain to Support Multitenancy using Fusion Middleware Control

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started creating the `jrf_domain`, this script will not work. To run the script in that case: delete the `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain` directory and delete the `/u01/fmw/wls1221_infra/user_projects/applications/jrf_domain` directory.

### Assumptions

WebLogic Server 12c (12.2.1) infrastructure is installed in `/u01/fmw/wls1221_infra`.

### Solution Tasks

1. Connect to `fmw1` as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.  

```
$ cd /u01/practices/part1/practice03-03
```
  - b. Run the solution script.  

```
$./solution.sh
```
  - c. Enter any passwords as you are prompted.

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain.py`, which creates the new domain for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>>Practice 03-03 solution applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

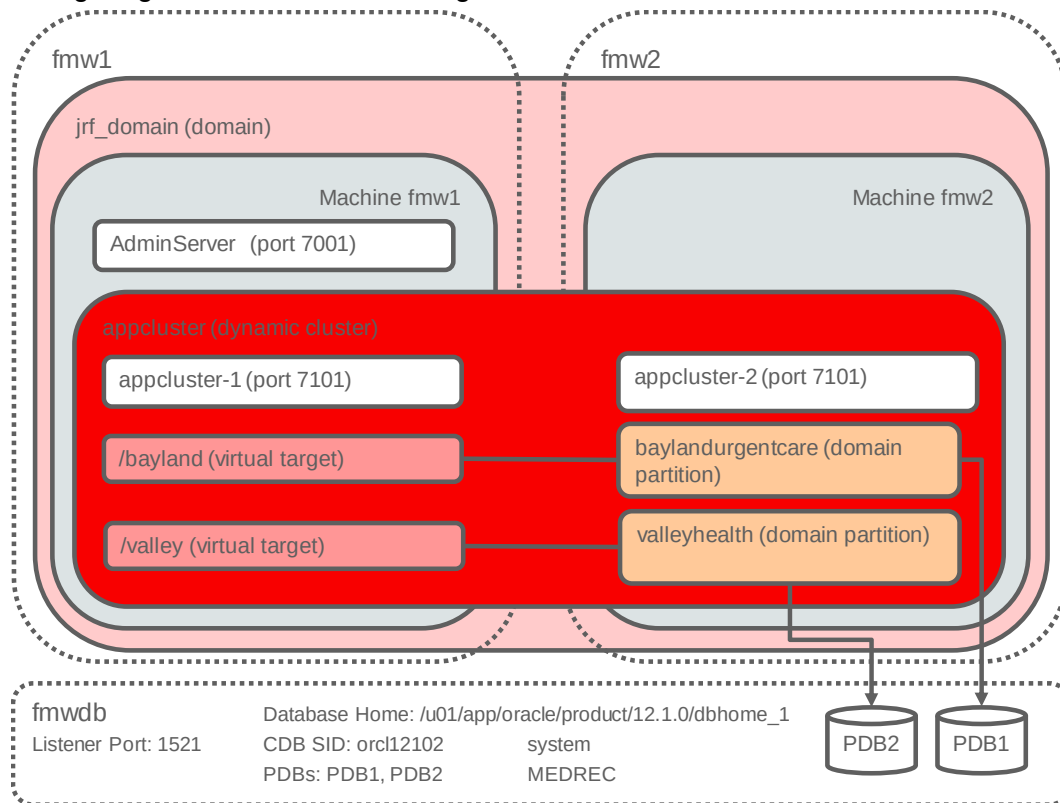
3. **Note:** this script starts a WebLogic Server domain in the background and may take several minutes start and make the console and Fusion Middleware Control available.
4. Close the Terminal window.

## Practice 3-4: Creating Domain Partitions Using Fusion Middleware Control

### Overview

In this practice, you create Domain Partitions for the Avitek Medical Records (MedRec) sample application using Fusion Middleware Control.

The following diagram illustrates the configuration:



### Assumptions

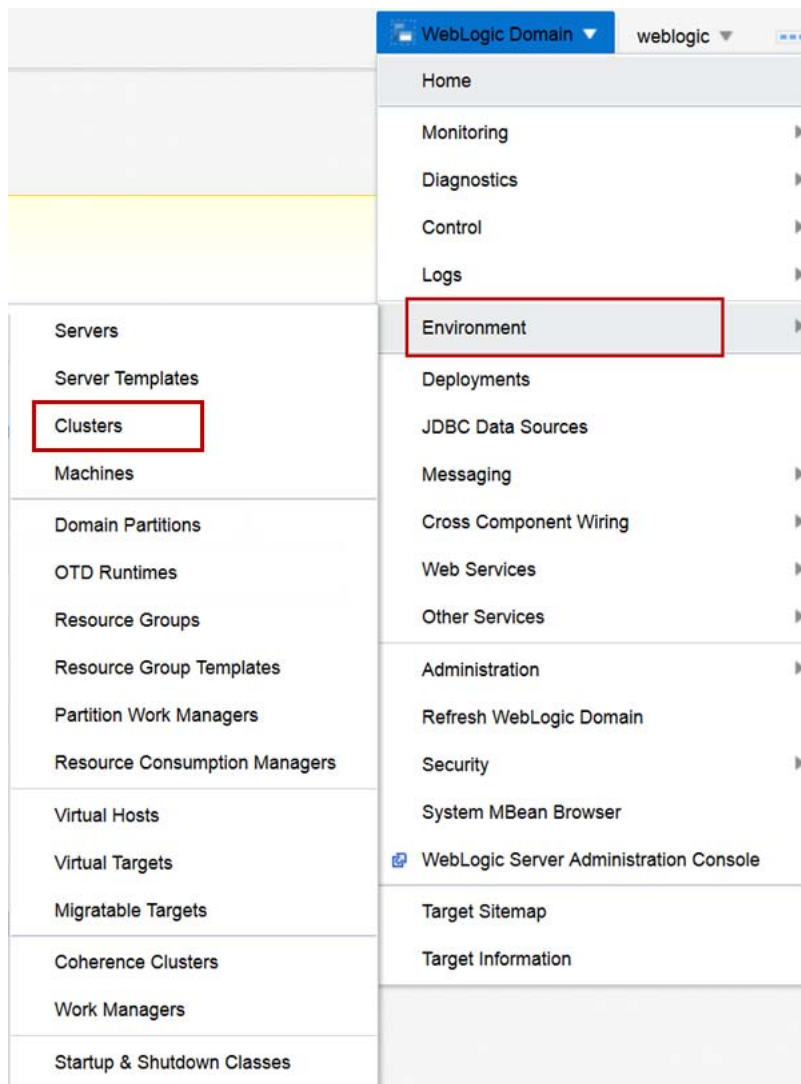
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

### Tasks

1. Start the Dynamic Cluster appcluster.
  - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter **weblogic** and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.

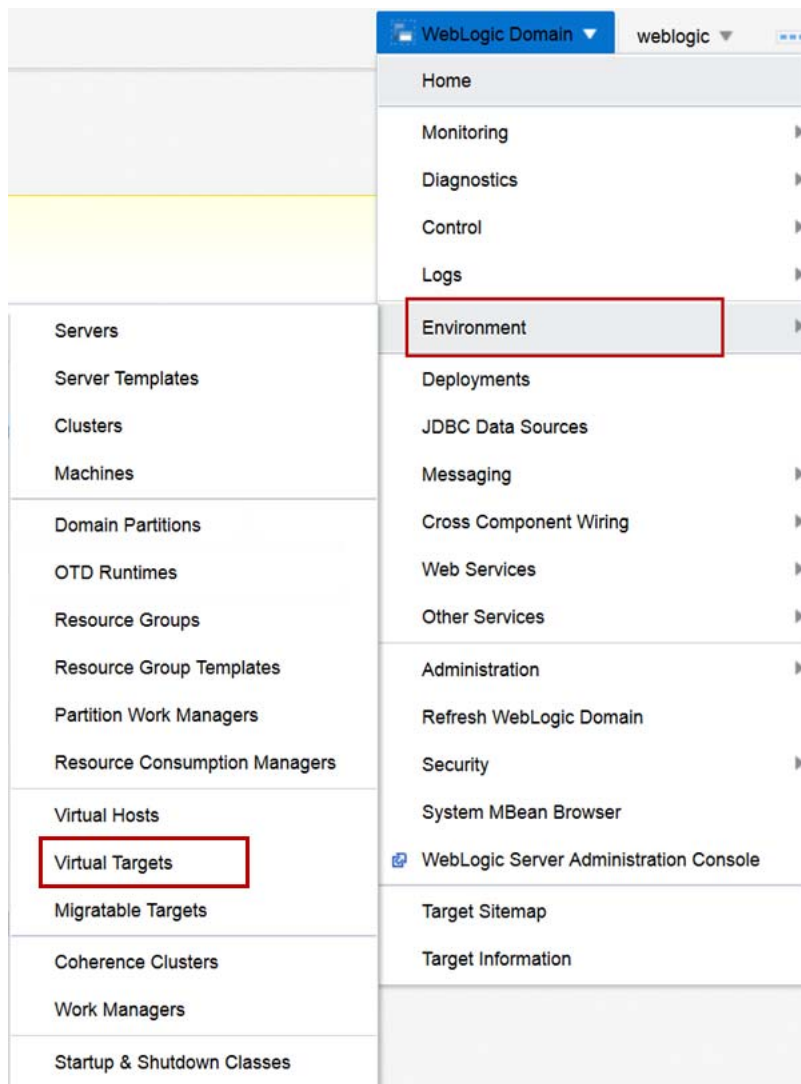
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

- c. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Clusters**.



- d. Select the **appcluster** entry, click **Control** and select **Start**.
- e. Click **Start Servers**.
- The request is sent to Node Manager to start the servers in appcluster.

2. Create a Virtual Target for baylandurgentcare.
  - a. From the **WebLogic Domain** drop-down menu, select **Environment** and then select **Virtual Targets**.



- b. Click **Create**.
    - c. For **Name**, enter baylandurgentcare and for **URI Prefix** enter /bayland. Click **Next**.

Create Virtual Target: General

BackStep 1 of 2NextCancel

Name \*baylandurgentcare

Uri Prefix/bayland

Hosts

Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

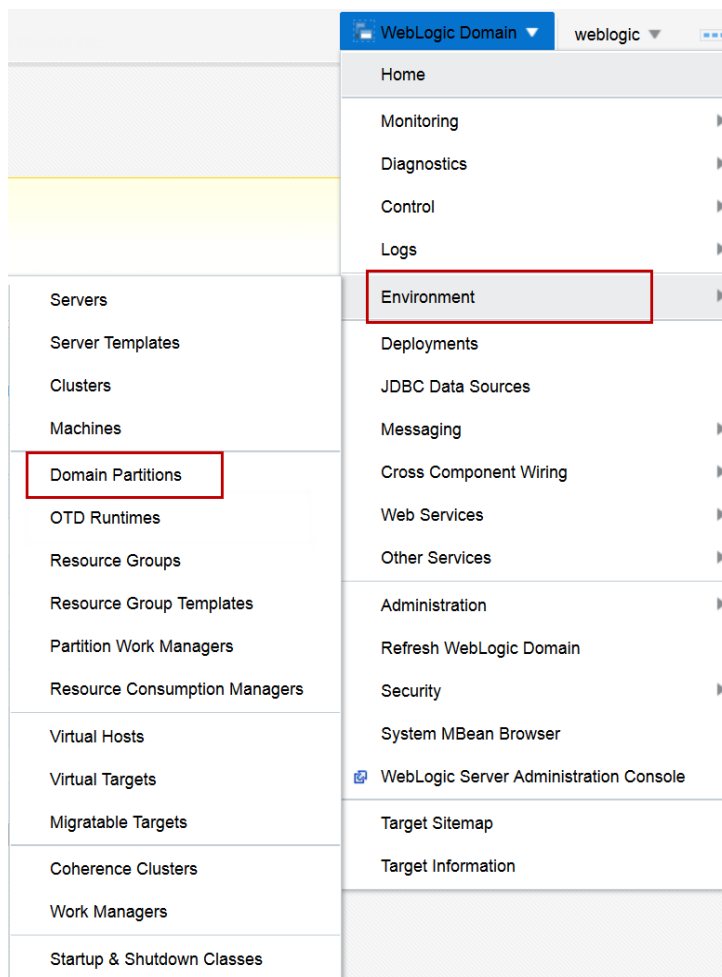
View ▾+ Add X Delete

| Host Name                                        |
|--------------------------------------------------|
| No host names specified for this virtual target. |

▶ Advanced Options

- d. From the **Cluster** menu select **appcluster** . Click **Create**.  
The Virtual Target **baylandurgentcare** creates successfully.

3. Create a Domain Partition for baylandurgentcare.
  - a. In the Browser window, from the **WebLogic Domain** drop-down menu, select **Environment** and then select **Domain Partitions**.



**Tip:** If the Lifecycle Manager is not enabled, click the button to enable it for this domain.

- b. Click **Create**.
- c. For **Name**, enter `baylandurgentcare` and for **Security Realm** select **myrealm**. Click **Next**.
- d. Select the check boxes for the **Virtual Target baylandurgentcare** in the **Select** column and in the **Set as Default** column. Click **Next**.
- e. For **Resource Group Name**, enter `baylandurgentcare`.
- f. Move the **baylandurgentcare** target from **Available Targets** to **Selected Targets**. Click **Next**.

Create Domain Partition: Summary

Back

Step 4 of 4

Next

Create

Cancel

Information

This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

|                                |                   |
|--------------------------------|-------------------|
| Name                           | baylandurgentcare |
| Realm                          | myrealm           |
| Primary Identity Domain        | None              |
| OTD Runtime                    | None              |
| Available Targets              | baylandurgentcare |
| Default Targets                | baylandurgentcare |
| Resource Group                 | baylandurgentcare |
| Resource Group Template        | None              |
| Targets for the Resource Group | baylandurgentcare |

- g. Review the summary and click **Create**.  
The Domain Partition **baylandurgentcare** creates successfully.
4. Create a Data Source for the baylandurgentcare Domain Partition.
  - a. In the Browser window, click the **baylandurgentcare** Domain Partition.
  - b. From the **Domain Partition** menu, select **JDBC Data Sources**.
  - c. Click **Create** and select **Generic Data Source**.
  - d. For **Data Source Name**, enter MedRecGlobalDataSourceXA.
  - e. For **JNDI Name**, enter jdbc/MedRecGlobalDataSourceXA.
  - f. Click **Next**.
  - g. Click **Generate URL and Properties...**
  - h. For **Database Host Name**, enter fmwdb.
  - i. For **Database Listener Port**, enter 1521 .
  - j. For **Database Name**, enter PDB1 .
  - k. For **Database User Name**, enter MEDREC .
  - l. For **Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
  - m. For **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Database product.
  - n. Click **OK**.
  - o. Click **Test Database Connection**.  
The test should succeed.

- p. If the test fails, validate that the listener is started, the database is started, and the pluggable databases are open on host fmwdb.
- 1) Connect to fmwdb as the `oracle` user.
    - a) Refer to the instructions provided by your instructor to connect to your machine.
    - b) Open a Terminal window to the oracle user home directory.
  - 2) Execute the `lsnrctl` utility with the `start` command.
 

```
$ lsnrctl start
```

Note: your output may vary – if the listener is running you will receive a message that it's already started.
  - 3) Execute the `sqlplus` utility, using `sys as sysdba` for user-name. For password, enter the password from the Course Practice Environment: Security Credentials document for the `sys` user for the Oracle Database product.
 

```
$ sqlplus
Enter user-name: sys as sysdba
Enter password:
```
  - 4) If the connection message to the database includes:
 

```
Connected to an idle instance.
```

 then execute startup to start the database:
 

```
SQL> startup
```

 the output includes database memory usage and:
 

```
Database mounted.
Database opened.
```
  - 5) Execute the command to open the PDB1 database.
 

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;
```
  - 6) Execute the command to open the PDB2 database.
 

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;
```
  - 7) Execute the query to list the mode for pluggable databases and validate that PDB1 and PDB2 are open for READ WRITE.
 

```
SQL> SELECT NAME, OPEN_MODE, RESTRICTED FROM vpdb;
```

| NAME      | OPEN_MODE  | RES |
|-----------|------------|-----|
| PDB\$SEED | READ ONLY  | NO  |
| PDBORCL   | MOUNTED    |     |
| NEW_PDB   | MOUNTED    |     |
| PDB0      | MOUNTED    |     |
| PDB1      | READ WRITE | NO  |
| PDB2      | READ WRITE | NO  |

6 rows selected.
  - 8) In the Browser, click **Test Database Connection**.  
The test should succeed.



- q. In the Browser, click **Next**.
- r. Accept the **Transaction Properties** defaults and click **Next**.

**Create a JDBC Data Source: Review** Back Step 5 of 5 Next Create Cancel

Review the settings for your new JDBC data source.

**Data Source Name** MedRecGlobalDataSourceXA

**Scope** Resource group "baylandurgentcare" in domain partition "baylandurgentcare"

**Type** Generic

**Driver Class Name** oracle.jdbc.OracleDriver

**JNDI Name** jdbc/MedRecGlobalDataSourceXA

**Database URL** jdbc:oracle:thin:@//fmwdb:1521/PDB1

**Properties**

| Name | Value  |
|------|--------|
| user | MEDREC |

**System Properties**

| Name                         | Value |
|------------------------------|-------|
| (No system properties found) |       |

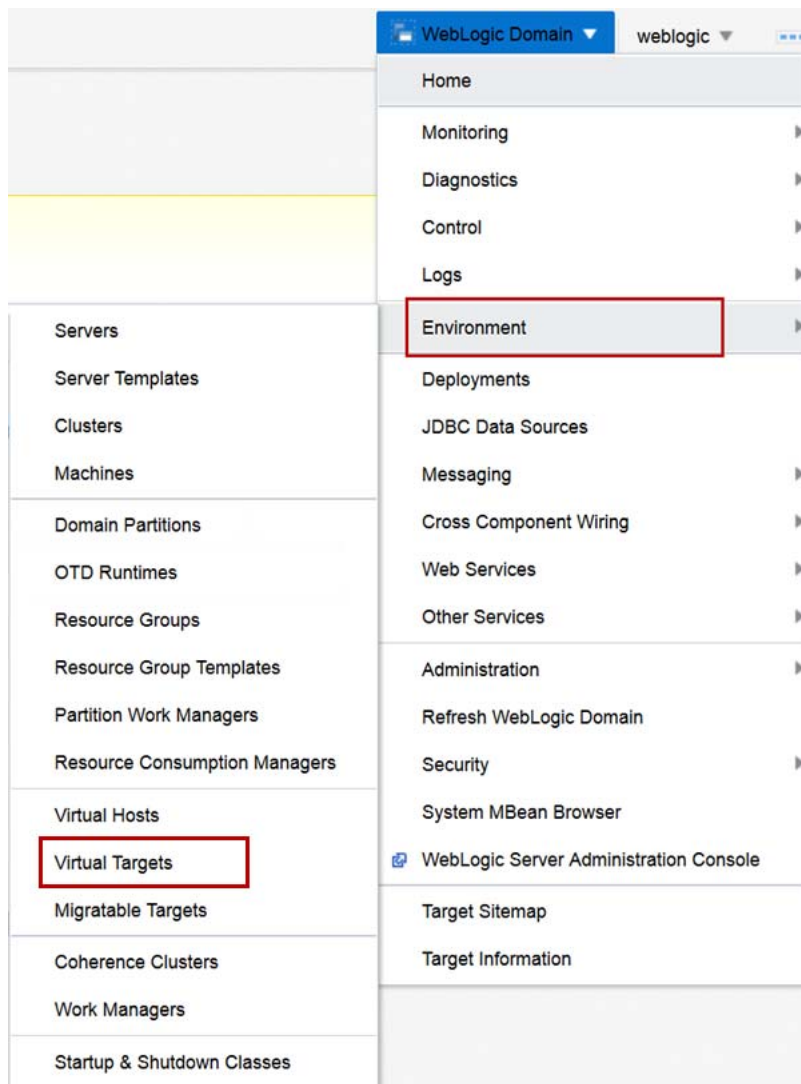
**Transaction Options** Supports global transactions and use the 'One-Phase Commit' global transaction protocol.

**Targets**

| Name                    | Type |
|-------------------------|------|
| (No targets specified). |      |

- s. Review the settings and click **Create**.
5. Deploy applications to the baylandurgentcare Domain Partition.
  - a. From the **Domain Partition** menu, select **Deployments**.
  - b. From the **Deployment** menu, select **Deploy**.
  - c. In the **Deploy Java EE Application: Select Archive** page, select **Archive or exploded directory is on the server where Enterprise Manager is running** and click **Browse...**
  - d. Navigate to the  
 /u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/stand  
 alone directory, select the medrec.ear file and click **OK**.
  - e. Click **Next**.
  - f. In the **Distribution** section, select **Install only. Do not start**.
  - g. Click **Next**.
  - h. Review the Deployment Settings and click **Deploy**.
  - i. After the Deployment succeeds, click **Close**.
6. Start the baylandurgentcare Domain Partition.
  - a. Next to the **Domain Partition** menu, click **Start Up**.
  - b. After the **Domain Partition** starts, click **Close**.
7. Test the baylandurgentcare Virtual Target.
  - a. Open a new Browser tab, navigate to <http://fmw1:7101/bayland/medrec>.

8. Create a Virtual Target for valleyhealth.
  - a. In the Browser window, from the **WebLogic Domain** drop-down menu, select **Environment** and then select **Virtual Targets**.



- b. Click **Create**.

- c. For **Name**, enter `valleyhealth` and for **URI Prefix** enter `/valley`. Click **Next**.

**Create Virtual Target: General** Back Step 1 of 2 Next Cancel

Name \*

Uri Prefix

**Hosts**

Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

View + Add X Delete

| Host Name                                        |
|--------------------------------------------------|
| No host names specified for this virtual target. |

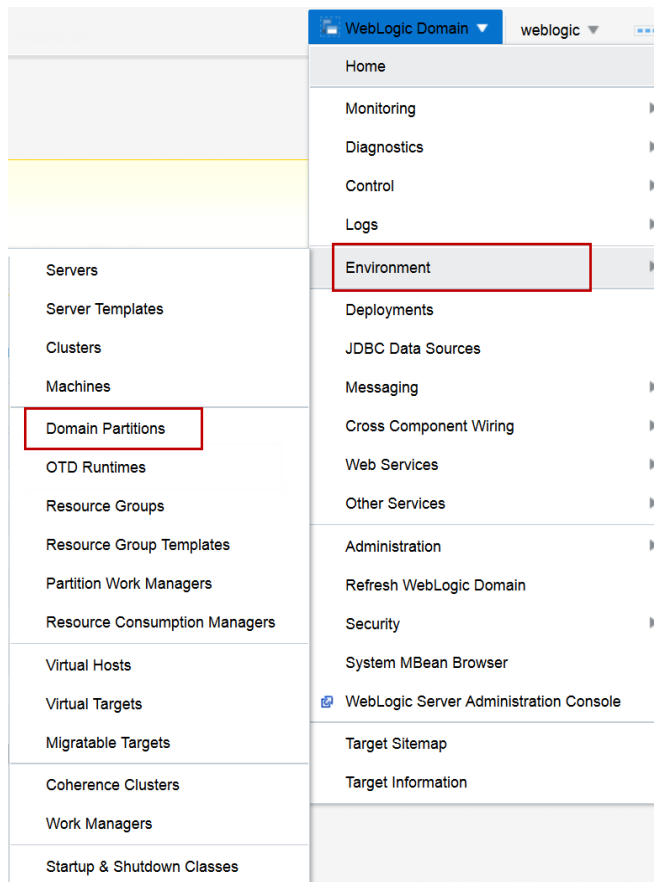
[Advanced Options](#)

- d. From the **Cluster** menu select **appcluster**. Click **Create**.

The Virtual Target **valleyhealth** creates successfully.

9. Create a Domain Partition for valleyhealth.

- a. In the Browser window, from the **WebLogic Domain** dropdown menu, select **Environment** and then select **Domain Partitions**.



**Tip:** If the Lifecycle Manager is not enabled, click the button to enable it for this domain.

- b. Click **Create**.
- c. For **Name**, enter `valleyhealth` and for **Security Realm** select **myrealm**. Click **Next**.
- d. Select the check boxes for the **Virtual Target valleyhealth** in the **Select** column and in the **Set as Default** column. Click **Next**.
- e. For Resource Group Name, enter `valleyhealth`.
- f. Move the **valleyhealth** target from **Available Targets** to **Selected Targets**. Click **Next**.

**Create Domain Partition: Summary**    Back    Step 4 of 4    Next    Create    Cancel

**Information**

This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

|                                |              |
|--------------------------------|--------------|
| Name                           | valleyhealth |
| Realm                          | myrealm      |
| Primary Identity Domain        | None         |
| OTD Runtime                    | None         |
| Available Targets              | valleyhealth |
| Default Targets                | valleyhealth |
| Resource Group                 | valleyhealth |
| Resource Group Template        | None         |
| Targets for the Resource Group | valleyhealth |

- g. Review the summary and click **Create**.  
The domain partition **valleyhealth** creates successfully.
10. Create a Data Source for the valleyhealth domain partition.
- a. In the Browser window, click the **valleyhealth** domain partition.
  - b. From the **Domain Partition** menu, select **JDBC Data Sources**.
  - c. Click **Create** and select **Generic Data Source**.
  - d. For **Data Source Name**, enter `MedRecGlobalDataSourceXA`.
  - e. For **JNDI Name**, enter `jdbc/MedRecGlobalDataSourceXA`.
  - f. Click **Next**.
  - g. Click **Generate URL and Properties...**
  - h. For **Database Host Name**, enter `fmwdb`.
  - i. For **Database Listener Port**, enter `1521`.
  - j. For **Database Name**, enter `PDB2`.
  - k. For **Database User Name**, enter `MEDREC`.
  - l. For **Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.

- m. For **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
- n. Click **OK**.
- o. Click **Test Database Connection**.  
The test should succeed.
- p. Click **Next**.
- q. Accept the **Transaction Properties** defaults and click **Next**.

**Create a JDBC Data Source: Review** [Back] [Step 5 of 5] [Next] [Create] [Cancel]

Review the settings for your new JDBC data source.

**Data Source Name** MedRecGlobalDataSourceXA

**Scope** Resource group "valleyhealth" in domain partition "valleyhealth"

**Type** Generic

**Driver Class Name** oracle.jdbc.OracleDriver

**JNDI Name** jdbc/MedRecGlobalDataSourceXA

**Database URL** jdbc:oracle:thin:@//fmwdb:1521/PDB2

**Properties**

| Name | Value  |
|------|--------|
| user | MEDREC |

**System Properties**

| Name                         | Value |
|------------------------------|-------|
| (No system properties found) |       |

**Transaction Options** Supports global transactions and use the 'One-Phase Commit' global transaction protocol.

**Targets**

| Name                    | Type |
|-------------------------|------|
| (No targets specified). |      |

- r. Review the settings and click **Create**.
11. Deploy applications to the valleyhealth Domain Partition.
    - a. From the **Domain Partition** menu, select **Deployments**.
    - b. From the **Deployment** menu, select **Deploy**.
    - c. In the **Deploy Java EE Application: Select Archive** page, select **Archive is on the machine where this Web browser is running** and click **Browse...**
    - d. Navigate to the  
/u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/stand  
alone directory, select the medrec.ear file and click **OK**.
    - e. Click **Next**.
    - f. In the **Distribution** section, select **Install only. Do not start**.
    - g. Click **Next**.
    - h. Review the Deployment Settings and click **Deploy**.
    - i. After the Deployment succeeds, click **Close**.
  12. Start the valleyhealth domain partition.
    - a. Next to the **Domain Partition** menu, click **Start Up**.
    - b. After the **Domain Partition** starts, click **Close**.
  13. Test the valleyhealth Virtual Target.
    - a. Open a new Browser tab, navigate to <http://fmw1:7101/valley/medrec>.

Two tenants – baylandurgentcare and valleyhealth – have been created with domain partitions – using the MedRec sample application deployed to both partitions isolated by URI.

## Practice Solution: Creating Domain Partitions Using Fusion Middleware Control

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started creating the domain partitions or virtual targets, this script will not work. To run the script in that case: delete any domain partitions or virtual targets.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

### Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.  

```
$ cd /u01/practices/part1/practice03-04
```
  - b. Run the solution script.  

```
$./solution.sh
```
  - c. Enter any passwords as you are prompted.

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_partitions.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>>Practice 03-04 solution applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

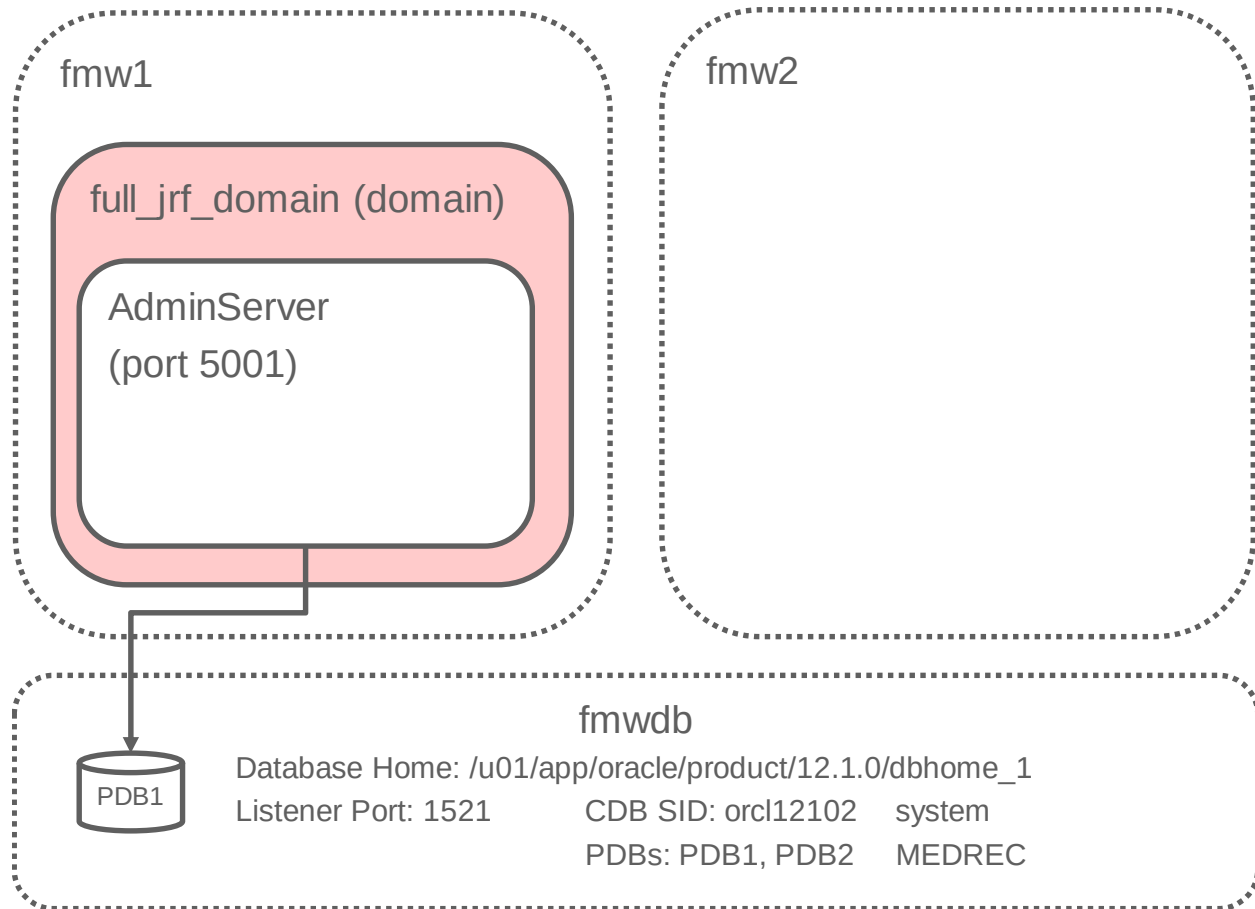
3. Close the Terminal window.

## Practice 3-5: Creating a Full JRF Domain to Support Multitenancy using Fusion Middleware Control

### Overview

In this practice, you create a WebLogic Server Domain that supports the full set of JRF functionality using the Fusion Middleware Configuration Wizard. To support full JRF, the WebLogic Server Domain requires a database and the Repository Creation Utility is used to create the required tables in a pluggable database. This practice creates a domain that uses Fusion Middleware Control for administration, Oracle Traffic Director and Oracle Coherence.

The following diagram illustrates the configuration:



### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed at `/u01/fmw/wls1221_infra/`.
- Oracle Database on host `fmwdb` is configured for JRF and is available.

### Tasks

1. Connect to `fmw1` as the `oracle` user.
  - a. Refer to the instructions provided by your instructor to connect to your machine.
  - b. Unless stated otherwise, you will be working within `fmw1` for the remainder of this practice.
2. Create the Full JRF schemas with the Repository Creation Utility.

- a. Open a Terminal window as the `oracle` user, and navigate to `/u01/fmw/wls1221_infra/`.

**Tip:** There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/
```

- b. Execute the Repository Creation Utility.

```
$ oracle_common/bin/rcu
```

3. After the command, the graphical Repository Creation Utility appears. Use the guidelines in the following table to configure the repository:

| Step | Window/Page Description                              | Choices or Values                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a.   | Step 1 of 8 – Welcome                                | Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| b.   | Step 2 of 8 – Create Repository                      | Select <b>Create Repository</b> and select <b>System Load and Product Load</b> .<br>Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                       |
| c.   | Step 3 of 8 – Database Connection Details            | For <b>Database Type</b> , select <b>Oracle Database</b> .<br>For <b>Host Name</b> , enter <code>fmwdb</code> .<br>For <b>Port</b> , enter <code>1521</code> .<br>For <b>Service Name</b> , enter <code>PDB1</code> .<br>For <b>Username</b> , enter <code>sys</code> .<br>For <b>Password</b> , enter the password from the Course Practice Environment: Security Credentials document for the <code>sys</code> user for the Oracle Database product.<br>Click <b>Next</b> . |
| d.   | Repository Creation Utility – Checking Prerequisites | After the operation has completed, click <b>OK</b> .                                                                                                                                                                                                                                                                                                                                                                                                                          |
| e.   | Step 4 of 8 – Select Components                      | Select <b>Create new prefix</b> .<br>For <b>prefix</b> , enter <code>FULL</code> .<br>Check <b>Oracle Platform Security Services</b> .<br>Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                 |
| f.   | Repository Creation Utility – Checking Prerequisites | After the operation has completed, click <b>OK</b> .                                                                                                                                                                                                                                                                                                                                                                                                                          |
| g.   | Step 5 of 8 – Schema Passwords                       | Select <b>Use same passwords for all schemas</b> .<br>For <b>Password</b> , enter the password from the Course Practice Environment: Security Credentials document for the <code>FULL_STB</code> user for the Oracle Database product..<br>For <b>Confirm Password</b> , enter the password from the Course Practice Environment: Security Credentials document for the <code>FULL_STB</code> user for the Oracle Database product.<br>Click <b>Next</b> .                    |



| Step | Window/Page Description                            | Choices or Values                                              |
|------|----------------------------------------------------|----------------------------------------------------------------|
| h.   | Step 6 of 8 – Map Tablespaces                      | Accept the default tablespace mappings and click <b>Next</b> . |
| i.   | Repository Creation Utility – Confirm              | Click <b>OK</b> to create tablespaces.                         |
| j.   | Repository Creation Utility – Creating Tablespaces | After the operation has completed, click <b>OK</b> .           |
| k.   | Step 7 of 8 - Summary                              | Click <b>Create</b> .                                          |
| l.   | Step 8 of 8 – Completion Summary                   | Click <b>Close</b> .                                           |

4. Create the Full JRF WebLogic Server Domain.

a. Open a Terminal window, and navigate to `/u01/fmw/wls1221_infra/`.

**Tip:** There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/
```

b. Execute the Fusion Middleware Configuration Wizard.

```
$ oracle_common/common/bin/config.sh
```

5. After the command, the graphical Fusion Middleware Configuration Wizard appears. Use the guidelines in the following table to configure the WebLogic Server Domain:

| Step | Window/Page Description              | Choices or Values                                                                                                                                                                                                                                                                                                              |
|------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a.   | Page 1 of 8 - Create Domain          | Click <b>Create a new domain</b> .<br>For <b>Domain Location</b> , enter <code>/u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain</code> and click <b>Next</b> .                                                                                                                                                     |
| b.   | Page 2 of 8 - Templates              | Select these <b>Available Templates</b> : <ul style="list-style-type: none"> <li>• Oracle Traffic Director – 12.2.1 [otd]</li> <li>• Oracle Enterprise Manager – 12.2.1 [em]</li> <li>• Oracle JRF – 12.2.1 [oracle_common]</li> <li>• WebLogic Coherence Cluster Extension – 12.2.1 [wlserver]</li> </ul> Click <b>Next</b> . |
| c.   | Page 3 of 12 – Application Location  | For <b>Application Location</b> , enter <code>/u01/fmw/wls1221_infra/user_projects/applications/full_jrf_domain</code> and click <b>Next</b> .                                                                                                                                                                                 |
| d.   | Page 4 of 12 – Administrator Account | For <b>Name</b> , enter <code>weblogic</code> .<br>For <b>Password</b> , enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product.                                                                                                  |

| Step | Window/Page Description                    | Choices or Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                            | For <b>Confirm Password</b> , enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                     |
| e.   | Page 5 of 12 – Domain Mode and JDK         | For <b>Domain Mode</b> select <b>Development</b> , and click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| f.   | Page 6 of 12 – Database Configuration Type | <p>Select <b>RCU Data</b>.</p> <p>For <b>Vendor</b>, select <b>Oracle</b>.</p> <p>For <b>Driver</b>, select <b>Oracle's Driver (Thin) for Service connections</b>.</p> <p>For <b>DMBS/Servers</b>, enter PDB1 .</p> <p>For <b>Host Name</b>, enter fmwdb .</p> <p>For <b>Port</b>, enter 1521 .</p> <p>For <b>Schema Owner</b>, enter FULL_STB .</p> <p>For <b>Schema Password</b>, enter the password from the Course Practice Environment: Security Credentials document for the FULL_STB user for the Oracle Database product.</p> <p>Click <b>Get RCU Configuration</b>.</p> <p>Click <b>Next</b>.</p> |
| g.   | Page 7 of 12 – Component Datasources       | Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| h.   | Page 8 of 12 – JDBC Test                   | Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| i.   | Page 9 of 13 – Advanced Configuration      | <p>Select <b>Administration Server</b>.</p> <p>Click <b>Next</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| j.   | Page 10 of 13 – Administration Server      | <p>For <b>Listen Port</b>, enter 5001 .</p> <p>Click <b>Next</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| k.   | Page 11 of 13 – Configuration Summary      | Click <b>Create</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| l.   | Page 12 of 13 – Configuration Progress     | Click <b>Next</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| m.   | Page 13 of 13 – End of Configuration       | Click <b>Finish</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

6. Log in to the WebLogic Server Domain using Fusion Middleware Control.
  - a. In the Terminal window navigate to  
`/u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain.`  
`$ cd`  
`/u01/fmw/wls1221_infra/user_projects/domains/full_jrf_domain`
  - b. Execute `startWebLogic.sh` to start the WebLogic Administration Server.  
`$ ./startWebLogic.sh`
  - c. Open a Browser window and navigate to **http://fmw1:5001/em.**
  - d. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the `weblogic` user for the Oracle WebLogic Server product. Click **Login**.  
The home page for the **full\_jrf\_domain** is displayed.
- The **full\_jrf\_domain** is created and running.
7. Stop the WebLogic Server Administration Server because the restricted JRF domain will be used for later practices.
  - a. From the **WebLogic Domain** menu, select **Environment** and then select **Servers**.
  - b. Click **AdminServer**.
  - c. Click **Control**, click **Shutdown**, and select **Force Shutdown Now**.
  - d. Click **Forcibly Shutdown Servers**.



# **Practices for Lesson 4 - Resource Groups**

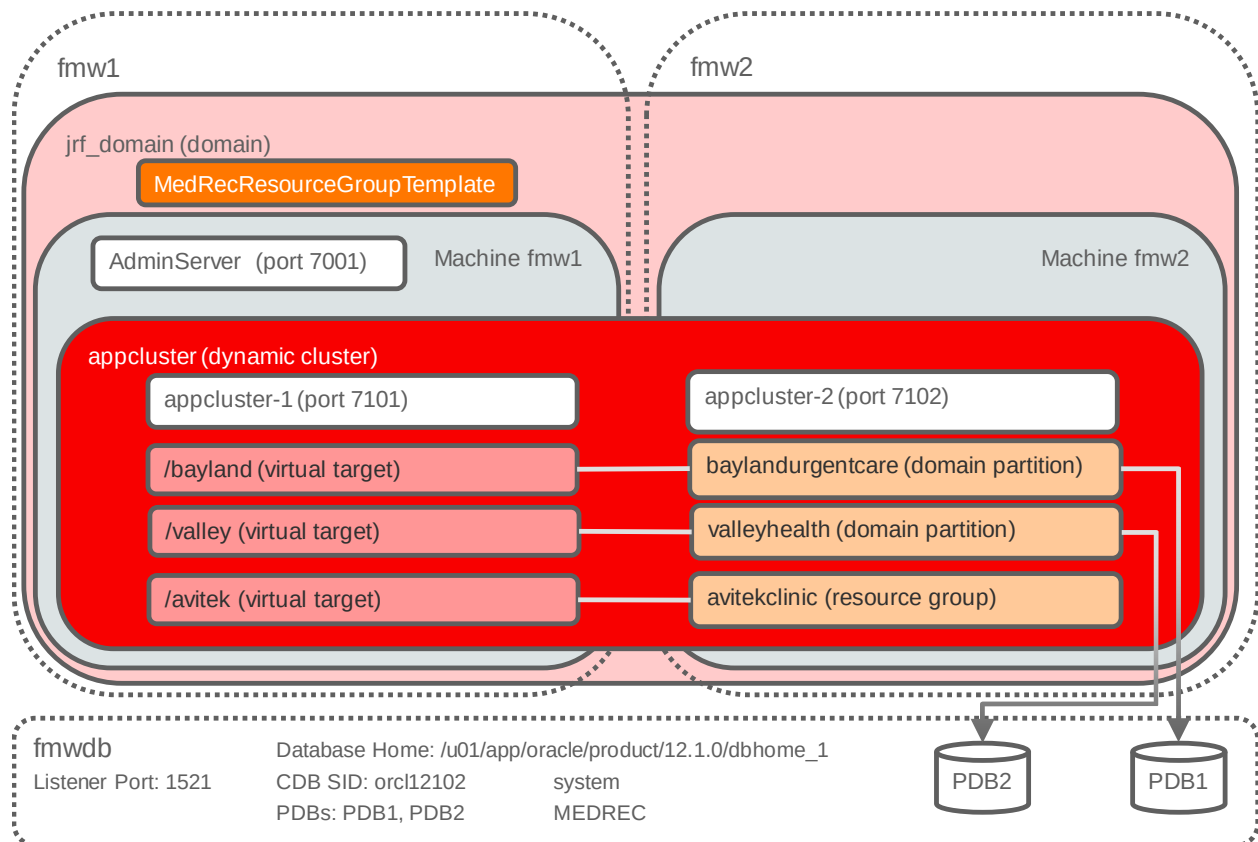
## **Chapter 4**

## Practices for Lesson 4: Overview

### Overview

In these practices, you create a Resource Group Template with a data source, JMS module and applications for the Avitek Medical Records (MedRec) sample application. Then, you create a Domain Partition that uses the Resource Group Template to create a Resource Group and override properties to meet the tenant requirements.

The following diagram illustrates the configuration:



## Practice 4-1: Creating a Resource Group Template

---

### Overview

In this practice, using Fusion Middleware Control, you create a new Resource Group Template and Java application resources including JDBC Data Sources, JMS resources and mail sessions for MedRec sample application deployments.

Because many WebLogic Server customers have requirements for this type of configuration—a Java application with a similar set of resources for each deployment configuration—this is a useful practice to execute the steps to create a Resource Group Template.

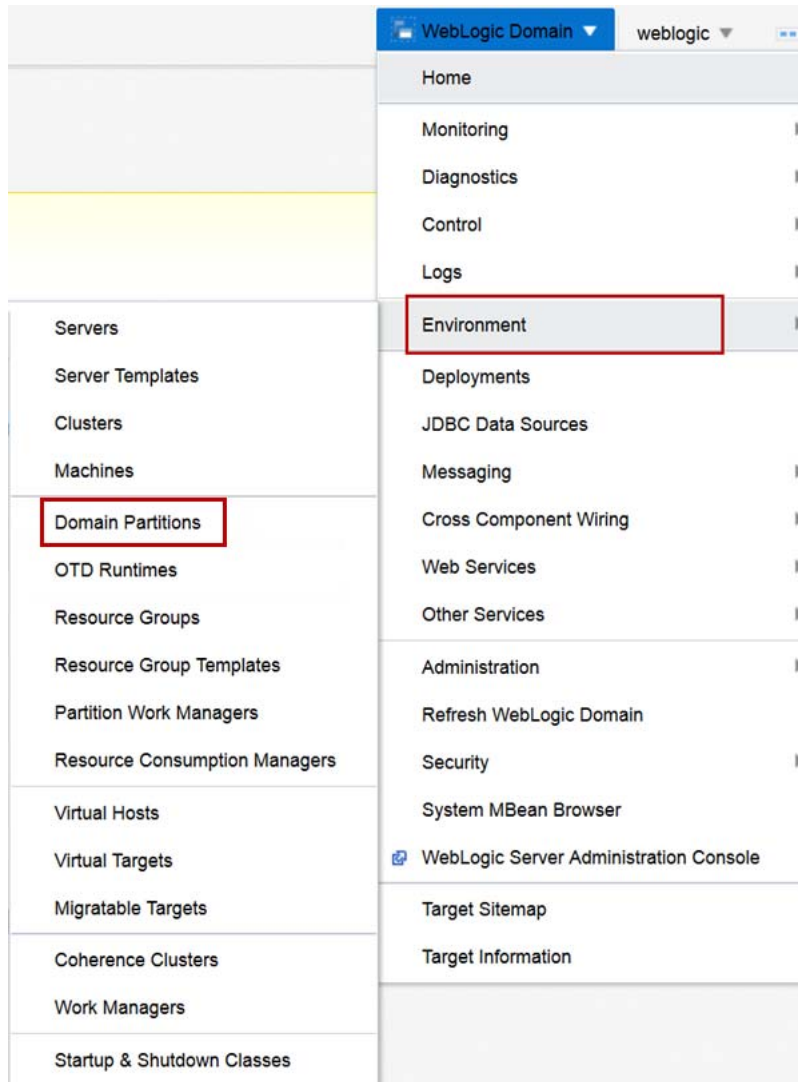
### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fmw1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

### Tasks

1. Log in to the jrf\_domain WebLogic Server Domain using Fusion Middleware Control.
  - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

2. If the domain partitions baylandurgentcare and valleyhealth are still configured and running, stop and delete both partitions.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.

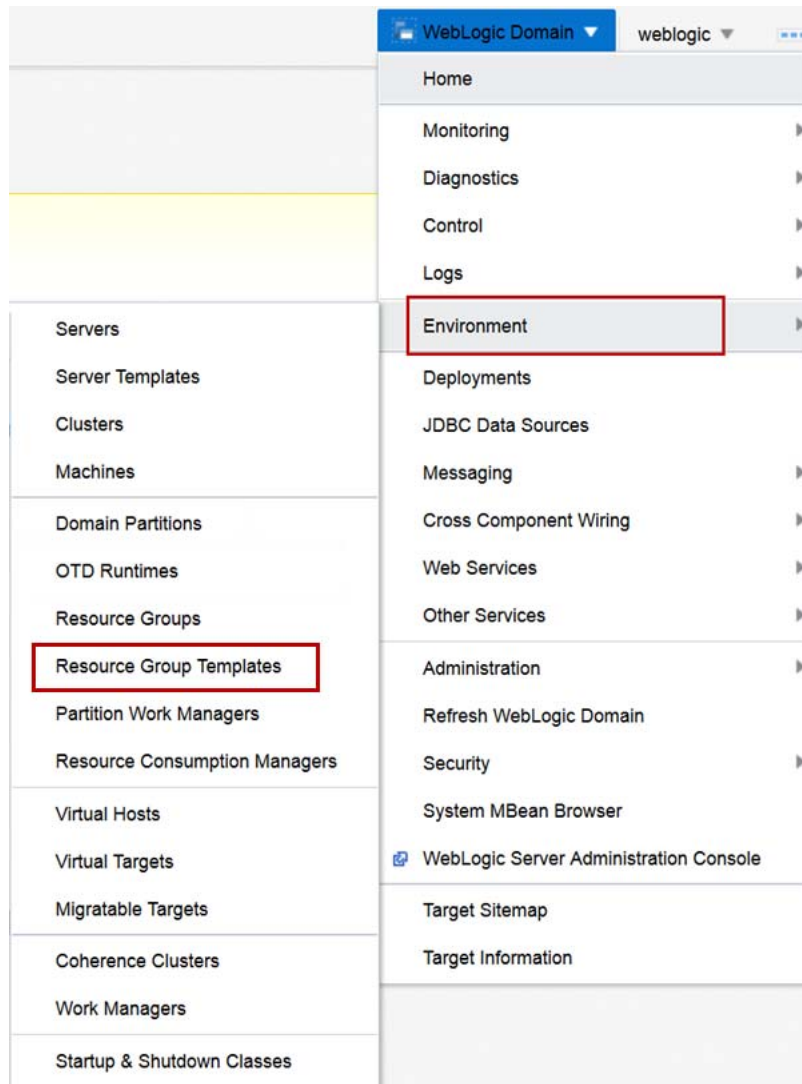


- b. Click the table entry for the **baylandurgentcare** domain partition.
- c. Click **Control**, select **Stop**, and select **Force stop now**.
- d. Click **Close**.
- e. Click the table entry for the **baylandurgentcare** domain partition.
- f. Click **Delete**.
- g. In the **Delete Domain Partition** dialog box, click **OK**.
- h. Click the table entry for the **valleyhealth** domain partition.
- i. Click **Control**, select **Stop**, and select **Force stop now**.
- j. Click **Close**.
- k. Click the table entry for the **valleyhealth** domain partition.
- l. Click **Delete**.
- m. In the **Delete Domain Partition** dialog box, click **OK**.



The baylandurgentcare and valleyhealth domain partitions are deleted—note that you will create these partitions again in these practices—but with a Resource Group Template. Also note that the Virtual Targets for both domain partitions will be used again in these practices.

3. Create a Resource Group Template for a MedRec type application.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates**.



- b. Click **Create**.
  - c. For **Name**, enter `MedRecResourceGroupTemplate` and click **Create**.  
The Resource Group Template **MedRecResourceGroupTemplate** creates successfully.
4. Create the JDBC Data Source for the `MedRecResourceGroupTemplate`.
  - a. In the Fusion Middleware Control window, click the **MedRecResourceGroupTemplate**.
  - b. Click **Services**.
  - c. In the **JDBC** tab, click **Create** and select **Generic Data Source**.
  - d. For **Name**, enter `MedRecGlobalDataSourceXA`.

- e. For **JNDI Name**, enter `jdbc/MedRecGlobalDataSourceXA`.
- f. For **Driver Class Name**, select `oracle.jdbc.OracleDriver`.
- g. Click **Next**.
- h. Click **Generate URL and Properties...**
- i. For **Database Host Name**, enter `fmwdb`.
- j. For **Database Listener Port**, enter `1521`.
- k. For **Database Name**, enter `PDB1`.
- l. For **Database User Name**, enter `MEDREC`.
- m. For **Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
- n. For **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
- o. Click **OK**.
- p. Click **Test Database Connection**.  
The test should succeed.
- q. Click **Next**.
- r. Accept the **Transaction Properties** defaults and click **Next**.

**Create a JDBC Data Source: Review** Back Step 5 of 5 Next Create Cancel

Review the settings for your new JDBC data source.

**Data Source Name** MedRecGlobalDataSourceXA

**Scope** Resource group template "MedRecResourceGroupTemplate"

**Type** Generic

**Driver Class Name** oracle.jdbc.OracleDriver

**JNDI Name** jdbc/MedRecGlobalDataSourceXA

**Database URL** jdbc:oracle:thin:@fmwdb:1521/PDB1

**Properties**

| Name | Value  |
|------|--------|
| user | MEDREC |

**System Properties**

| Name                         | Value |
|------------------------------|-------|
| (No system properties found) |       |

**Transaction Options** Supports global transactions and use the 'One-Phase Commit' global transaction protocol.

**Targets**

| Name                   | Type |
|------------------------|------|
| (No targets specified) |      |

- s. Review the settings and click **Create**.  
The JDBC Data Source MedRecGlobalDataSourceXA creates successfully.
5. Create the JMS Messaging resources for the MedRecResourceGroupTemplate.
- a. In the Fusion Middleware Control window, click the **Messaging** subtab in the **Services** tab of the **MedRecResourceGroupTemplate**.
  - b. Click the **JMS Servers** tab.
  - c. Click **Create**.
  - d. For **Name**, enter `MedRecJMSServer` and click **Create a Store** and select **File Store**.

- e. For **Name**, enter MedRec-fs and click **Next**.

**Create a File Store: Review** Back Step 3 of 3 Next Create Cancel

Review the settings for your new file store.

**Name** MedRec-fs

**Scope** Resource group template "MedRecResourceGroupTemplate"

**Directory**

- f. Click **Create**.
- g. Select the **MedRec-fs(File Store)** as the **Persistent Store** and click **Next**.

**Create a JMS Server: Review** Back Step 3 of 3 Next Create Cancel

Review the settings for your new JMS server.

**Name** MedRecJMSServer

**Scope** Resource group template "MedRecResourceGroupTemplate"

**Persistent Store** MedRec-fs

- h. Review the JMS Server settings and click **Create**.
- i. Click the **JMS Modules** tab.
- j. Click **Create**.
- k. For **Name**, enter MedRecModule and click **Next**.

**Create a JMS System Module: Review** Back Step 3 of 3 Next Create Cancel

Review the settings for your new JMS module.

**Name** MedRecModule

**Scope** Resource group template "MedRecResourceGroupTemplate"

**Descriptor File** Not specified

- l. Click **Create**.
- m. Click the JMS Module MedRecModule.
- n. Click the **Subdeployments** tab.
- o. Click **Create**.

- p. Enter **MedRecJMS** for **SubDeployment Name**, select the **MedRecJMSServer** and click **Create**.

**Create a Subdeployment**

Create Cancel

Subdeployment Name: MedRecJMS

Select targets for this subdeployment.

**JMS Servers**

| Select                           | Name            |
|----------------------------------|-----------------|
| <input checked="" type="radio"/> | MedRecJMSServer |

**SAF Agents**

| Select             | Name |
|--------------------|------|
| No data to display |      |

- q. Click the **General** tab in the JMS Module **MedRecModule**.
- r. Click **Create**.
- s. Select **Connection Factory** and click **Next**.
- t. Click **Next**.
- u. Enter **MedRecConnectionFactory** for **Name** and **com.oracle.medrec.jms.connectionFactory** for **JNDI Name**. Click **Next**.

**Create a Connection Factory: Review**

Back Step 5 of 5 Next Create Cancel

Review the settings for the new JMS resource that you are going to create.

JMS Module: MedRecModule

Scope: Resource group template "MedRecResourceGroupTemplate"

Resource Type: Connection Factory

Name: MedRecConnectionFactory

JNDI Name: com.oracle.medrec.jms.connectionFactory

Notes:

Client ID Policy: Restricted

Subscription Sharing Policy: Exclusive

Prefetch Mode for Synchronous Consumer: disabled

Maximum Messages per Session: 10

Load Balancing Enabled: true

Server Affinity Enabled: true

Enable XA connection factory (enables JTA transaction support): true

Transacted session transaction timeout: 3600

- v. Review the Connection Factory settings and click **Create**.
- w. Click the **MedRecConnectionFactory**.

- x. In the **General** tab, scroll down to the **Targeting** section. Select **Subdeployment targeting** for **Targeting Policy** and select the **MedRecJMS** for **Subdeployment** and click **Apply**.

**Connection Factory: MedRecConnectionFactory**

Configuration Notes

**General** Advanced Configuration Default Delivery Flow Control

**Apply** **Revert**

Use this page to define the general configuration parameters for this JMS connection factory, which includes various client connection, default delivery, load balancing, and security parameters.

**Name** MedRecConnectionFactory

**Scope** Resource group template "MedRecResourceGroupTemplate"

**JNDI Name** com.oracle.medrec.jms.connectionFactory

**Client ID Policy** Restricted

**Subscription Sharing Policy** Exclusive

**Prefetch Mode for Synchronous Consumer** Disabled

**Maximum Messages per Session** 10

**Distributed Destination Load Balancing**

**Load Balancing Enabled** ☒

**Server Affinity Enabled** ☒

**Transaction**

**XA Connection Factory Enabled** ☒

**Transaction Timeout** 3600

**Targeting**

Oracle recommends using default targeting for connection factories under most circumstances.

**Targeting Policy**

☒ Subdeployment targeting

☐ Default to the parent module's target

**Subdeployment** MedRecJMS

- y. Click the **General** tab in the JMS Module **MedRecModule**.
- z. Click **Create**.
- aa. Select **Uniform Distributed Queue** and click **Next**.
- bb. Click **Next**.
- cc. Enter PatientNotificationQueue for **Name** and com.oracle.medrec.jms.PatientNotificationQueue for **JNDI Name**. Click **Next**.

**Create a Uniform Distributed Queue: Review** Back Step 5 of 5 Next Create Cancel

Review the settings for the new JMS resource that you are going to create.

JMS Module MedRecModule

Scope Resource group template "MedRecResourceGroupTemplate"

Resource Type Uniform Distributed Queue

Name PatientNotificationQueue

Notes com.oracle.medrec.jms.PatientNotificationQueue

JNDI Name

Template

Load Balancing Policy Round-Robin

Unit-of-Order Routing Hash

Forward Delay -1

- dd. Review the Uniform Distributed Queue settings and click **Create**.
- ee. Click the **PatientNotificationQueue**.
- ff. Scroll down to the **Targeting** section, select **Subdeployment targeting** for **Targeting Policy**, and select the **MedRedJMS** for **Subdeployment** and click **Apply**.

**Uniform Distributed Queue: PatientNotificationQueue**

[To configure and manage this Uniform Distributed Queue, use the WebLogic Server Administration Console.](#)

**Configuration** Notes

**General** Advanced Configuration Delivery Failure Delivery Overrides Logging Thresholds and Quotas

Apply Revert

Use this page to define the general configuration parameters for a distributed queue, such as selecting a destination key for sorting messages as they arrive on the distributed queue members.

Name PatientNotificationQueue

Scope Resource group template "MedRecResourceGroupTemplate"

JNDI Name

Template None

**Distributed Configuration**

Load Balancing Policy Round-Robin

Unit-of-Order Routing Hash

Set the Forward Delay ☐ Forward ☒ No messages are forwarded

delay (secs) -1

Reset Delivery Count On Forward ☒

**Targeting**

Oracle strongly recommends avoiding default targeting with distributed destination. Instead, use a subdeployment target that in turn references one or more JMS Servers.

Targeting Policy ☒ Subdeployment targeting ☐ Default to the parent module's target

Subdeployment MedRecJMS

The JMS Server and resources have been created.

6. Create the Mail session for the MedRecResourceGroupTemplate.
  - a. In the Fusion Middleware Control window, click the **Mail** subtab in the **Services** tab of the **MedRecResourceGroupTemplate**.
  - b. Click **Create**.
  - c. Enter `MedRecMailSession` for **Name** and enter `mail/MedRecMailSession` for **JNDI Name**.
  - d. Select **Resource Group Template** for **Scope** and select the **MedRecResourceGroupTemplate**.
  - e. Click **Next**.

**Create a Mail Session: Review**    Back    Step 3 of 3    Next    Create    Cancel

Review the settings for your new mail session.

|                  |                                                       |
|------------------|-------------------------------------------------------|
| Name             | MedRecMailSession                                     |
| Scope            | Resource group template "MedRecResourceGroupTemplate" |
| JNDI Name        | mail/MedRecMailSession                                |
| Session Username |                                                       |
| Target           |                                                       |

- f. Click **Create**.  
The mail session for the MedRecResourceGroupTemplate is created.
7. Deploy the MedRec Java applications to the MedRecResourceGroupTemplate.
  - a. In the Fusion Middleware Control window, click the **Deployments** tab of the **MedRecResourceGroupTemplate**.
  - b. Click **Deployment** and select **Deploy**.
  - c. Click **Archive is on the machine where this Web browser is running**.
  - d. Click **Browse...**
  - e. Navigate to the `/u01/fmw/wls1221_infra/wlserver/samples/server/medrec/dist/standalone` directory, select the `medrec.ear` file and click **Open**.
  - f. Click **Next**.

Deploy Java EE Application: Deployment Settings
Back
Step 4 of 4
Next
Deploy
Cancel

Hide Deployment Summary

Archive Type

Java EE Application (EAR file)

Archive Location

/u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/standalone/medrec.ear

Deployment Plan

Create a new plan

Scope

Resource group template \*MedRecResourceGroupTemplate\*

Deployment Type

Application

Application Name

medrec

Version

Not versioned

Context Root

[medrec-web.war]; medrec<BR>[medrec-jaxrs-facade-broker.war]; medrec-jaxrs-services<BR>[medrec-jaxws-facade-broker.war]; medrec-jaxws-services<BR>

Deployment Mode

Install only. Do not start.

Deployment Tasks

The table below lists common tasks that you may wish to do before deploying the application.

| Name                           | Go To Task | Description                                                                               |
|--------------------------------|------------|-------------------------------------------------------------------------------------------|
| Configure Web Modules          |            | Configure the Web modules in your application.                                            |
| Configure EJBs                 |            | Configure the Enterprise Java Beans in your application.                                  |
| Configure Application Security |            | Configure application policy migration, credential migration and other security behavior. |
| Configure Persistence          |            | Configure JPA persistence units.                                                          |

Deployment Plan

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

Edit Deployment Plan

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

Save Deployment Plan

- g. Review the Application Attributes and click **Deploy**.
- h. After the application successfully deploys, click **Close**.
- i. Click **Deployment** and select **Deploy**.
- j. Click **Archive is on the machine where this Web browser is running**.
- k. Click **Browse...**
- l. Navigate to the  
/u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/standalone directory, select the chat.war file and click **Open**.
- m. Click **Next**.



Deploy Java EE Application: Deployment Settings
Back
Step 4 of 4
Next
Deploy
Cancel

Hide Deployment Summary

Archive TypeWeb Module (WAR file)

Archive Location/u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/standalone/chat.war

Deployment PlanCreate a new plan

ScopeResource group template 'MedRecResourceGroupTemplate'

Deployment TypeApplication

Application Namechat

VersionNot versioned

Context Rootmedrec/chat

Deployment ModeInstall only. Do not start.

Deployment Tasks

The table below lists common tasks that you may wish to do before deploying the application.

| Name                           | Go To Task | Description                                                                               |
|--------------------------------|------------|-------------------------------------------------------------------------------------------|
| Configure Application Security |            | Configure application policy migration, credential migration and other security behavior. |

Deployment Plan

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

Edit Deployment Plan

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

Save Deployment Plan

- n. Review the Application Attributes and click **Deploy**.
- o. After the application is successfully deployed, click **Close**.
- p. Click **Deployment** and select **Deploy**.
- q. Click **Archive is on the machine where this Web browser is running**.
- r. Click **Browse...**
- s. Navigate to the  
**/u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/standalone**  
directory, select the **physician.ear** file and click **Open**.
- t. Click **Next**.

### Deploy Java EE Application: Deployment Settings

Back Step 4 of 4 Next Deploy Cancel

**Hide Deployment Summary**

Archive Type Java EE Application (EAR file)

Archive Location /u01/fmw/wls1221\_infra/wlserver/samples/server/medrec/dist/standalone/physician.ear

Deployment Plan Create a new plan

Scope Resource group template \*MedRecResourceGroupTemplate\*

Deployment Type Application

Application Name physician

Version Not versioned

Context Root medrec/physician

**Deployment Mode** Install only. Do not start.

**Deployment Tasks**

The table below lists common tasks that you may wish to do before deploying the application.

| Name                           | Go To Task | Description                                                                               |
|--------------------------------|------------|-------------------------------------------------------------------------------------------|
| Configure Web Modules          |            | Configure the Web modules in your application.                                            |
| Configure EJBs                 |            | Configure the Enterprise Java Beans in your application.                                  |
| Configure Application Security |            | Configure application policy migration, credential migration and other security behavior. |

**Deployment Plan**

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

[Edit Deployment Plan](#)

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

[Save Deployment Plan](#)

- u. Review the Application Attributes and click **Deploy**.
- v. After the application is successfully deployed, click **Close**.

### Resource Group Template : MedRecResourceGroupTemplate

General **Deployments** Services Notes

This page displays a list of Java EE applications and standalone application modules that have been deployed to this resource group template.

View  Deployment

|  | Name      | Status | State     | Health | Type                   |
|--|-----------|--------|-----------|--------|------------------------|
|  | chat      | N/A    | Installed | N/A    | Web Application        |
|  | medrec    | N/A    | Installed | N/A    | Enterprise Application |
|  | physician | N/A    | Installed | N/A    | Enterprise Application |

Columns Hidden 4 Columns Frozen 1 Deployments 3 of 3

The MedRec applications are successfully deployed to the MedRecGroupTemplate.

## Practice Solution: Creating a Resource Group Template

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started to create the Resource Group Template, delete the partially created Resource Group Template before executing the solution.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

### Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.  

```
$ cd /u01/practices/part1/practice04-01
```
  - b. Run the solution script.  

```
$./solution.sh
```

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_rgt.py`, which creates the resource group template for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>>Solution for Practice 04-01 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

## Practice 4-2: Creating a Domain Partition from a Resource Group Template

---

### Overview

In this practice, you create the security realms for baylandurgentcare and valleyhealth – because it is required that different administrators work with each application. Because the virtual targets for baylandurgentcare and valleyhealth were created in the practices for the lesson titled “Domain Partitions,” they are reused here. Finally, you create the domain partitions using the MedRecResourceGroupTemplate.

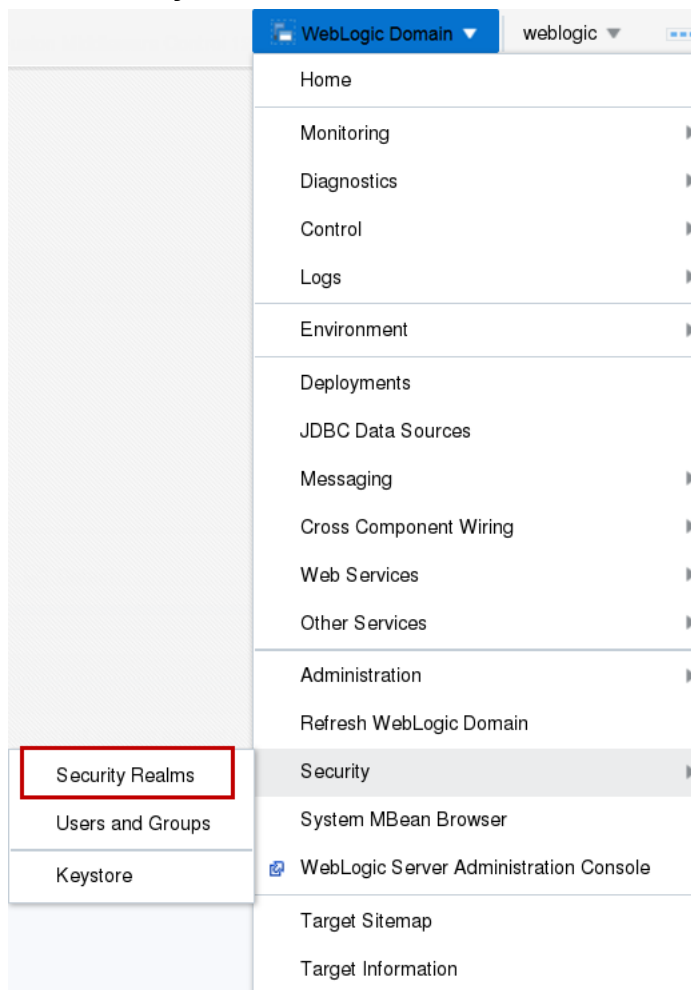
### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Tasks

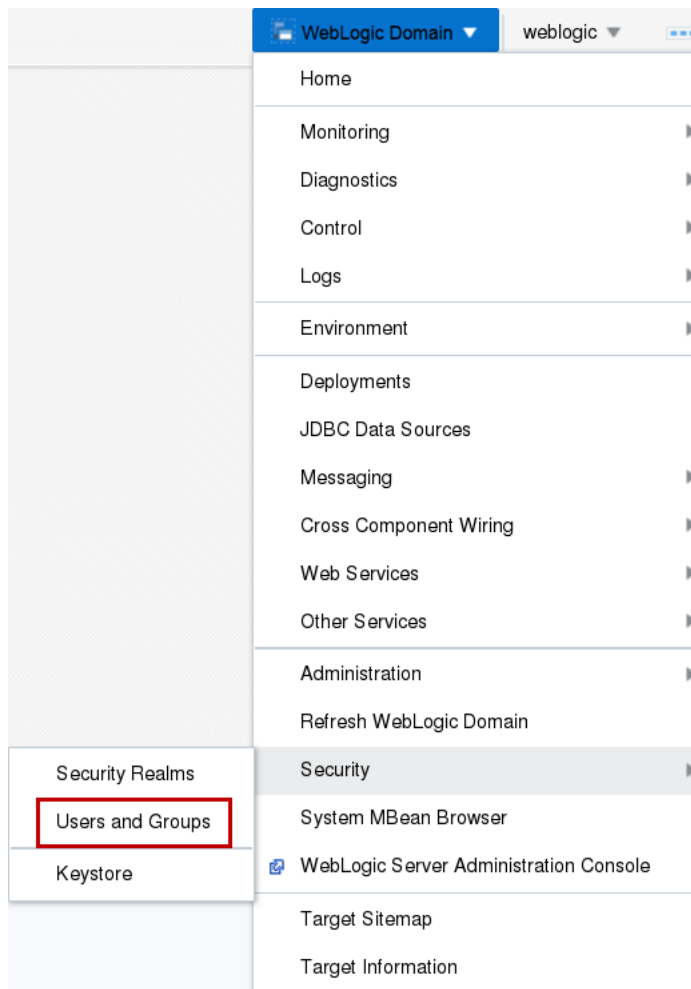
1. Log in to the jrf\_domain WebLogic Server domain using Fusion Middleware Control.
  - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

2. Create a Security Realm for baylandurgentcare.
  - a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Security** and click **Security Realms**.



- b. Click **Create**.
    - c. For **Name**, enter bayland and click **Create**.

- d. From the **WebLogic Domain** menu, select **Security** and click **Users and Groups**.

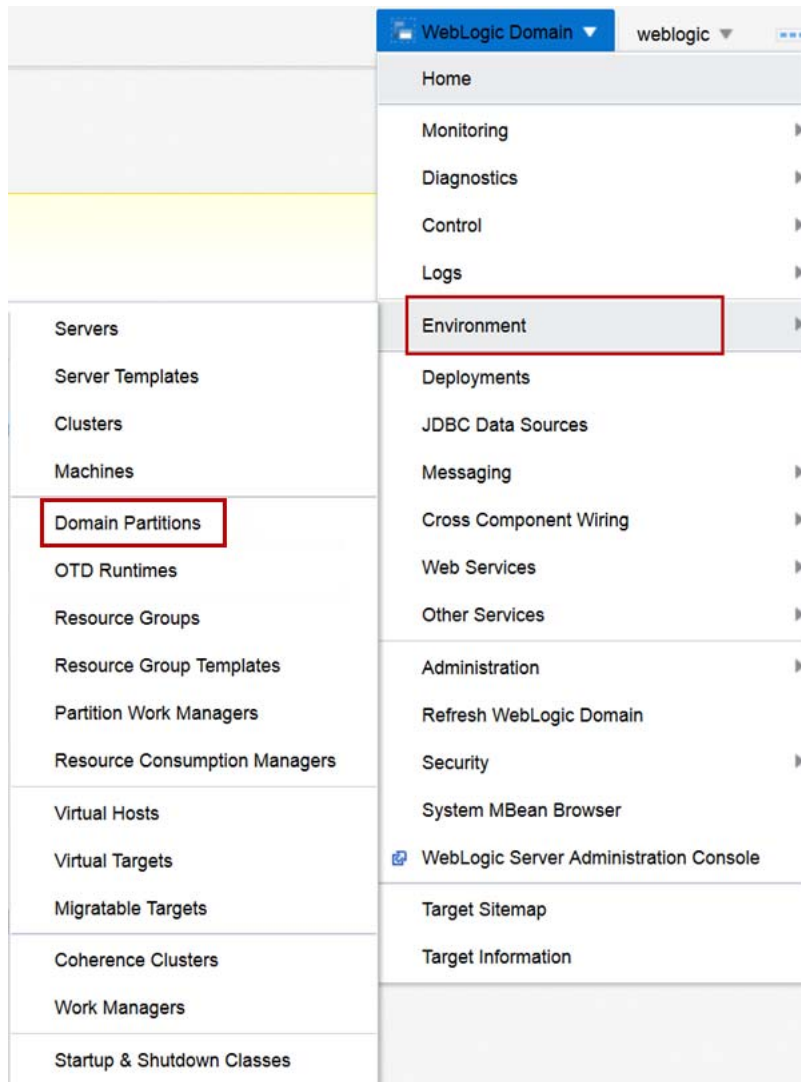


- e. In the **Select a Realm** drop-down menu, select the **bayland** realm.



- f. Click **Create**.
- g. For **Name**, enter `administrator`.
- h. For **Password** and **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the administrator user for the Oracle WebLogic Server product (bayland Security Realm).
- i. Click **Create**.
- The Security Realm **bayland** and the user **administrator** for the MedRec are created successfully.

3. Create the baylandurgentcare domain partition.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter baylandurgentcare.
- d. For **Security Realm**, select bayland.
- e. Click **Next**.
- f. For the Virtual Target baylandurgentcare, choose both **Select** and **Set as Default** and click **Next**.
- g. For **Resource Group Name**, enter baylandurgetcare.
- h. For **Resource Group Template**, select **MecRedResourceGroupTemplate**.
- i. Move the baylandurgentcare target to **Selected Targets**.
- j. Click **Next**.

Create Domain Partition: Summary

Back

Step 4 of 4

Next

Create

Cancel

Information

This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

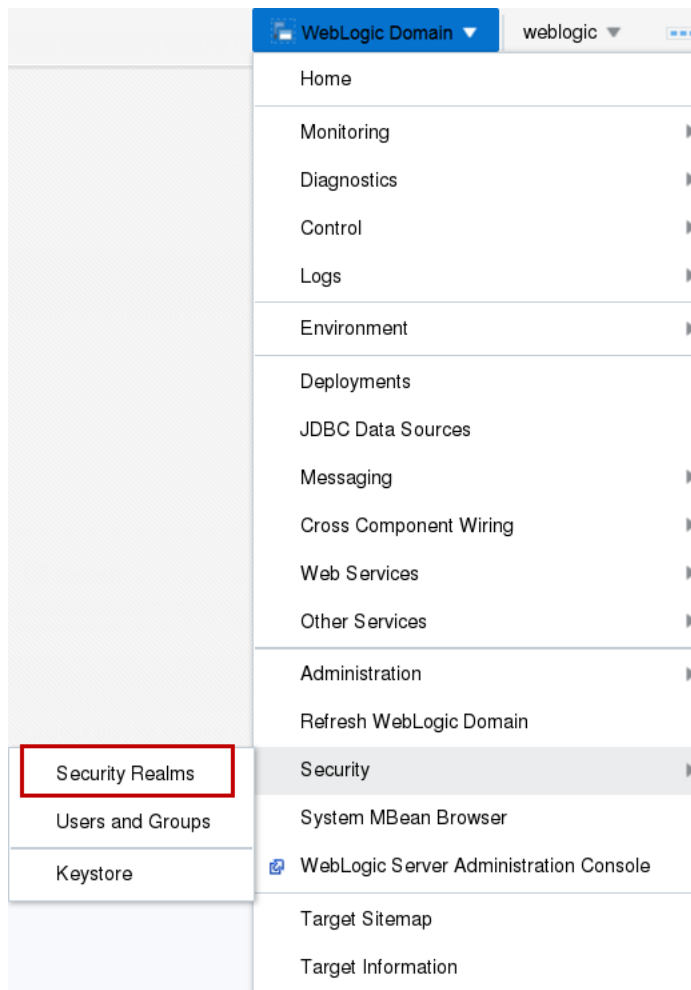
A WebLogic domain partition will be created with the following attributes.

|                                |                             |
|--------------------------------|-----------------------------|
| Name                           | baylandurgentcare           |
| Realm                          | bayland                     |
| Primary Identity Domain        | None                        |
| OTD Runtime                    | None                        |
| Available Targets              | baylandurgentcare           |
| Default Targets                | baylandurgentcare           |
| Resource Group                 | baylandurgentcare           |
| Resource Group Template        | MedRecResourceGroupTemplate |
| Targets for the Resource Group | baylandurgentcare           |

- k. Review the domain partition summary and click **Create**.  
The baylandurgentcare domain partition is created.
4. Start the baylandurgentcare domain partition.
  - a. In the list of Domain Partitions, click the **baylandurgentcare** entry.
  - b. Click **Control** and click **Start**.
  - c. When the Confirmation dialog box indicates that the Start Operation is successful, click **Close**.  
The baylandurgentcare domain partition successfully starts.
5. Test the baylandurgentcare domain partition.
  - a. In the Browser window, navigate to <http://fmw1:7101/bayland/medrec>.

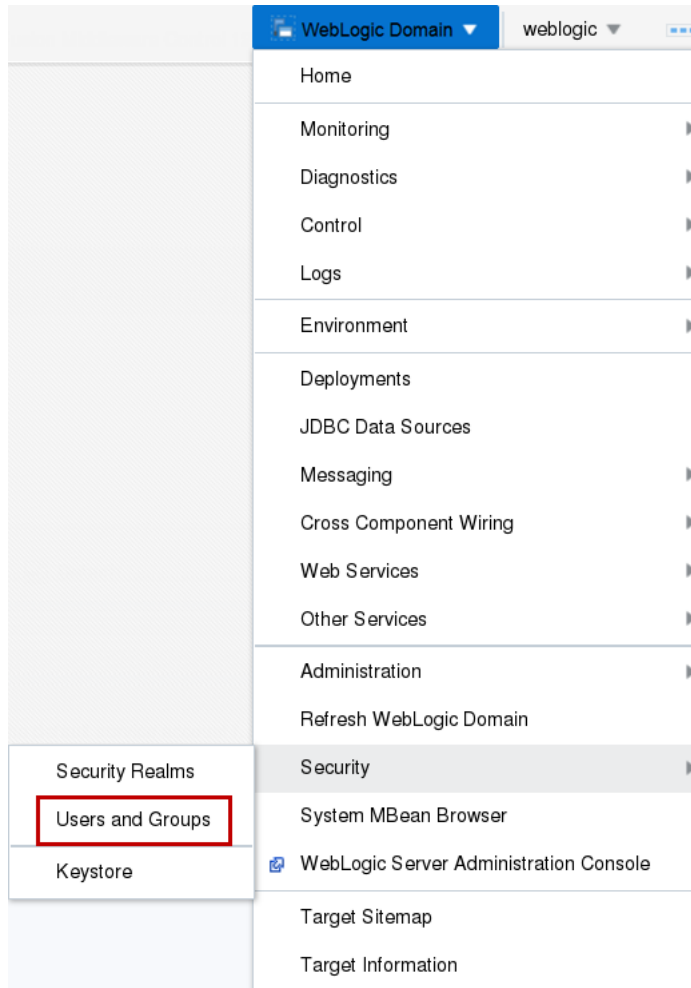


6. Create a Security Realm for valleyhealth.
- a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Security** and click **Security Realms**.



- b. Click **Create**.
- c. For **Name**, enter valley and click **Create**.

- d. From the **WebLogic Domain** menu, select **Security** and click **Users and Groups**.

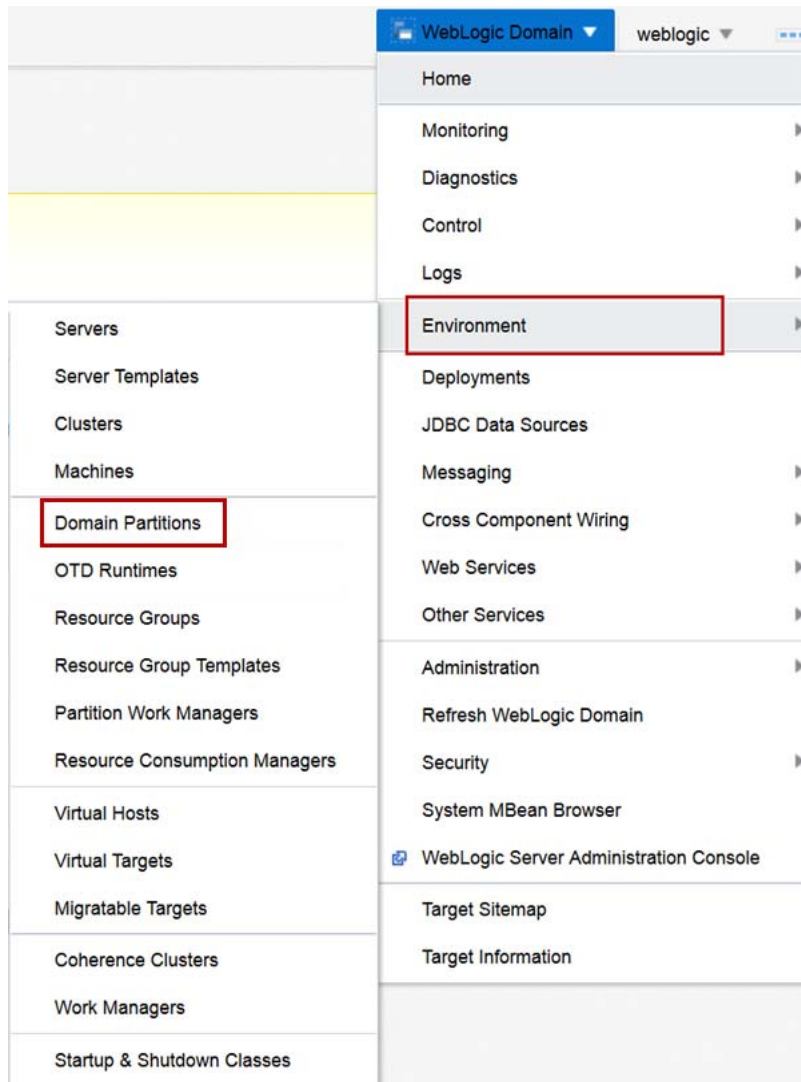


- e. In the **Select a Realm** dropdown, select the **valley** realm.



- f. Click **Create**.
- g. For **Name**, enter `administrator`.
- h. For **Password** and **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the administrator user for the Oracle WebLogic Server product (valley Security Realm).
- i. Click **Create**.
- The Security Realm **valley** and the user **administrator** for the MedRec application are created successfully.

7. Create the valleyhealth domain partition.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter valleyhealth.
- d. For **Security Realm**, select valley.
- e. Click **Next**.
- f. For the Virtual Target **valleyhealth**, check both **Select** and **Set as Default** and click **Next**.
- g. For **Resource Group Name**, enter valleyhealth.
- h. For **Resource Group Template**, select MecRedResourceGroupTemplate.
- i. Move the **valleyhealth** target to **Selected Targets**.
- j. Click **Next**.

Create Domain Partition: Summary

BackStep 4 of 4NextCreateCancel

Information

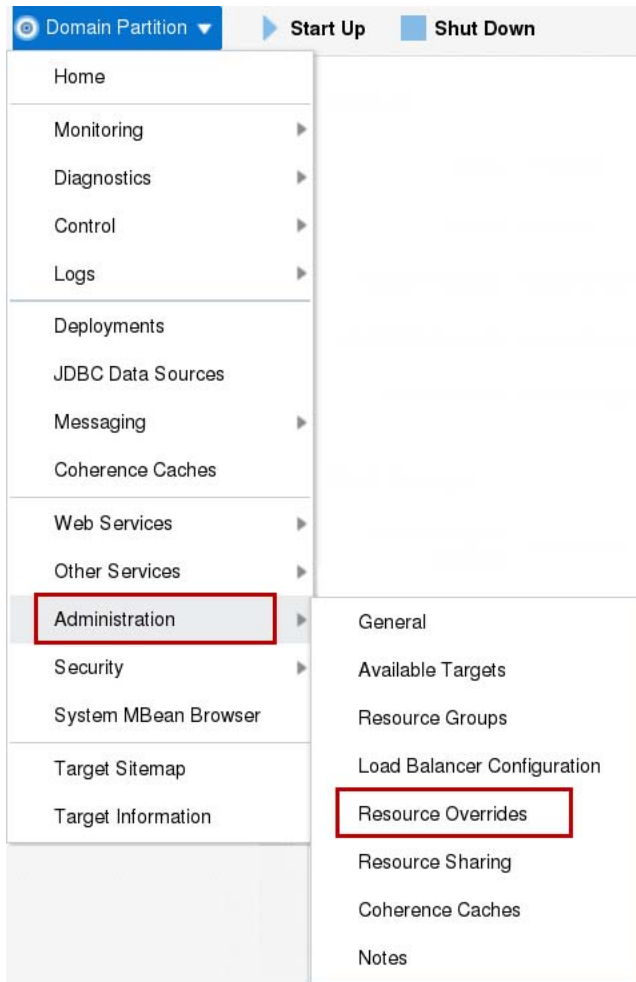
This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

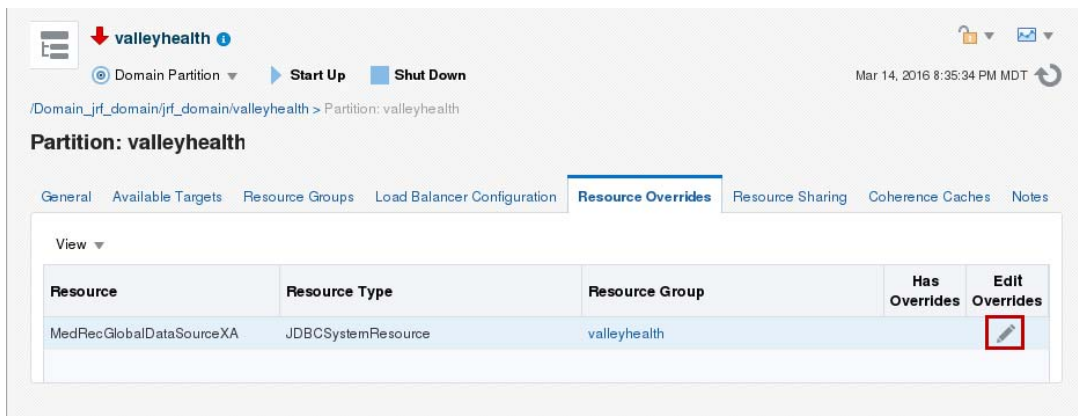
|                                |                             |
|--------------------------------|-----------------------------|
| Name                           | valleyhealth                |
| Realm                          | valley                      |
| Primary Identity Domain        | None                        |
| OTD Runtime                    | None                        |
| Available Targets              | valleyhealth                |
| Default Targets                | valleyhealth                |
| Resource Group                 | valleyhealth                |
| Resource Group Template        | MedRecResourceGroupTemplate |
| Targets for the Resource Group | valleyhealth                |

- k. Review the domain partition summary and click **Create**.  
The valleyhealth domain partition is created.  
However, before starting the valleyhealth domain partition, change the JDBC connection to use a different pluggable database, namely PDB2.
8. Override the JDBC connection settings for valleyhealth domain partition Resource Group.
  - a. In the list of **Domain Partitions**, click the valleyhealth domain partition.

- b. From the **Domain Partition** menu, select **Administration** and then select **Resource Overrides**.



- c. Click the **Edit Overrides** pencil icon for the **MedRecGlobalDataSourceXA** entry.



- d. Override the **User** property with the value MEDREC.
- e. Override the **URL** property with the value  
jdbc:oracle:thin:@//fmwdb:1521/PDB2.

- f. Override the **Password** property with value for the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.

Overrides: MedRecGlobalDataSourceXA

Resource Group valleyhealth

| Name           | Original Value                      | Override Value                      | Copy                                                                                |
|----------------|-------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------|
| User           | MEDREC                              | MEDREC                              |  |
| URL            | jdbc:oracle:thin:@//fmwdb:1521/PDB1 | jdbc:oracle:thin:@//fmwdb:1521/PDB2 |  |
| Password       | *****                               | *****                               |                                                                                     |
| DataSourceName |                                     | MedRecGlobalDataSourceXA            |                                                                                     |

OK Cancel

- g. Click **OK**.
9. Start the **valleyhealth** domain partition.
- a. Next to the **Domain Partition** menu, click **Start Up**.
- b. After the **Domain Partition** starts, click **Close**.
10. Test the **valleyhealth** domain partition.
- a. In the Browser window, navigate to **http://fmw1:7101/valley/medrec**.

## Practice Solution: Creating a Domain Partition from a Resource Group Template

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started creating domain partitions, this script will not work. To run the script in that case, delete the partially created domain partitions before running this script.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice04-02
```
  - b. Run the solution script.

```
$./solution.sh
```

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_domain_partitions_from_rgt.py`, which creates the domain partitions for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>> Solution for Practice 04-02 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

## Practice 4-3: Creating a Resource Group from a Resource Group Template

---

### Overview

In this practice, you create a resource group at the WebLogic Server domain level using a resource group template. Resource groups can be used within domain partitions or standalone, within a WebLogic Server domain, to contain a related set of resources and access them from a virtual target.

### Assumptions

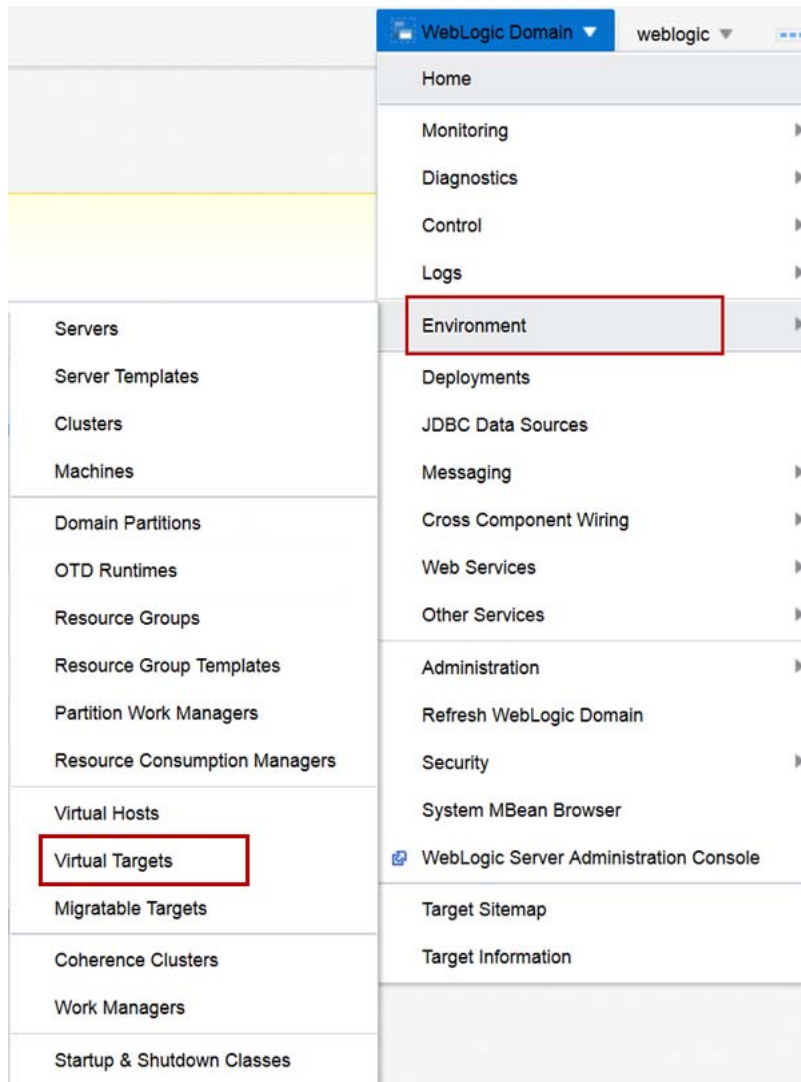
- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Tasks

1. Log in to the jrf\_domain WebLogic Server domain using Fusion Middleware Control.
  - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

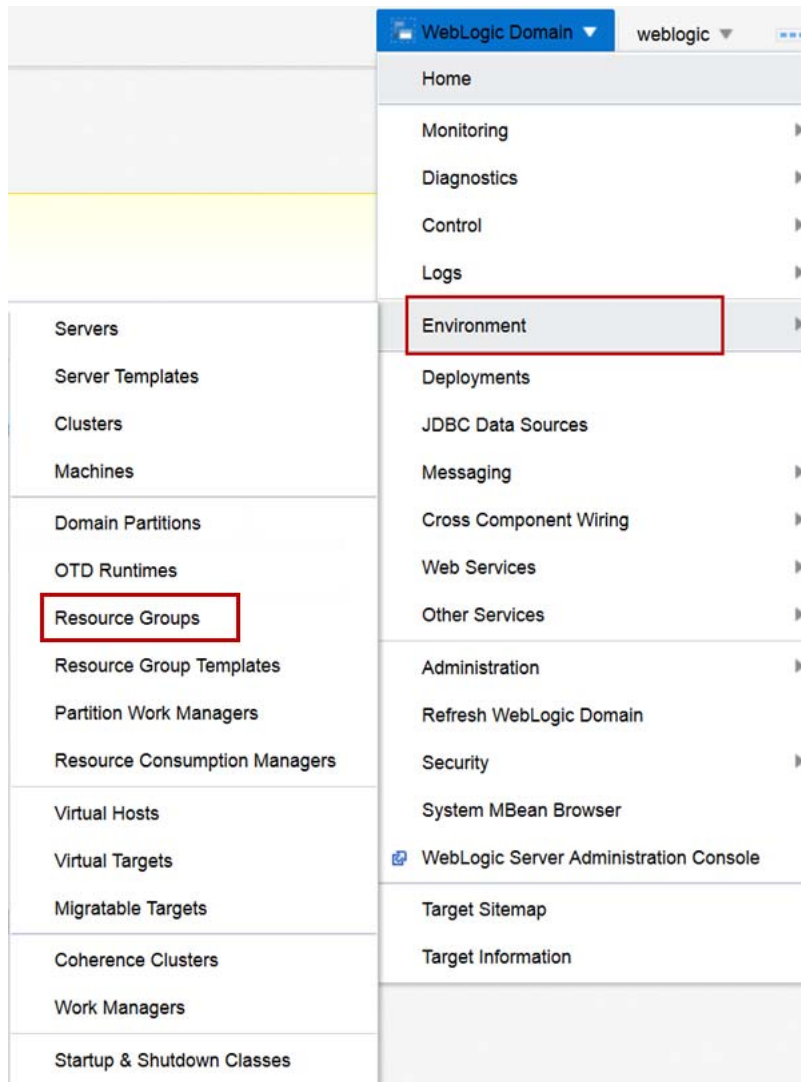


2. Create a Virtual Target for avitekclinic.
  - a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets**.



- b. Click **Create**.
- c. For **Name**, enter avitek.
- d. For **Jri Prefix**, enter /avitek.
- e. Click **Next**.
- f. From the **Cluster** menu, select **appcluster**.
- g. Click **Create**.

3. Create a Resource Group for avitekclinic.
  - a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Environment** and click **Resource Groups**.



- b. Click **Create**.
    - c. For **Name**, enter `avitekclinic`.
    - d. For **Scope**, click **Create a domain level resource group**.
    - e. For **Resource Group Template**, select `MedRecResourceGroupTemplate`.

- f. In **Targets**, select **avitekclinic** and move it to the **Chosen Targets**.

**Create a Resource Group**

\* Name: avitekclinic

Scope: ☐ Create a partition level resource group ☒ Create a domain level resource group

Domain Partition: baylandurgentcare

Resource Group Template: MedRecResourceGroupTemplate

**Targets**

| Available | Chosen |
|-----------|--------|
|           | avitek |

Notes:

Create Cancel

- g. Click **Create**.

The Resource Group **avitekclinic** is created successfully and is started.

4. Test the avitekclinic Resource Group.

- a. In the Browser window, navigate to **<http://fmw1:7101/avitek/medrec>**.

## Practice Solution: Creating a Resource Group from a Resource Group Template

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started creating a resource group, this script will not work. To run the script in that case, delete the partially created resource group prior to running this script.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice04-03
```
  - b. Run the solution script.

```
$./solution.sh
```

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_rg_from_rgt.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>> Solution for Practice 04-03 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

# **Practices for Lesson 5 - Oracle Traffic Director**

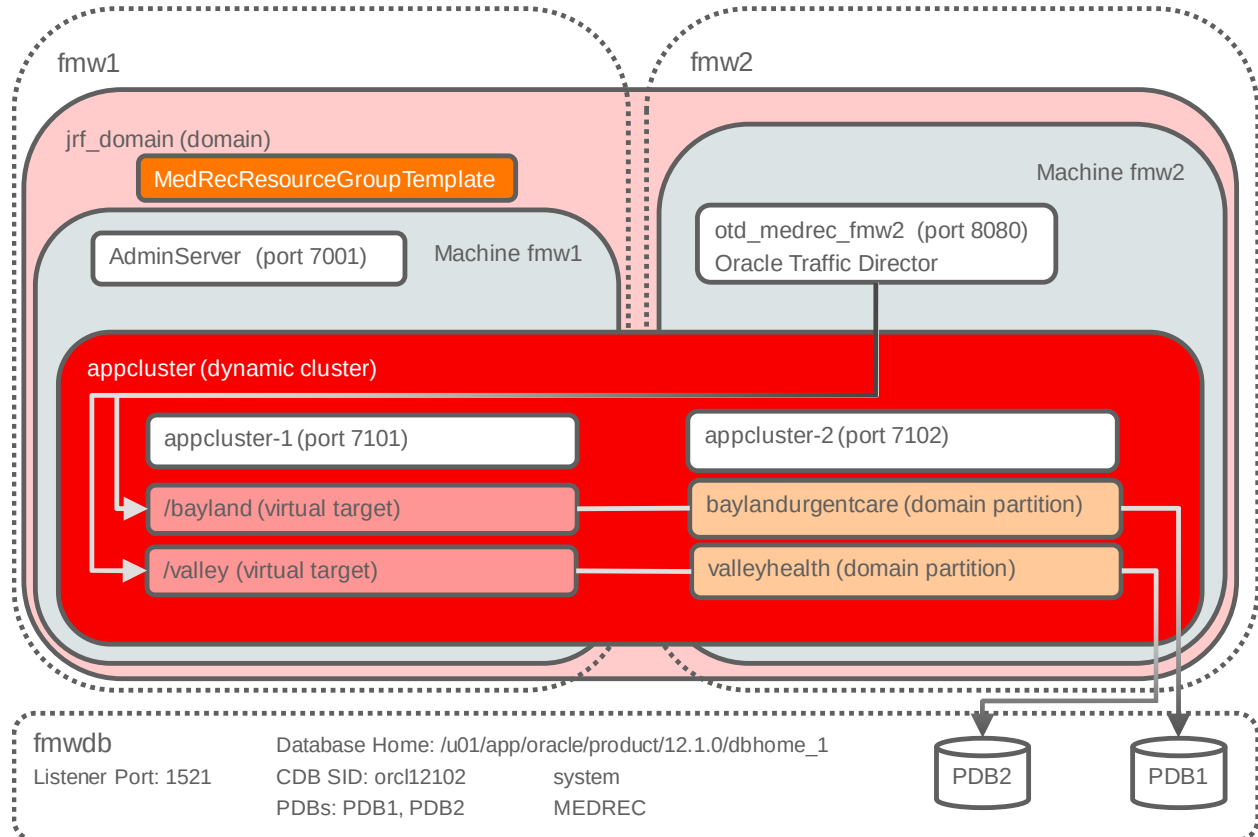
## **Chapter 5**

## Practices for Lesson 5: Overview

### Overview

In these practices, you create an Oracle Traffic Director configuration and register the Oracle Traffic Director Runtime with WebLogic Server. Then, you create a domain partition that uses the Resource Group Template and Oracle Traffic Director configuration. Finally, you set the Oracle Traffic Director logging levels.

The following diagram illustrates the configuration:



## Practice 5-1: Creating an Oracle Traffic Director Configuration

---

### Overview

In this practice, using Fusion Middleware Control, you create an Oracle Traffic Director configuration that you use in the MedRec applications.

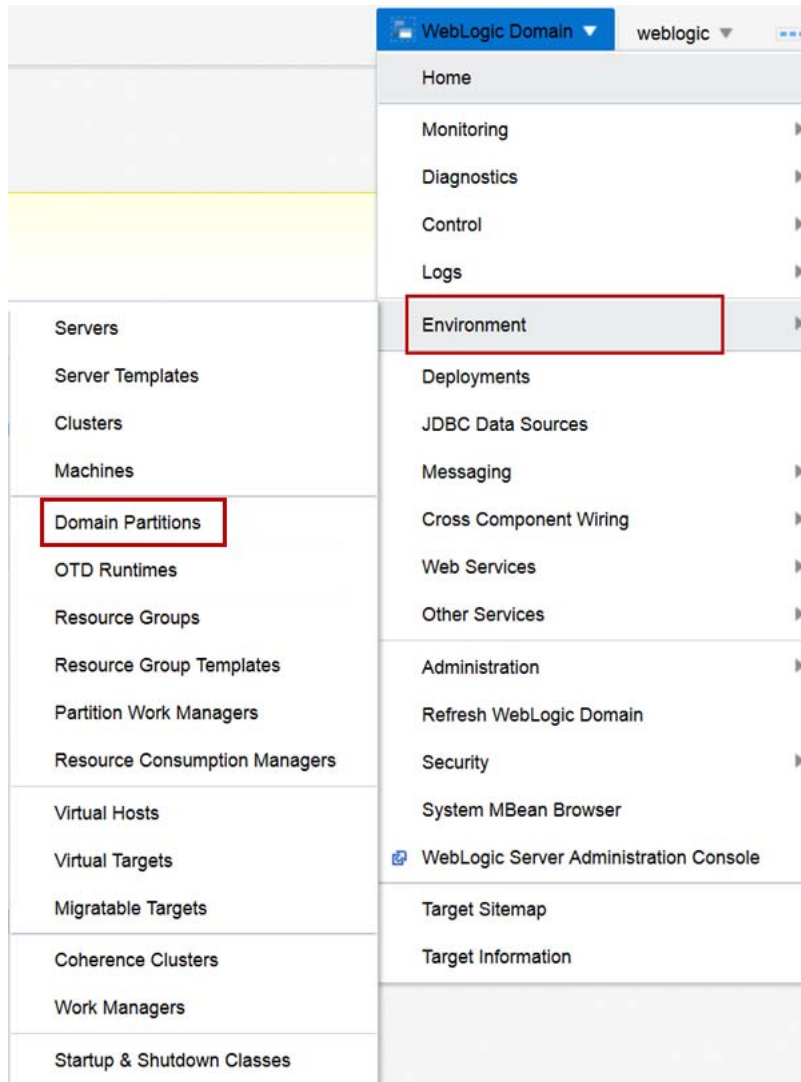
### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Tasks

1. Log in to the jrf\_domain WebLogic Server Domain using Fusion Middleware Control.
  - a. Open a browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

2. If the domain partitions baylandurgentcare and valleyhealth are still configured and running, stop and delete them.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.

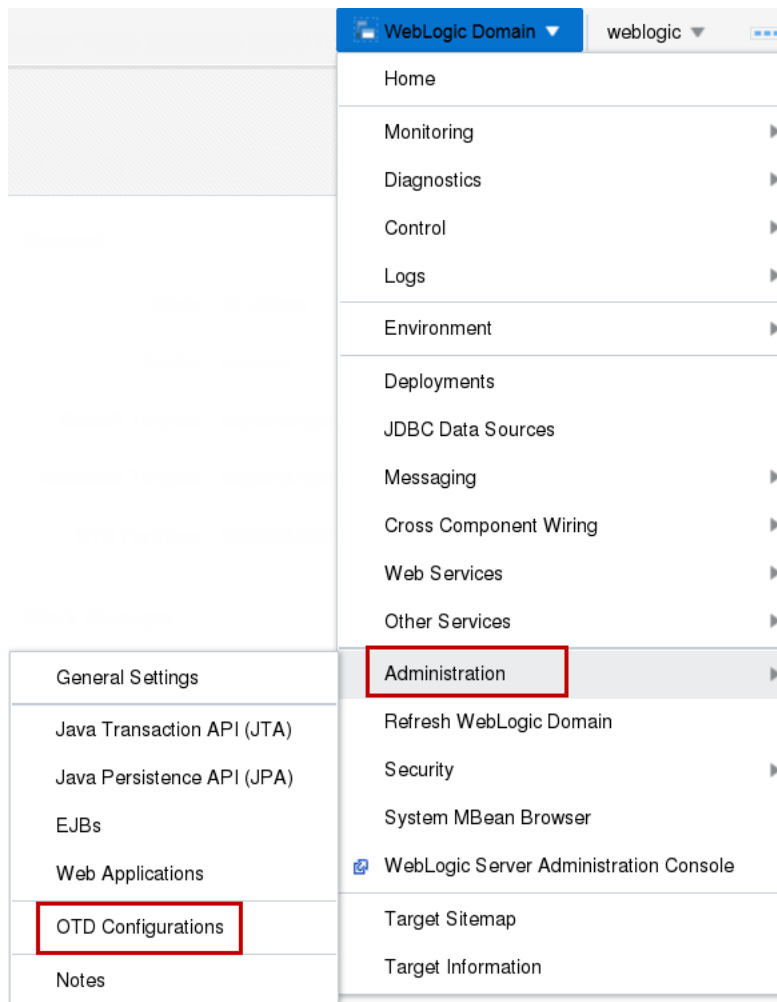




- b. Click the table entry for the **baylandurgentcare** domain partition.
- c. Click **Control**, select **Stop**, and select **Force stop now**.
- d. Click **Close**.
- e. Click the table entry for the **baylandurgentcare** domain partition.
- f. Click **Delete**.
- g. In the **Delete Domain Partition** dialog box, click **OK**.
- h. Click the table entry for the **valleyhealth** domain partition.
- i. Click **Control**, select **Stop**, and select **Force stop now**.
- j. Click **Close**.
- k. Click the table entry for the **valleyhealth** domain partition.
- l. Click **Delete**.
- m. In the **Delete Domain Partition** dialog box, click **OK**.

The baylandurgentcare and valleyhealth domain partitions are deleted—note that you will create these partitions again in these practices—but with an Oracle Traffic Director configuration and a Resource Group Template. Also, note that the Virtual Targets for both domain partitions will be used again in these practices.

3. Create an Oracle Traffic Director Configuration.
  - a. From the **WebLogic Domain** menu, select **Administration** and then click **OTD Configurations**.



- b. Click **Create**.
- c. For **Name**, enter `medrec` and click **Next**.
- d. For **Port**, enter `8080`.
- e. For **IP Address**, enter `*`.
- f. For **Server Name**, enter `fmw2` and click **Next**.
- g. In the **Server Pool** configuration, click **Next**.  
Note that the Origin Server pool will be created with the domain partitions by the Lifecycle Manager.
- h. For **Deployment**, select the **fmw2** machine and click **Next**.

**New Configuration Wizard**

General Listener Server Pool Deployment **Review**

**Create Configuration: Review** Back Step 5 of 5 Next Create Configuration Cancel

Review the information. Click 'Create Configuration' to proceed. Click 'Back' to go back to the previous screen and make changes.

**Configuration Properties**

Name medrec

**Listener Information**

Port 8080

IP Address \*

Server Name fmw2

**Server Pool Properties**

Type http

SSL/TLS Enabled No

Origin Servers

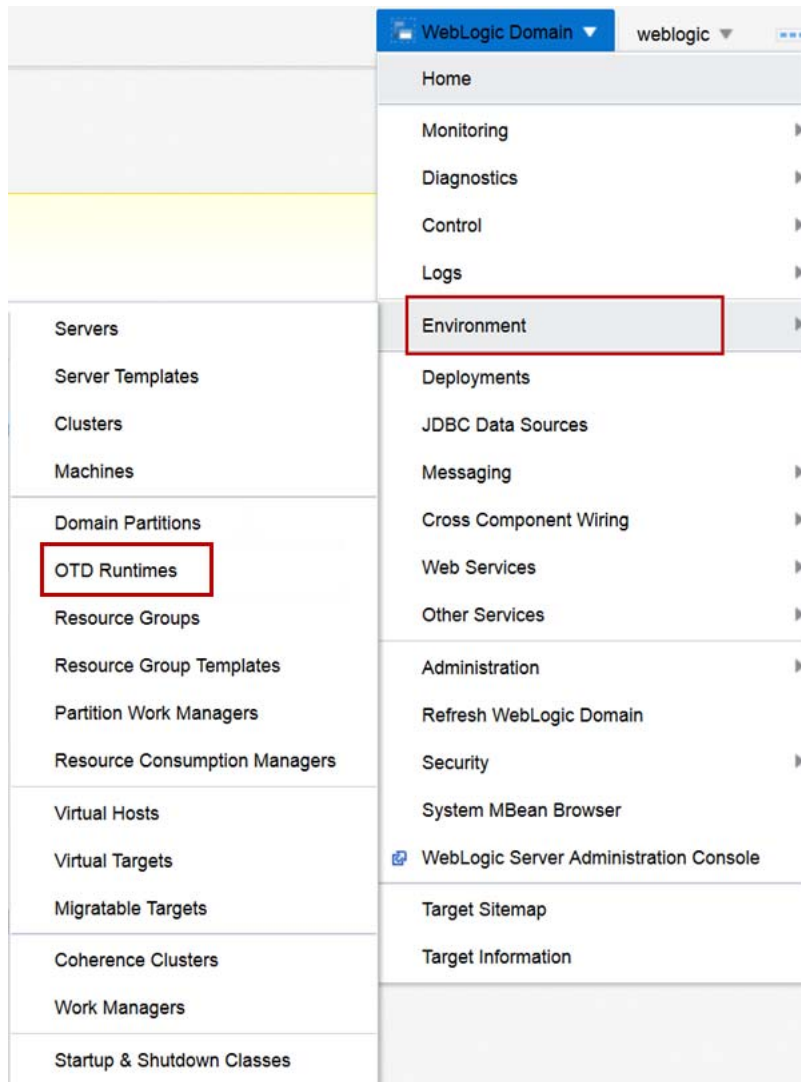
| Host               | Port |
|--------------------|------|
| No data to display |      |

**Deployment Information**

Machines fmw2

- i. Review the configuration and click **Create Configuration**.  
The Oracle Traffic Director configuration **medrec** creates successfully.
4. Start the medrec Oracle Traffic Director configuration.
  - a. In the Fusion Middleware Control window, click the **medrec** configuration entry.
  - b. Click **Start Instances**.
  - c. In the Confirmation dialog box, click **Close**.  
The Oracle Traffic Director instances for the **medrec** configuration start successfully.

5. Register the OTD Runtime.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **OTD Runtimes**.



- b. Click **Register Runtime**.
- c. For **Runtime Name**, enter `medrec`.
- d. For **OTD Configuration Name**, enter `medrec`.
- e. For **Admin Server Host**, enter `fmw1`.
- f. For **Admin Server Port**, enter `7001`.
- g. For **Username**, enter `weblogic`.
- h. For **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product.
- i. For **OTD Domain Name**, enter `jrf_domain` and click **OK**.

Register Runtime

An OTD configuration will be used to front-end domain partitions. Provide the OTD configuration name and connection information (host, port and credentials) for the WebLogic Admin Server where this configuration is located.

\* Runtime Name

medrec

\* OTD Configuration Name

medrec

\* Admin Server Host

fmw1

\* Admin Server Port

7001

\* Username

weblogic

\* Password

.....

\* OTD Domain Name

jrf\_domain

OK

Cancel

The Oracle Traffic Director Runtime **medrec** registers successfully.

## Practice Solution: Creating an Oracle Traffic Director Configuration

---

Perform the following tasks if you did not complete this practice and want to use the finished solution.

**Note:** If you started creating Oracle Traffic Director configurations, this script will not work. To run the script in that case, delete the partially created Oracle Traffic Director configurations before running this script.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.

### Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
  - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.  

```
$ cd /u01/practices/part1/practice05-01
```
  - b. Run the solution script.  

```
$./solution.sh
```
  - c. Enter any passwords as you are prompted.

**Note:** This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_otd_configuration.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>> Solution for Practice 05-01 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

## Practice 5-2: Creating Domain Partitions That Use Oracle Traffic Director

---

### Overview

In this practice, you create domain partitions that use the Oracle Traffic Director for load balancing.

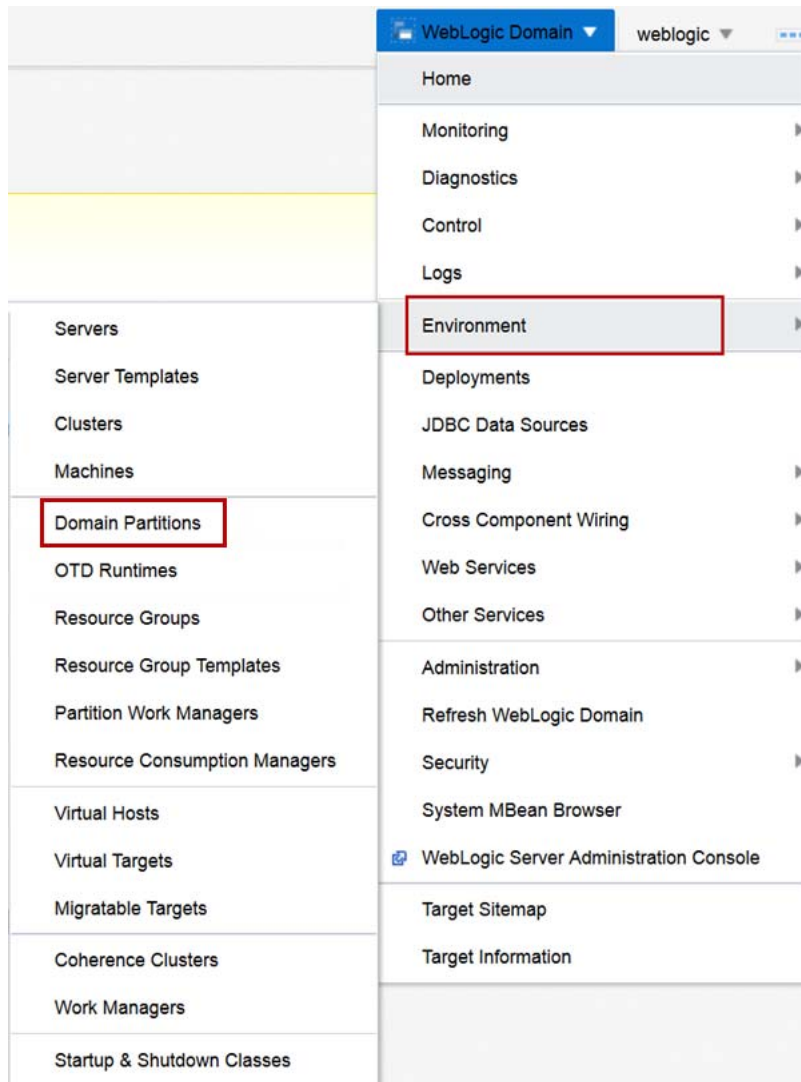
### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.
- The Oracle Traffic Director configuration medrec has been created.
- The Oracle Traffic Director server otd\_medrec\_fmw2 is running.

### Tasks

1. Log in to the jrf\_domain WebLogic Server Domain using Fusion Middleware Control.
  - a. Open a browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

2. Create the baylandurgentcare domain partition.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter baylandurgentcare.
- d. For **Security Realm**, select bayland.
- e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing**.
- f. For **OTD Runtime**, select medrec.
- g. Click **Next**.
- h. For the Virtual Target baylandurgentcare, choose both **Select** and **Set as Default** and click **Next**.
- i. For **Resource Group Name**, enter baylandurgentcare.
- j. For **Resource Group Template**, select MecRedResourceGroupTemplate.
- k. Move the baylandurgentcare target to **Selected Targets**.
- l. Click **Next**.



Create Domain Partition: Summary

Back

Step 4 of 4

Next

Create

Cancel

Information

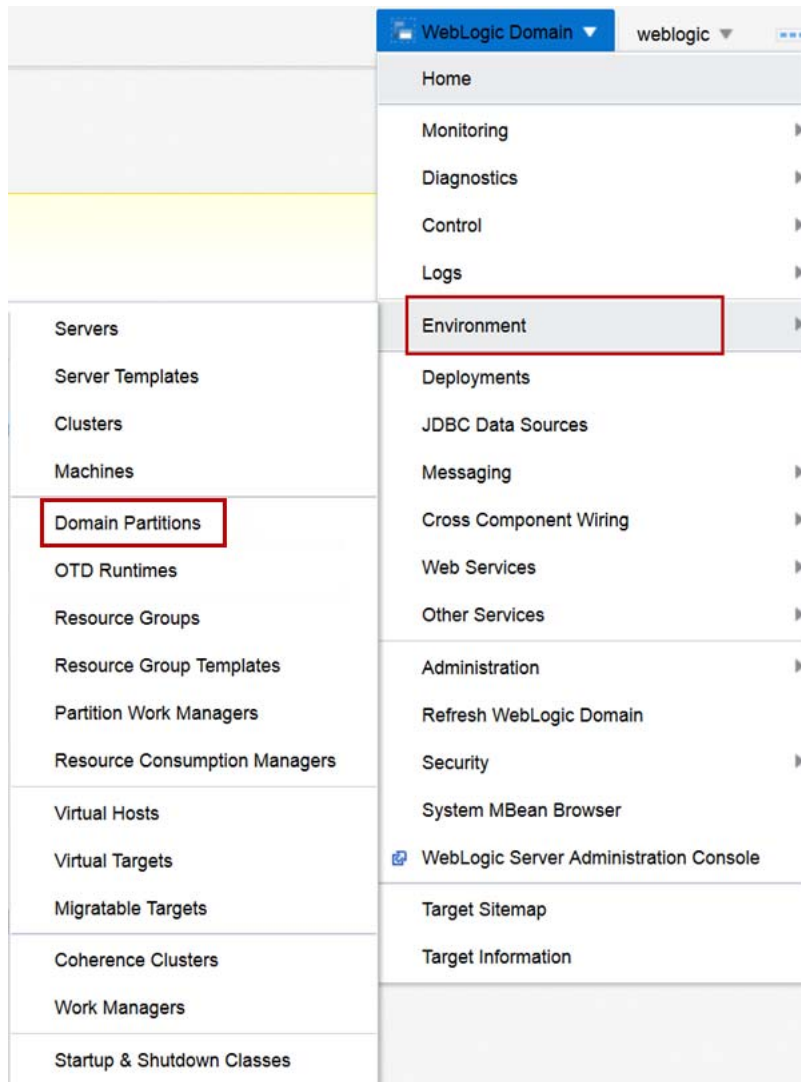
This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

|                                |                             |
|--------------------------------|-----------------------------|
| Name                           | baylandurgentcare           |
| Realm                          | bayland                     |
| Primary Identity Domain        | None                        |
| OTD Runtime                    | medrec                      |
| Available Targets              | baylandurgentcare           |
| Default Targets                | baylandurgentcare           |
| Resource Group                 | baylandurgentcare           |
| Resource Group Template        | MedRecResourceGroupTemplate |
| Targets for the Resource Group | baylandurgentcare           |

- m. Review the domain partition summary and click **Create**.  
The baylandurgentcare domain partition is created.
3. Start the baylandurgentcare domain partition.
  - a. In the list of Domain Partitions, click the **baylandurgentcare** entry.
  - b. Click **Control** and click **Start**.
  - c. Once the Confirmation dialog box indicates the Start Operation is successful, click **Close**.  
The baylandurgentcare domain partition successfully starts.
4. Test the baylandurgentcare domain partition.
  - a. In the browser window, navigate to <http://fmw2:8080/bayland/medrec>.  
Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the baylandurgentcare deployment.

5. Create the valleyhealth domain partition.
  - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter valleyhealth.
- d. For **Security Realm**, select valley.
- e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing**.
- f. For **OTD Runtime**, select medrec.
- g. Click **Next**.
- h. For the Virtual Target **valleyhealth**, choose both **Select** and **Set as Default** and click **Next**.
- i. For **Resource Group Name**, enter valleyhealth.
- j. For **Resource Group Template**, select MecRedResourceGroupTemplate.
- k. Move the **valleyhealth** target to **Selected Targets**.
- l. Click **Next**.

Create Domain Partition: Summary

BackStep 4 of 4NextCreateCancel

Information

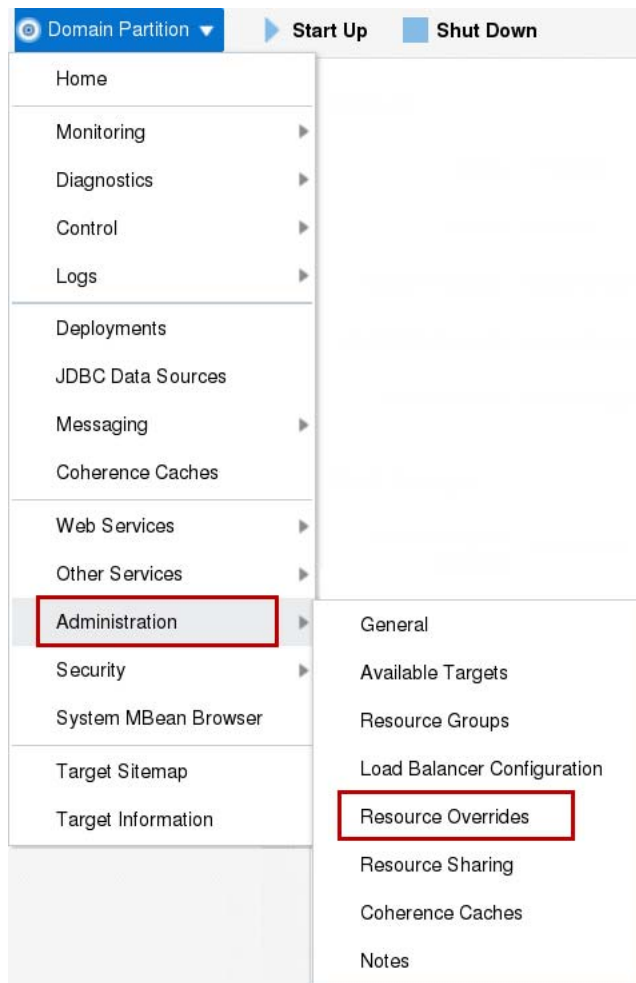
This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

|                                |                             |
|--------------------------------|-----------------------------|
| Name                           | valleyhealth                |
| Realm                          | valley                      |
| Primary Identity Domain        | None                        |
| OTD Runtime                    | medrec                      |
| Available Targets              | valleyhealth                |
| Default Targets                | valleyhealth                |
| Resource Group                 | valleyhealth                |
| Resource Group Template        | MedRecResourceGroupTemplate |
| Targets for the Resource Group | valleyhealth                |

- m. Review the domain partition summary and click **Create**.  
The valleyhealth domain partition is created.  
However, before starting the valleyhealth domain partition, change the JDBC connection to use a different pluggable database, namely PDB2.
6. Override the JDBC connection settings for the valleyhealth domain partition Resource Group.
  - a. In the list of **Domain Partitions**, click the valleyhealth domain partition.

- b. From the **Domain Partition** menu, select **Administration** and then select **Resource Overrides**.



- c. Click the **Edit Overrides** pencil icon for the **MedRecGlobalDataSourceXA** entry.
- d. Override the **User** property with the value `MEDREC`.
- e. Override the **URL** property with the value `jdbc:oracle:thin:@//fmwdb:1521/PDB2`.
- f. Override the **Password** property with value for the password from the Course Practice Environment: Security Credentials document for the MEDREC user for the Oracle Database product.
- g. Click **OK**.
7. Start the **valleyhealth** domain partition.
  - a. Next to the **Domain Partition** menu, click **Start Up**.
  - b. After the **Domain Partition** starts, click **Close**.
8. Test the **valleyhealth** domain partition.
  - a. In the browser window, navigate to <http://fmw2:8080/valley/medrec>.

## Practice 5-3: Configuring Oracle Traffic Director Logging Level

---

### Overview

In this practice, you work with the Oracle Traffic Director logs to locate and configure log files. First, you use Fusion Middleware Control to view log files, specifically the access log and server log. Next, you configure the logs to provide finer tracing and manually roll the log files. Finally, you configure and view the Traffic Director performance pages.

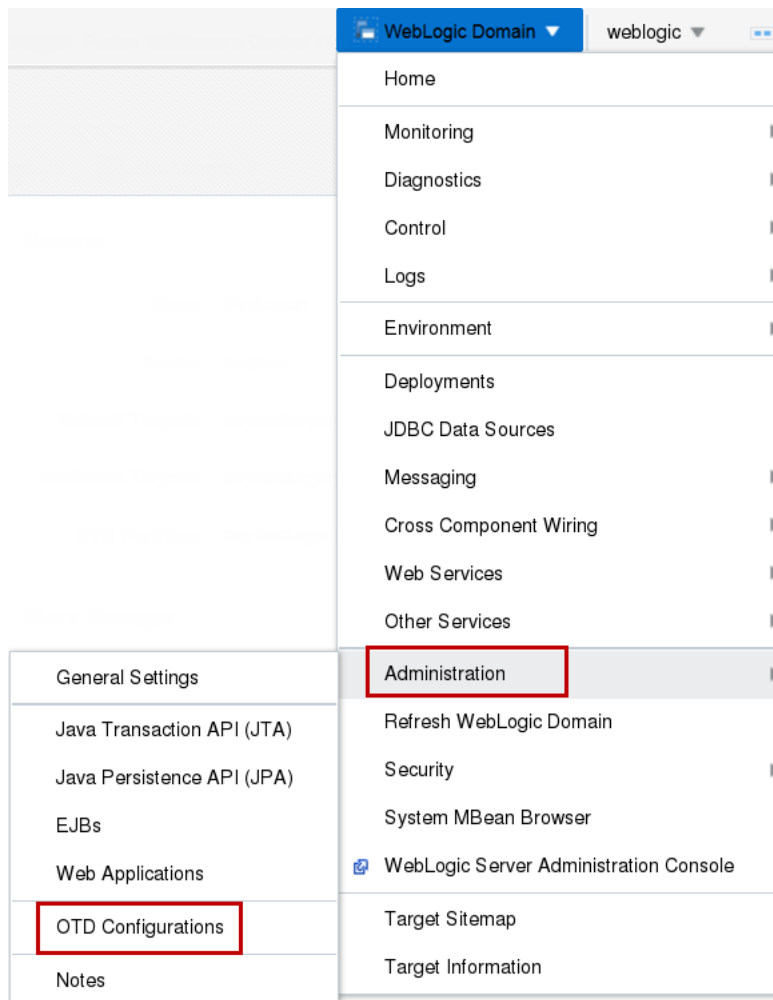
### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.
- The resource group template MedRecResourceGroupTemplate has been created.
- The Oracle Traffic Director configuration medrec has been created and registered.
- The Oracle Traffic Director server otd\_medrec\_fmw2 is running.
- Domain partitions baylandurgentcare and valleyhealth are running.

### Tasks

1. Log in to the jrf\_domain WebLogic Server Domain using Fusion Middleware Control.
  - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The Fusion Middleware Control home page for the **jrf\_domain** is displayed.

2. View the Oracle Traffic Director log files.
  - a. In Fusion Middleware Control, from the **WebLogic Domain** menu, select **Administration** and click **OTD Configurations**.



- b. Click the **medrec** configuration.
- c. From the **Traffic Director Configuration** menu, select **Logs** and select **View Log Messages**.
- d. If there are no log messages to display, click the **Search** icon and expand the date range – for example, from the most recent 10 minutes to the most recent 10 hours.
- e. In the Search dialog box, also include all the Message Types – check **Incident Error, Error, Warning, Notification, Trace** and **Unknown**.
- f. Click **Search**.
- g. In the rightmost column, click the **baylandurgentcare.log** link.  
The log entries for baylandurgentcare are displayed. Note that you can download the baylandurgentcare.log file.
- h. Click the **Log Files** link in the breadcrumbs display, just beneath the **Traffic Director Configuration** menu.
- i. Click the **access.log** file entry and click the **View Log File** button.
- j. Browse the **access.log**.

- k. Click the **Log Files** link in the breadcrumbs display, just beneath the **Traffic Director Configuration** menu.
  - l. Click the **server.log** file entry and click the **View Log File** button.
  - m. Browse the **server.log**.
3. Configure the Oracle Traffic Director logging level.
- a. From the **Traffic Director Configuration** menu, select **Administration** and select **Logging**.
  - b. For **Log Level**, select **TRACE: 1 (fine)** and click **Apply**.  
View the **server.log**. Now messages should be displayed that include Warning, Notification, and Error.
4. Manually roll the Oracle Traffic Director log files.
- a. From the **Traffic Director Configuration** menu, select **Administration** and select **Log Rotation**.
  - b. Click **Rotate Logs Now**.
5. Configure the Oracle Traffic Director monitoring and performance pages.
- a. From the **Traffic Director Configuration** menu, select **Administration** and select **Virtual Servers**.
  - b. Click the **medrec** Virtual Server.
  - c. In the **Virtual Server Settings** configuration page, scroll down and select **Advanced Settings**.
  - d. Select the check box for **XML Report** and **Plain Text Report**.

**Virtual Server : medrec**

[Settings](#)
[Routes](#)
[Caching](#)
[Compression](#)
[Request Limits](#)
[Bandwidth Limits](#)
[Content Serving](#)
[Web Applica](#)

**Virtual Server Settings** Apply Revert

**Advanced Settings**

Canonical Server Name

Default Language

Negotiate Client Language ☐

**Monitoring**

XML Report ☒

XML Report URI

Plain Text Report ☒

Plain Text Report URI

**Variables**

This table lists the variables created for this virtual server.

Create... Delete

| Name                                                              | Value |
|-------------------------------------------------------------------|-------|
| There are no variables defined. Click 'Create' to add a variable. |       |

- e. Click **Apply**.
6. View the Oracle Traffic Director monitoring and performance pages.
  - a. Open a browser window and navigate to **http://fmw2:8080/.perf**.
  - b. The performance-monitoring page for Oracle Traffic Director is displayed. Note that it displays information for queuing, keepalive, sessions, cache, pools, and origin servers.
  - c. Open a browser window and navigate to **http://fmw2:8080/stats.xml**.
  - d. The statistics page for Oracle Traffic Director is displayed. Note that it displays information for the server threads, CPU utilization, and server pool statistics.



# **Practices for Lesson 6 - Database Multitenancy**

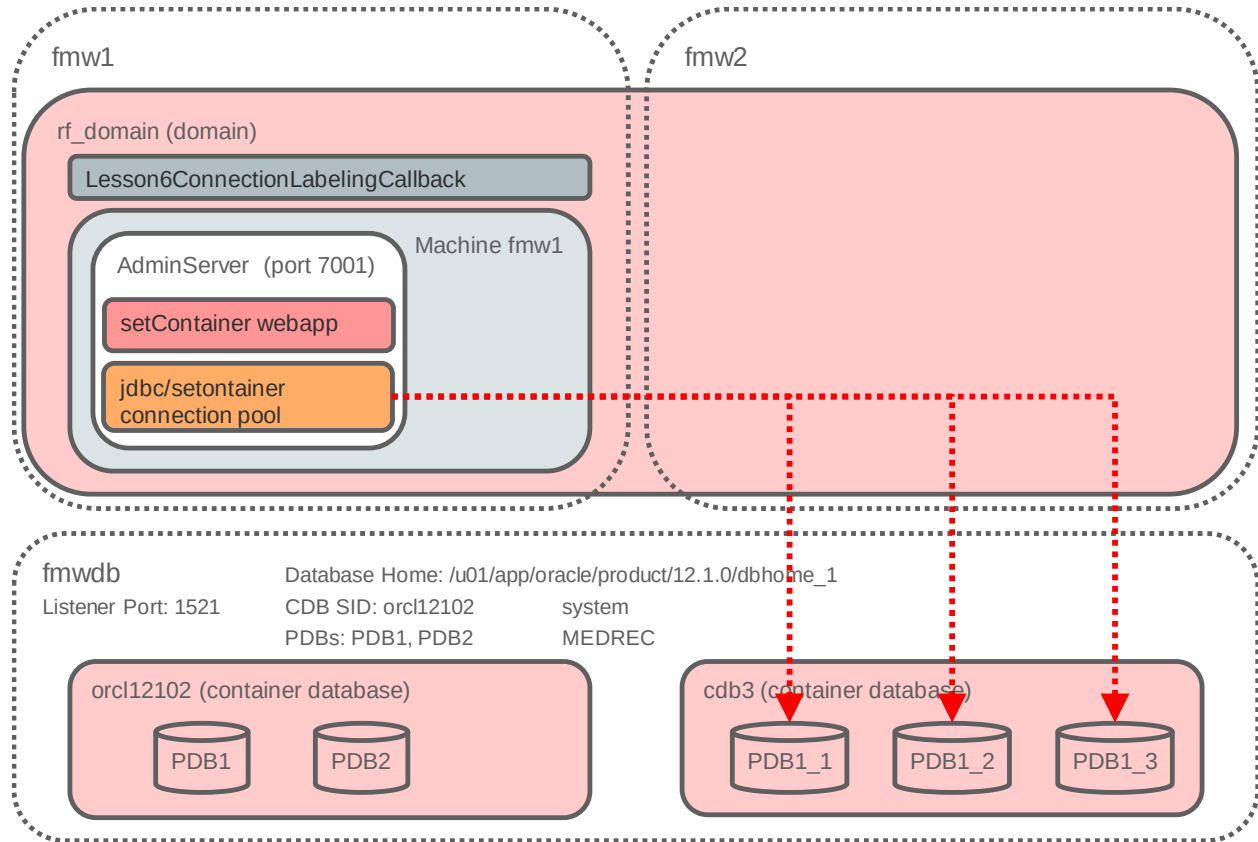
## **Chapter 6**

## Practices for Lesson 6: Overview

### Overview

In these practices, you create a new container database and then add pluggable databases to the new container database. Finally, you use a test WebLogic Server application to exercise the SET CONTAINER capability.

The following diagram describes the database and application configuration:



## Practice 6-1: Creating a Container Database

---

### Overview

In this practice, using the Database Configuration Assistant you create a new Container Database (CDB) to host new pluggable databases.

### Assumptions

- Oracle Database 12.1.0.2 is installed.

### Tasks

1. Connect to the machine fmwdb as the `oracle` user, in a VNC session. Open a terminal session as the `oracle` user.
2. Set the environment.
  - a. For the currently configured database `orcl12102`, on host fmwdb:

```
$. oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

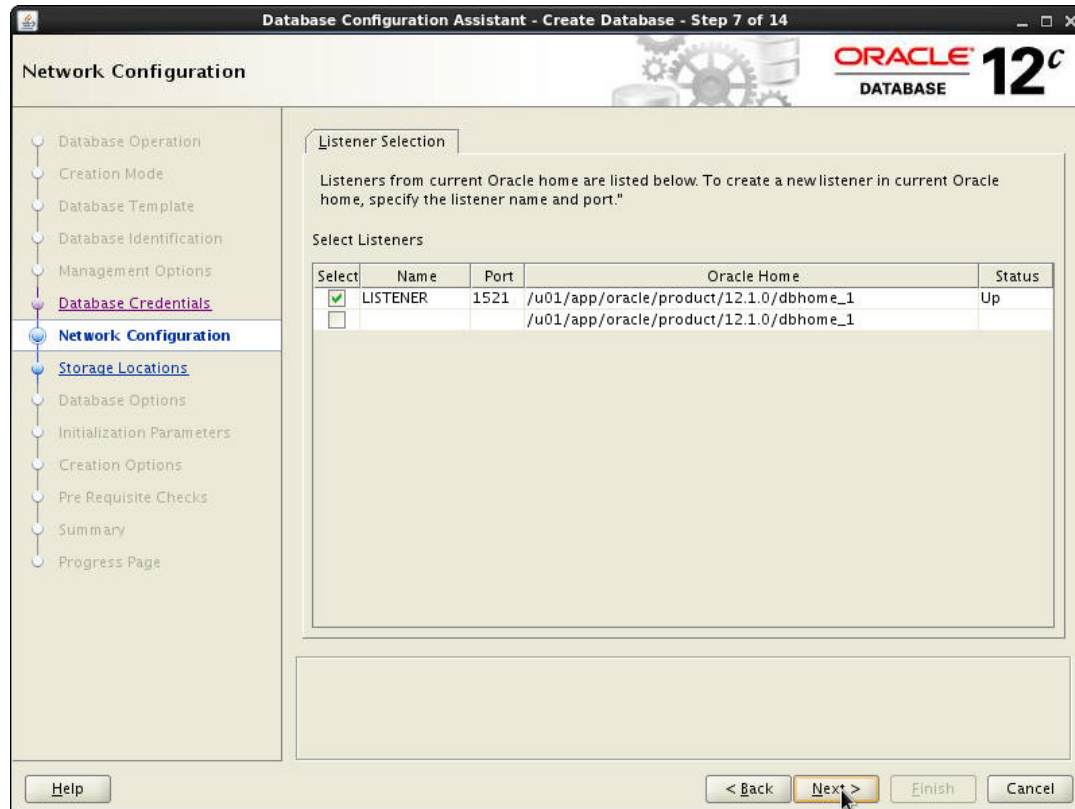
3. Create a Container Database (CDB) named `cdb2` using Database Configuration Assistant by starting `dbca` and performing the following steps in the graphical wizard.

```
$ dbca
```

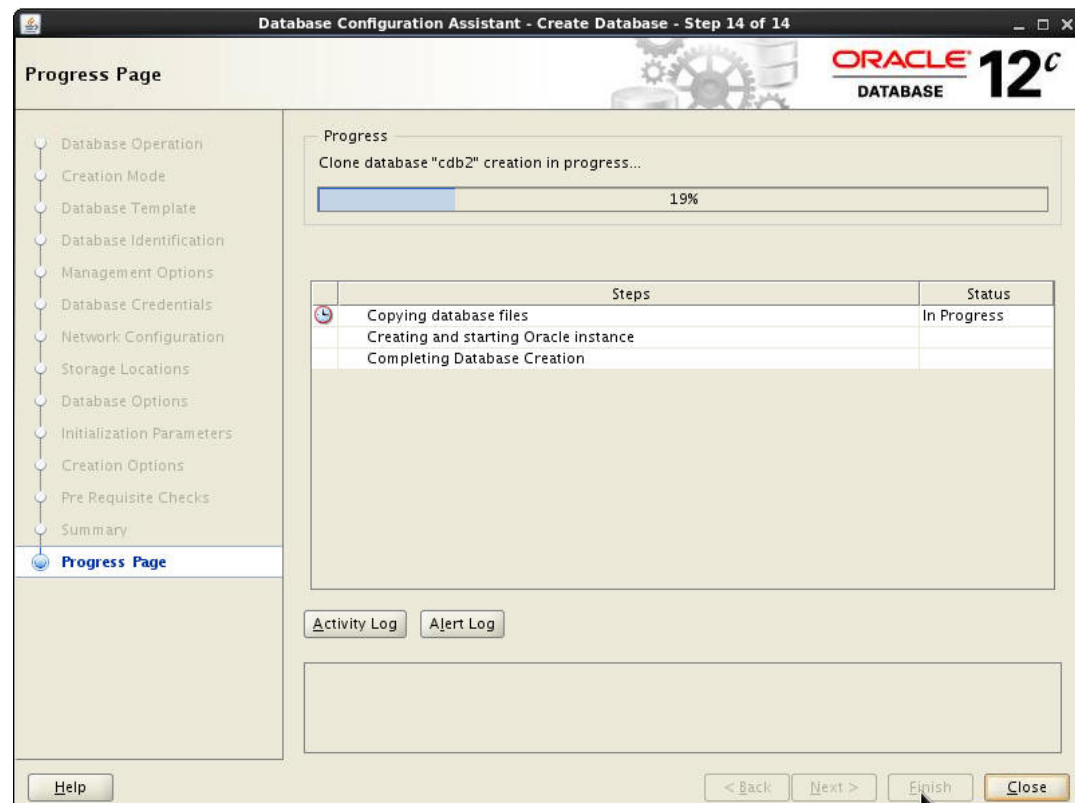
| Step | Window/Page Description         | Choices or Values                                                                                                                                                                                     |
|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a.   | Step 1: Database Operation      | Select "Create Database."<br>Click Next.                                                                                                                                                              |
| b.   | Step 2: Creation Mode           | Select "Advanced Mode."<br>Click Next.                                                                                                                                                                |
| c.   | Step 3: Database Template       | Select "General Purpose or Transaction Processing."<br>Click Next.                                                                                                                                    |
| d.   | Step 4: Database Identification | Enter<br>Global Database Name: <code>cdb2</code><br>SID: <code>cdb2</code><br>Select " <b>Create As Container Database.</b> "<br>Select " <b>Create An Empty Container Database.</b> "<br>Click Next. |
| e.   | Step 5: Management Options      | Deselect "Configure Enterprise Manager (EM) Database Express."<br>Click Next.                                                                                                                         |
| f.   | Step 6: Database Credentials    | Select "Use same Administrative password..."                                                                                                                                                          |

| Step | Window/Page Description            | Choices or Values                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                    | <p>Enter:</p> <p>Password: enter the password from the Course Practice Environment: Security Credentials document for the Administrative user for the Oracle Database product.</p> <p>Confirm password: enter the password from the Course Practice Environment: Security Credentials document for the Administrative user for the Oracle Database product.</p> <p>Click Next.</p> |
| g.   | Step 7: Network Configuration      | Click Next.                                                                                                                                                                                                                                                                                                                                                                        |
| h.   | Step 8: Storage Locations          | <p>Confirm Storage type is "File System."</p> <p>Select "Use Common Location for All Database Files."</p> <p>Click Next.</p>                                                                                                                                                                                                                                                       |
| i.   | Step 9: Database Options           | Click Next.                                                                                                                                                                                                                                                                                                                                                                        |
| j.   | Step 10: Initialization Parameters | <p>Select "Character Sets."</p> <p>Select "Use Unicode (<b>AL32UTF8</b>)."</p> <p>Click Next.</p>                                                                                                                                                                                                                                                                                  |
| k.   | Step 11: Creation Option           | <p>Select "Create Database."</p> <p>Click Next.</p>                                                                                                                                                                                                                                                                                                                                |
| l.   | Step 12: Pre Requisite Checks      | Click Next.                                                                                                                                                                                                                                                                                                                                                                        |
| m.   | Step 13: Summary                   | Click Finish.                                                                                                                                                                                                                                                                                                                                                                      |
| n.   | Step 14: Progress Page             | On the Database Configuration Assistant page (for password management), click Finish.                                                                                                                                                                                                                                                                                              |

The screenshot below corresponds to step g.



The screenshot below corresponds to step n.



## Practice 6-2: Creating Pluggable Databases

---

### Overview

In this practice, using SQLPlus, you unplug a current pluggable database named PDB1 (from database `orcl12102`) and use it to create new pluggable databases in database `cdb2`.

### Assumptions

- Oracle Database 12.1.0.2 is installed.
- The Oracle Database `orcl12102` is running.
- The Oracle Database `cdb2` is running.

### Tasks

1. Connect to the machine `fmwdb` as the `oracle` user, in a VNC session. Open a terminal session as the `oracle` user.
2. Set the environment for database `orcl12102`.
  - a. For the currently configured database `orcl12102`, on host `fmwdb`:

```
$. oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

3. Unplug the PDB1 database from `orcl12102`.
  - a. Execute `sqlplus`:

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

- b. Close the PDB1 pluggable database.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 CLOSE IMMEDIATE;

Pluggable database altered.

SQL>
```

- c. Unplug the PDB1 pluggable database in to the file PDB1.XML.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 UNPLUG INTO 'PDB1.XML';

Pluggable database altered.

SQL>
```

- d. Drop the PDB1 pluggable database.

```
SQL> DROP PLUGGABLE DATABASE PDB1;

Pluggable database dropped.

SQL>
```

- e. Exit SQLPlus.

```
SQL> EXIT;

Disconnected from Oracle Database 12c Enterprise Edition Release
12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options
$
```

4. Set the environment for database cdb2.

- a. For the currently configured database cdb2, on host fmwdb:

```
$. oraenv
ORACLE_SID = [] ? cdb2
The Oracle base has been set to /u01/app/oracle
$
```

5. Create the pluggable databases in the container database cdb2.

- a. Execute sqlplus:

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

- b. Ensure that you are connected to the correct database by viewing the data files – the set of data files for the cdb2 database are displayed.

```
SQL> SELECT FILE_NAME FROM DBA_DATA_FILES;
```

```
FILE_NAME
```

```


```

```
/u01/app/oracle/oradata/cdb2/users01.dbf
/u01/app/oracle/oradata/cdb2/undotbs01.dbf
/u01/app/oracle/oradata/cdb2/system01.dbf
/u01/app/oracle/oradata/cdb2/sysaux01.dbf
```

```
SQL>
```

- c. Create the new PDB1\_1 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_1 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT= ('PDB1', 'PDB1_1', 'DEVICIS', 'DEVICISPDB1_1', 'FULL'
, 'FULLPDB1_1');
```

```
Pluggable database created.
```

```
SQL>
```

- d. Open the new PDB1\_1 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_1 OPEN READ WRITE;
```

```
Pluggable database altered.
```

```
SQL>
```

- e. Create the new PDB1\_2 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_2 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT= ('PDB1', 'PDB1_2', 'DEVICIS', 'DEVICISPDB1_2', 'FULL'
, 'FULLPDB1_2');
```

```
Pluggable database created.
```

```
SQL>
```

- f. Open the new PDB1\_2 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_2 OPEN READ WRITE;
```

```
Pluggable database altered.
```

```
SQL>
```



- g. Create the new PDB1\_3 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE PDB1_3 AS CLONE USING 'PDB1.XML'
COPY STORAGE UNLIMITED TEMPFILE REUSE
FILE_NAME_CONVERT= ('PDB1','PDB1_3','DEVICIS','DEVICISPDB1_3','FULL
','FULLPDB1_3');
```

Pluggable database created.

SQL>

- h. Open the new PDB1\_3 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1_3 OPEN READ WRITE;
```

Pluggable database altered.

SQL>

- i. Ensure the pluggable databases are open for read write, by selecting from the CDB\_PDBS table and viewing the pluggable database names and status.

```
SQL> SELECT PDB_NAME, STATUS FROM CDB_PDBS;
```

PDB\_NAME

-----

STATUS

-----

PDB1\_1

NORMAL

PDB\$SEED

NORMAL

PDB1\_2

NORMAL

PDB1\_3

NORMAL

SQL>

- j. Exit SQLPlus.

```
SQL> EXIT;
Disconnected from Oracle Database 12c Enterprise Edition Release
12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options
$
```

6. Set the environment for database orcl12102.

- a. For the currently configured database orcl12102, on host fmwdb:

```
$. oraenv
ORACLE_SID = [] ? orcl12102
The Oracle base has been set to /u01/app/oracle
$
```

7. Create the pluggable database PDB1 in container database orcl12102.

- a. Execute sqlplus:

```
$ sqlplus / as sysdba

SQL*Plus: Release 12.1.0.2.0 Production on Mon Jan 25 15:12:47
2016

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 -
64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real
Application Testing options

SQL>
```

- b. Ensure that you are connected to the correct database by viewing the data files – the set of data files for the orcl12102 database are displayed.

```
SQL> SELECT FILE_NAME FROM DBA_DATA_FILES;

FILE_NAME

/u01/app/oracle/oradata/orcl12102/users01.dbf
/u01/app/oracle/oradata/orcl12102/undotbs01.dbf
/u01/app/oracle/oradata/orcl12102/system01.dbf
/u01/app/oracle/oradata/orcl12102/sysaux01.dbf

SQL>
```

- c. Create the PDB1 pluggable database.

```
SQL> CREATE PLUGGABLE DATABASE "PDB1" USING 'PDB1.XML'
SOURCE_FILE_NAME_CONVERT=NONE NOCOPY STORAGE UNLIMITED
TEMPFILE REUSE;
```

Pluggable database created.

```
SQL>
```

- d. Open the PDB1 pluggable database for read write.

```
SQL> ALTER PLUGGABLE DATABASE PDB1 OPEN READ WRITE;
```

Pluggable database altered.

```
SQL>
```

- e. Ensure the PDB1 pluggable database is open for read write, by selecting from the CDB\_PDBS table and viewing the pluggable database names and status.

```
SQL> SELECT PDB_NAME, STATUS FROM CDB_PDBS;
```

PDB\_NAME

-----  
-----

STATUS

-----

PDB1

NORMAL

PDB\$SEED

NORMAL

PDB2

NORMAL

```
SQL>
```

- f. Exit SQLPlus.

```
SQL> EXIT;
```

Disconnected from Oracle Database 12c Enterprise Edition Release  
12.1.0.2.0 - 64bit Production

With the Partitioning, OLAP, Advanced Analytics and Real  
Application Testing options

\$

## Practice 6-3: Using Pluggable Databases in a Java Application

---

### Overview

In this practice, you use a Java application that can utilize connection labeling and the SET CONTAINER directive to change the applications' JDBC connection pool to different pluggable databases.

### Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf\_domain admin server is running.

### Tasks

1. Connect to fmw1 as the `oracle` user.
  - a. Refer to the instructions provided by your instructor to connect to your machine.
  - b. Unless stated otherwise, you will be working within fmw1 for the remainder of this practice.
2. Copy the connection labeling class JAR file to the jrf\_domain.
  - a. Open a Terminal window as the `oracle` user, and navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/lib`.  
**Tip:** There is a launcher for a Terminal window in the panel at the top of the desktop.

```
$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/lib/
```
  - b. Copy the `Lesson6ConnectionLabelingCallback.jar` file into the domain `lib` directory.

```
$ cp /u01/practices/part1/practice06-03/Lesson6ConnectionLabelingCallback.jar .
```
3. Log in to the jrf\_domain WebLogic Server Domain using the WebLogic Server Administration Console.
  - a. Open a Browser window and navigate to **`http://fmw1:7001/console`**.
  - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.  
The WebLogic Server Administration Console home page for the **jrf\_domain** is displayed.
4. Create a Data Source that uses the connection labeling callback class.
  - a. In the WebLogic Server Administration Console, from the **Domain Structure** panel, click **Services** and click **Data Sources**.
  - b. Click **New**, and click **Generic Data Source**.
  - c. For **Name**, enter `ConnectionLabeling`.
  - d. For **JNDI Name**, enter `jdbc/setcontainer`.

**Create a New JDBC Data Source**

Back Next Finish Cancel

---

**JDBC Data Source Properties**

The following properties will be used to identify your new JDBC data source.

\* Indicates required fields

---

What would you like to name your new JDBC data source?

 **Name:**

---

What scope do you want to create your data source in ?

**Scope:**

---

What JNDI name would you like to assign to your new JDBC Data Source?

 **JNDI Name:**

---

What database type would you like to select?

**Database Type:**

---

Back Next Finish Cancel

- e. Click **Next**.
- f. In **JDBC Data Source Properties**, click **Next**.
- g. In **Transaction Options**, click **Next**.
- h. For **Database Name**, enter PDB1\_1.
- i. For **Host Name**, enter fmwdb.
- j. For **Port**, enter 1521.
- k. For **Database User Name**, enter system.
- l. For **Password**, enter the password from the Course Practice Environment: Security Credentials document for the system user for the Oracle Database product.
- m. For **Confirm Password**, enter the password from the Course Practice Environment: Security Credentials document for the system user for the Oracle Database product.

Create a New JDBC Data Source

Back Next Finish Cancel

Connection Properties

Define Connection Properties.

What is the name of the database you would like to connect to?

Database Name: PDB1\_1

What is the name or IP address of the database server?

Host Name: fmwdb

What is the port on the database server used to connect to the database?

Port: 1521

What database account user name do you want to use to create database connections?

Database User Name: system

What is the database account password to use to create database connections?

Password: ●●●●●●●●

Confirm Password: ●●●●●●●●

Additional Connection Properties:

oracle.jdbc.DRCPConnectionClass:

Back Next Finish Cancel

- n. Click **Next**.
  - o. Click **Test Configuration**.
- The connection test succeeds.

Messages

✔ Connection test succeeded.

Create a New JDBC Data Source

Test Configuration Back Next Finish Cancel

Test Database Connection

Test the database availability and the connection properties you provided.

What is the full package name of JDBC driver class used to create database connections in the connection pool?

(Note that this driver class must be in the classpath of any server to which it is deployed.)

Driver  
Class  
Name: oracle.jdbc.OracleDriver

What is the URL of the database to connect to? The format of the URL varies by JDBC driver.

URL: jdbc:oracle:thin:@//fmwdb:1521/PDB1\_1

What database account user name do you want to use to create database connections?

Database  
User  
Name: system

What is the database account password to use to create database connections?

(Note: for secure password management, enter the password in the Password field instead of the Properties field below)

Password: ●●●●●●●●●●●●●●●●

Confirm  
Password: ●●●●●●●●●●●●●●●●

What are the properties to pass to the JDBC driver when creating database connections?

Properties:  
user=system

The set of driver properties whose values are derived at runtime from the named system property.

System Properties:

What table name or SQL statement would you like to use to test database connections?

Test Table Name:  
SQL ISVALID

Test Configuration Back Next Finish Cancel

- p. Click **Next**.
- q. In **Select Targets**, select the check box for **AdminServer**.

The screenshot shows a dialog box titled "Create a New JDBC Data Source". At the top, there are four buttons: "Back", "Next", "Finish", and "Cancel". Below these buttons is a section titled "Select Targets". Inside this section, there is a paragraph of text: "You can select one or more targets to deploy your new JDBC data source. If you don't select a target, the data source will be created but not deployed. You will need to deploy the data source at a later time." Below the text are two sections: "Servers" and "Clusters". The "Servers" section has a single entry "AdminServer" with a checked checkbox. The "Clusters" section has two entries: "appcluster" with an unchecked checkbox, and "All servers in the cluster" with a selected radio button. At the bottom of the dialog box, there are four buttons: "Back", "Next", "Finish", and "Cancel".

- r. Click **Finish**.
5. Configure the Connection Labeling Callback.
- a. In the Summary of **JDBC Data Sources** table, click the **ConnectionLabeling** Data Source.
  - b. Click the **Connection Pool** tab.
  - c. Scroll down and click **Advanced**.
  - d. For **Connection Labeling Callback**, enter `com.oracle.fmw12cmt.Lesson6ConnectionLabelingCallback`.



Settings for ConnectionLabeling

Configuration Targets Monitoring Control Security Notes

General Connection Pool Oracle ONS Transaction Diagnostics Identity Options

Save

The connection pool within a JDBC data source contains a group of JDBC connections that applications reserve, use, and then return to the pool. The connection pool and the connections within it are created when the connection pool is registered, usually when starting up WebLogic Server or when deploying the data source to a new target.

Use this page to define the configuration for this data source's connection pool.

URL:  The URL of the database to connect to. The format of the URL varies by JDBC driver. [More Info...](#)

Driver Class Name:  The full package name of JDBC driver class used to create the physical database connections in the connection pool. (Note that this driver class must be in the classpath of any server to which it is deployed.) [More Info...](#)

Properties:  The list of properties passed to the JDBC driver that are used to create physical database connections. For example: server=dbserver1. List each property=value pair on a separate line. [More Info...](#)

Fatal Error Codes:  Specifies a comma-separated list of error codes that are treated as fatal errors. These errors include deployment errors that cause a server to fail to boot and connection errors that prevent a connection from being put back in the connection pool. [More Info...](#)

Connection Labeling Callback:  The class name of the connection labeling callback. This is automatically passed to registerConnectionLabelingCallback when the datasource is created. The class must implement oracle.ucp.ConnectionLabelingCallback. [More Info...](#)

Connection Harvest Max Count:  The maximum number of connections that may be harvested when the connection harvesting occurs. The range of valid values is 1 to MaxCapacity. [More Info...](#)

Connection Harvest Trigger Count:  Specifies the number of available connections (trigger value) used to determine when connection harvesting occurs. [More Info...](#)

Connection Count of Refresh Failures Till Disable:  Specifies the number of reconnect failures allowed before WebLogic Server disables a connection pool to minimize the delay in handling the connection request caused by a database failure. Zero means it is disabled. [More Info...](#)

Count of Test Failures Till Flush:  Specifies the number of test failures allowed before WebLogic Server closes all unused connections in a connection pool to minimize the delay caused by further database testing. Zero means it is disabled. [More Info...](#)

Save

e. Click **Save**.

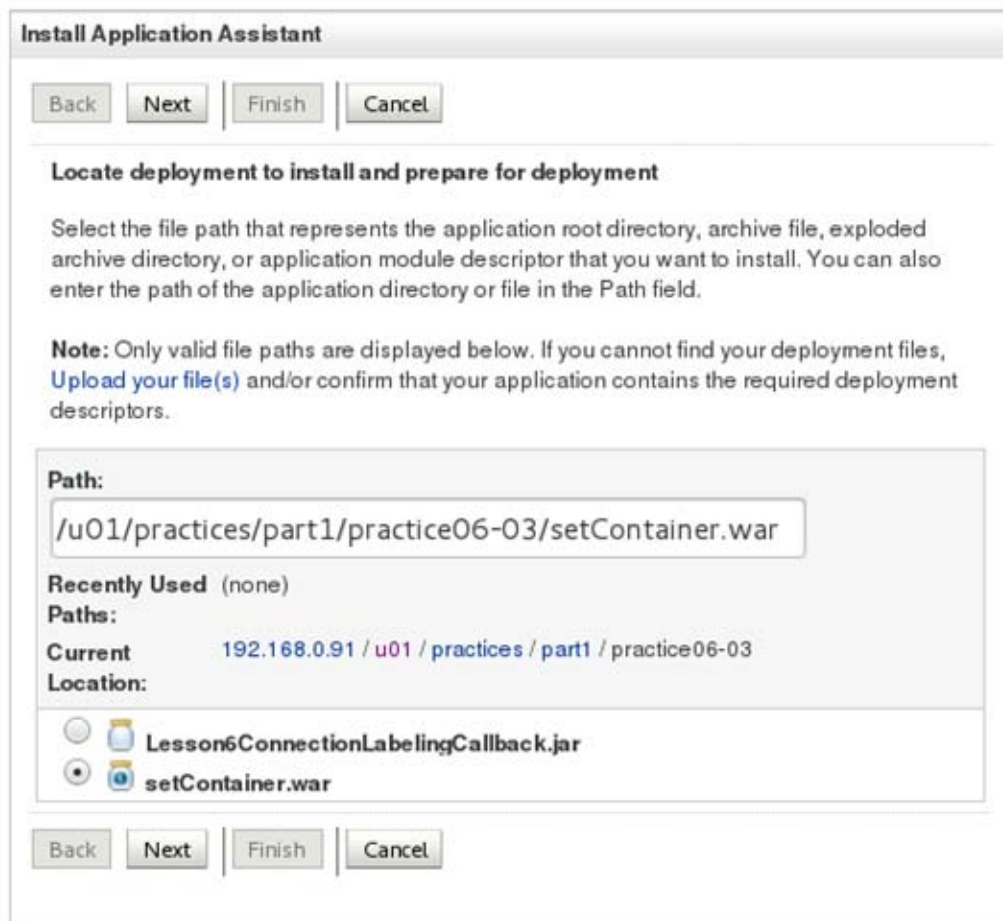
The connection pool changes are activated; however, the AdminServer must be restarted for the changes to take effect.

6. Restart the AdminServer.

- In the WebLogic Server Administration Console, from the **WebLogic Domain Structure** panel, click **Environment** and click **Servers**.
- Click the **Control** tab.
- Select the **AdminServer**.
- Click **Shutdown** and select **Force Shutdown Now**.
- On the fmw1 machine, open a Terminal window as the `oracle` user and navigate to `/u01/fmw/wls1221_infra/user_projects/domains/jrf_domain`

- ```
$ cd /u01/fmw/wls1221_infra/user_projects/domains/jrf_domain/
```
- f. Execute `startWebLogic.sh` to start the WebLogic Administration Server.


```
$ ./startWebLogic.sh
```
 - g. Keep this terminal window open to view the output from the connection labeling callback class.
7. Log in to the jrf_domain WebLogic Server Domain using the WebLogic Server Administration Console.
 - a. Open a Browser window and navigate to <http://fmw1:7001/console>.
 - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.
The WebLogic Server Administration Console home page for the **jrf_domain** is displayed.
 8. Deploy the setContainer web application.
 - a. In the WebLogic Server Administration Console, from the **Domain Structure** panel, click **Deployments**.
 - b. Click **Install**.
 - c. In the **Install Application Assistant**, navigate to `/u01/practices/part1/practice06-03/setContainer.war`, select **setContainer.war** and click **Next**.



- d. Select **Install the deployment as an application** and click **Next**.

The screenshot shows the 'Install Application Assistant' dialog box. At the top, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'. The main heading is 'Choose installation type and scope'. Below this, a text box explains: 'Select if the deployment should be installed as an application or library. Also decide the scope of this deployment.' Another text box states: 'The application and its components will be targeted to the same locations. This is the most common usage.' There are two radio buttons: 'Install this deployment as an application' (which is selected) and 'Install this deployment as a library'. Below the radio buttons, another text box says: 'Select a scope in which you want to install the deployment.' There is a 'Scope:' label followed by a dropdown menu currently showing 'Global'. At the bottom, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

- e. For deployment targets, select **AdminServer** and click **Next**.

The screenshot shows the 'Install Application Assistant' dialog box. At the top, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'. The main heading is 'Select deployment targets'. Below this, a text box explains: 'Select the servers and/or clusters to which you want to deploy this application. (You can reconfigure deployment targets later).' Another text box states: 'Available targets for setContainer :'. There are two sections: 'Servers' and 'Clusters'. Under 'Servers', there is a checkbox labeled 'AdminServer' which is checked. Under 'Clusters', there is a checkbox labeled 'appcluster' which is unchecked, and below it, a radio button labeled 'All servers in the cluster' which is selected. At the bottom, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

- f. In the **Optional Settings** page, click **Next**.

The screenshot shows the 'Install Application Assistant' dialog box, specifically the 'Optional Settings' page. At the top, there are navigation buttons: 'Back', 'Next', 'Finish', and 'Cancel'. The page title is 'Optional Settings'. Below the title, it says 'You can modify these settings or accept the defaults.' and '* Indicates required fields'. The settings are organized into sections: 'General', 'Security', 'Source Accessibility', and 'Plan Source Accessibility'. In the 'General' section, the question is 'What do you want to name this deployment?' and the 'Name' field is filled with 'setContainer'. In the 'Security' section, the question is 'What security model do you want to use with this application?' and the 'DD Only' option is selected. In the 'Source Accessibility' section, the question is 'How should the source files be made accessible?' and the 'Use the defaults defined by the deployment's targets' option is selected. Below this, it says 'Recommended selection.' and the 'Copy this application onto every target for me' option is selected. In the 'Plan Source Accessibility' section, the question is 'How should the plan source files be made accessible?' and the 'Use the same accessibility as the application' option is selected. Below this, it says 'Recommended selection.' and the 'Copy this plan onto every target for me' option is selected. At the bottom, there are navigation buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

Install Application Assistant

Back Next Finish Cancel

Optional Settings

You can modify these settings or accept the defaults.

* Indicates required fields

General

What do you want to name this deployment?

* Name: setContainer

Security

What security model do you want to use with this application?

☒ **DD Only:** Use only roles and policies that are defined in the deployment descriptors.

☐ **Custom Roles:** Use roles that are defined in the Administration Console; use policies that are defined in the deployment descriptor.

☐ **Custom Roles and Policies:** Use only roles and policies that are defined in the Administration Console.

☐ **Advanced:** Use a custom model that you have configured on the realm's configuration page.

Source Accessibility

How should the source files be made accessible?

☒ **Use the defaults defined by the deployment's targets**

Recommended selection.

☐ **Copy this application onto every target for me**

During deployment, the files will be copied automatically to the Managed Servers to which the application is targeted.

☐ **I will make the deployment accessible from the following location**

Location: /u01/practices/part1/practice06-03/setContainer.war

Provide the location from where all targets will access this application's files. This is often a shared directory. You must ensure the application files exist in this location and that each target can reach the location.

Plan Source Accessibility

How should the plan source files be made accessible?

☒ **Use the same accessibility as the application**

Recommended selection.

☐ **Copy this plan onto every target for me**

During deployment, the plan files will be copied automatically to the Managed Servers to which the application is targeted.

☐ **Do not copy this plan to targets**

You must ensure the plan files exist in the shared location and that each target can reach the location.

Back Next Finish Cancel

- g. Click **Finish**.

The web application is deployed. It uses the connection labeling callback class to switch database connections to other pluggable databases.

9. Test the setContainer web application.

a. Open a Browser window and navigate to <http://fmw1:7001/setContainer>.

The page displays the SQL results and fields to enter connection parameters.

The screenshot shows a web browser window with the address bar displaying `fmw1:7001/setContainer/`. The page content includes:

- SQL Query Results:** A table with columns **ID**, **NAME**, **PRICE**, and **VERSION**. The data rows are:

ID	NAME	PRICE	VERSION
8	Drixoral 12	1	1
12	Codeine 8	1	1
7	Advil 5	1	1
- Connect to DataSource :**
- Connect to PDB :**
- SQL to Execute :**
- Connect to URL :**
- Username :**
- Password :**
- Submit** button
- Server information:
 - Server: appcluster-1
 - Partition: baylandurgentcare
 - Partition: valleyhealth
 - Server: appcluster-2
 - Partition: baylandurgentcare
 - Partition: valleyhealth
 - Server: AdminServer

b. For **Connect to PDB**, enter `PDB1_2`.

c. Click **Submit**.

The connection labeling callback class uses the SET CONTAINER capability to use the `PDB1_2` pluggable database and the results from executing the SQL query are displayed.

SQL Query Results

ID NAME PRICE VERSION

8	Drixoral 12		1
12	Codeine 8		1
7	Advil 5		1

Connect to DataSource : jdbc/setcontainer

Connect to PDB : PDB1_2

SQL to Execute : select * from medrec.drugs

Connect to URL : t3://localhost:7001

Username : weblogic

Password : Welcome1

Submit

Server: appcluster-1
Partition: baylandurgentcare
Partition: valleyhealth
Server: appcluster-2
Partition: baylandurgentcare
Partition: valleyhealth
Server: AdminServer

- For **Connect to PDB**, enter PDB1_3 .
- Click **Submit**.

The connection labeling callback class uses the SET CONTAINER capability to use the PDB1_2 pluggable database and the results from executing the SQL query are displayed.

Login - Oracle Enterprise Mana... x Settings for setContainer - jrf_d... x WebLogic PDB Connection Demo x

fmw1:7001/setContainer/

SQL Query Results

ID	NAME	PRICE	VERSION
8	Drixoral 12		1
12	Codeine 8		1
7	Advil 5		1

Connect to DataSource : jdbc/setcontainer

Connect to PDB : PDB1_3

SQL to Execute : select * from medrec.drugs

Connect to URL : t3://localhost:7001

Username : weblogic

Password : Welcome1

Submit

Server: appcluster-1
 Partition: baylandurgentcare
 Partition: valleyhealth
 Server: appcluster-2
 Partition: baylandurgentcare
 Partition: valleyhealth
 Server: AdminServer

- f. View the terminal window running startWebLogic.sh.
The connection labeling callback class messages to the system display.


```

Admin Server(JRF Domain)
File Edit View Search Terminal Help
]" is now listening on 192.168.0.160:7001 for protocols iiop, t3, ldap, snmp, ht
tp.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <Server> <BEA-002613> <Channel "Default[2
]" is now listening on 0:0:0:0:0:0:1%lo:7001 for protocols iiop, t3, ldap, snm
p, http.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <WebLogicServer> <BEA-000360> <The server
started in RUNNING mode.>
<Feb 18, 2016 3:39:44 PM UTC> <Notice> <WebLogicServer> <BEA-000365> <Server sta
te changed to RUNNING.>
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_1' != 'null' ##
###--> Lesson6ConnectionLabelingCallback: In configure for value PDB1_1
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_2' != 'PDB1_1' ##
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_2' != 'null' ##
###--> Lesson6ConnectionLabelingCallback: In configure for value PDB1_2
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_1' != 'PDB1_2' ##
###--> Lesson6ConnectionLabelingCallback: Exact match found = PDB1_1
getting server runtimes
printing servers
server: appcluster-1
2
partition: baylandurgentcare
partition: valleyhealth
server: appcluster-2
2
partition: baylandurgentcare
partition: valleyhealth
server: AdminServer
0
setContainer -->
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_3' != 'PDB1_1' ##
###--> Lesson6ConnectionLabelingCallback: No match 'PDB1_3' != 'PDB1_2' ##

```


Practices for Lesson 7 - Coherence with Multitenancy

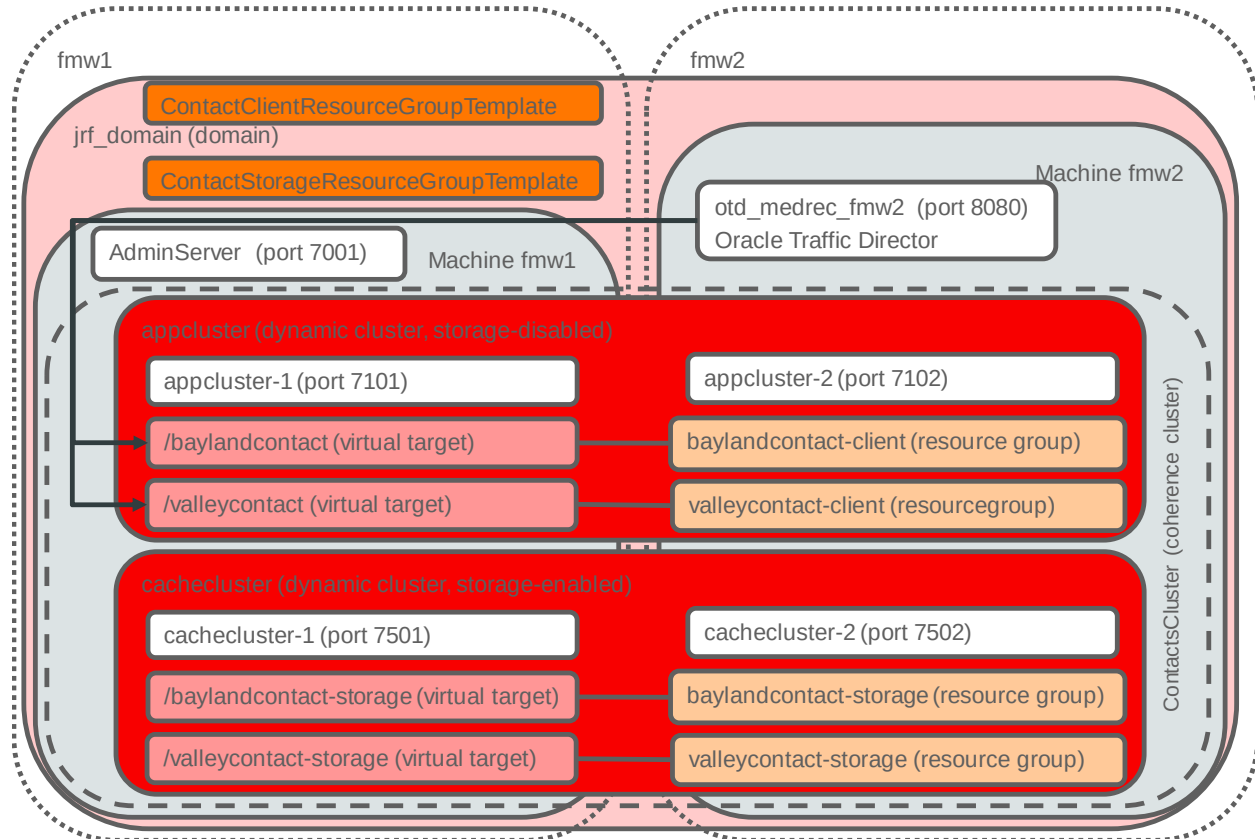
Chapter 7

Practices for Lesson 7: Overview

Overview

In these practices, you create Coherence Clusters managed by WebLogic Server, Resource Group Templates for the coherence application and storage, and create domain partitions for the coherence applications. You also configure one cache as a shared cache for both applications and then test the applications to demonstrate the shared cache.

The diagram below illustrates the configuration:



Practice 7-1: Creating Coherence Clusters Managed by WebLogic Server

Overview

In this practice, using Fusion Middleware Control, you create a new dynamic cluster and coherence cluster for the contacts' sample application. Finally, you restart both dynamic clusters to ensure they are running for subsequent practices.

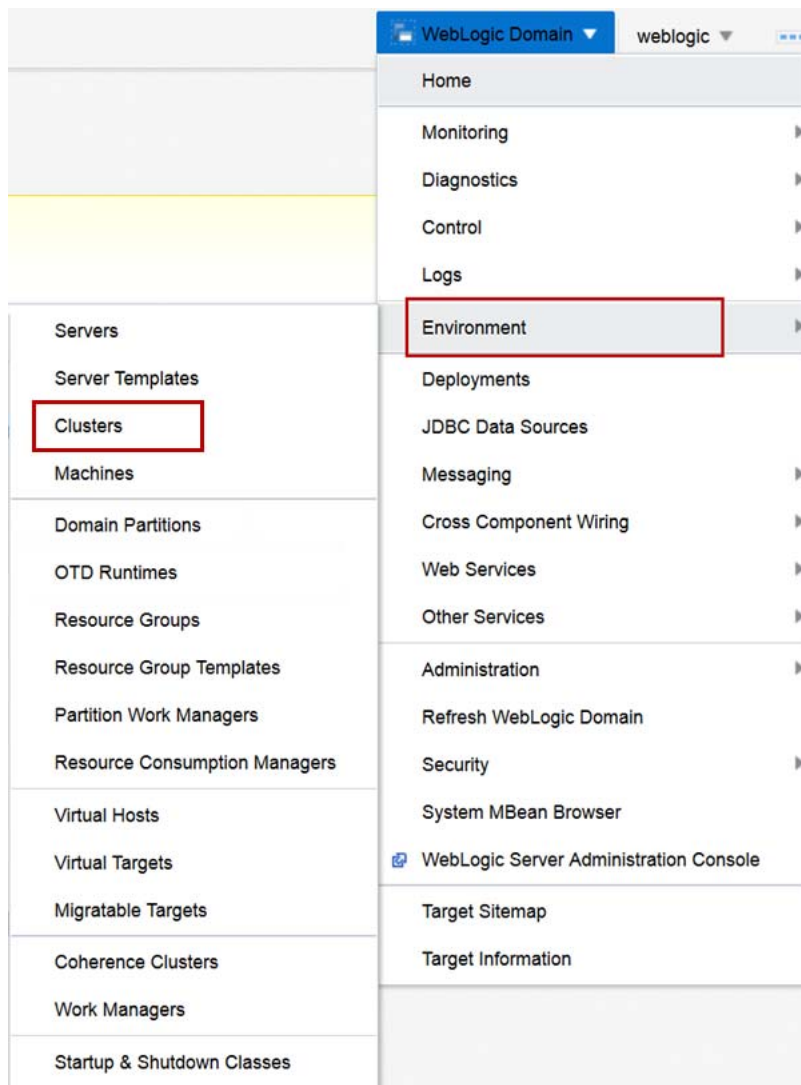
Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
 - a. Open a browser window and navigate to **http://fmw1:7001/em**.
 - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.
The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the Dynamic Cluster cachecluster.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters**.



- b. Click **Create** and select **Dynamic Cluster**.
- c. For **Name**, enter `cachecluster` and click **Next**.
- d. For **Dynamic Cluster Size**, enter **2**.
- e. For **Max Dynamic Cluster Size**, enter **4**, and click **Next**.
- f. Select **Use any machine configured in this domain** and click **Next**.
- g. Select **Assign each dynamic server unique listen ports**.
- h. For **Base Listen Port**, enter **7500**.
- i. For **SSL Base Listen Port**, enter **8500** and click **Next**.

Create a Dynamic Cluster: Review

Back

Step 5 of 5

Next

Create

Cancel

You have elected to create a new cluster with the following configuration.

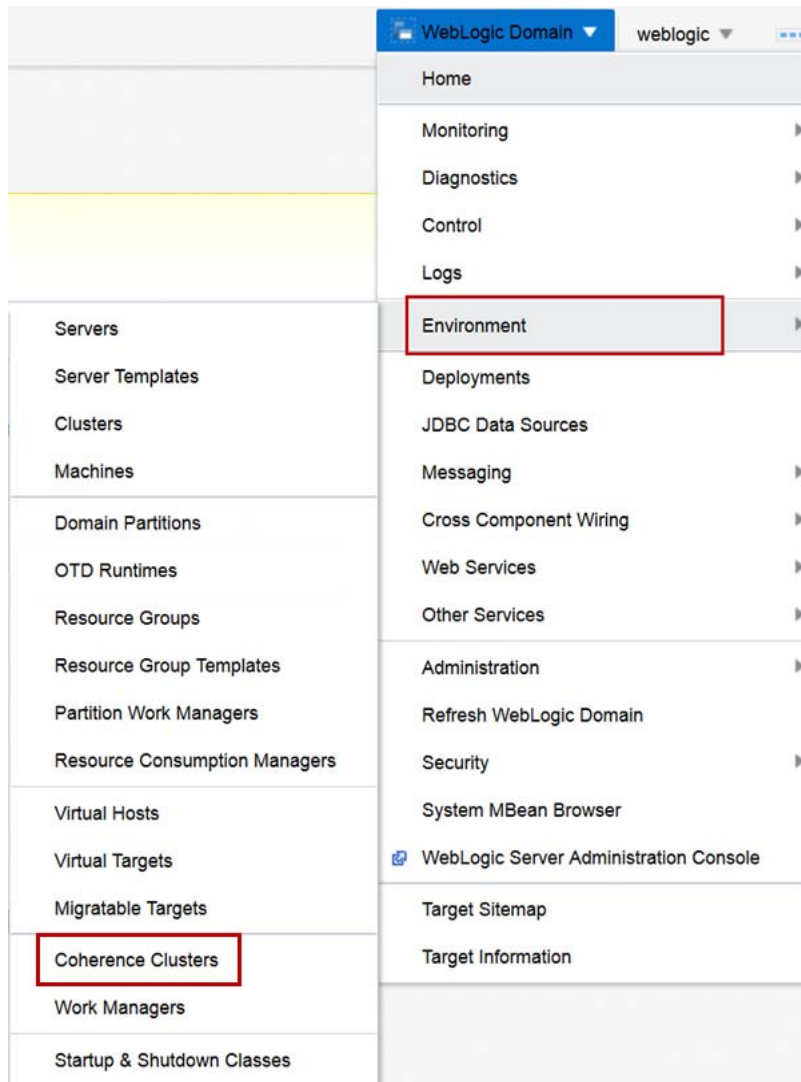
Name	cachecluster
Messaging Mode	unicast
Unicast Broadcast Channel	
Multicast Address	239.192.0.0
Multicast Port	7001

Dynamic Cluster Size	2
Max Dynamic Cluster Size	4
Server Name Prefix	cachecluster-
Enable Calculated Listen Ports	Yes
Enable Calculated Machine Associations	Yes

Name of New Server Template	cachecluster-Template
Listen Port	7500
SSL Listen Port	8500

j. Click **Create**.

3. Create the Coherence Cluster ContactsCluster.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Coherence Clusters**.



- b. Click **Create**.
 - c. For **Name**, enter `ContactsCluster` and click **Next**.
 - d. Select **appcluster** and **cachecluster** as members of this Coherence Cluster and click **Next**.

Create a Coherence Cluster Configuration: Review

Back

Step 3 of 3

Next

Create

Cancel

Review the properties of your new Coherence cluster configuration.

Name

ContactsCluster2

Use a Custom Cluster Configuration File

false

Clustering Mode

unicast

Cluster Listen Port

7574

Multicast Listen Address

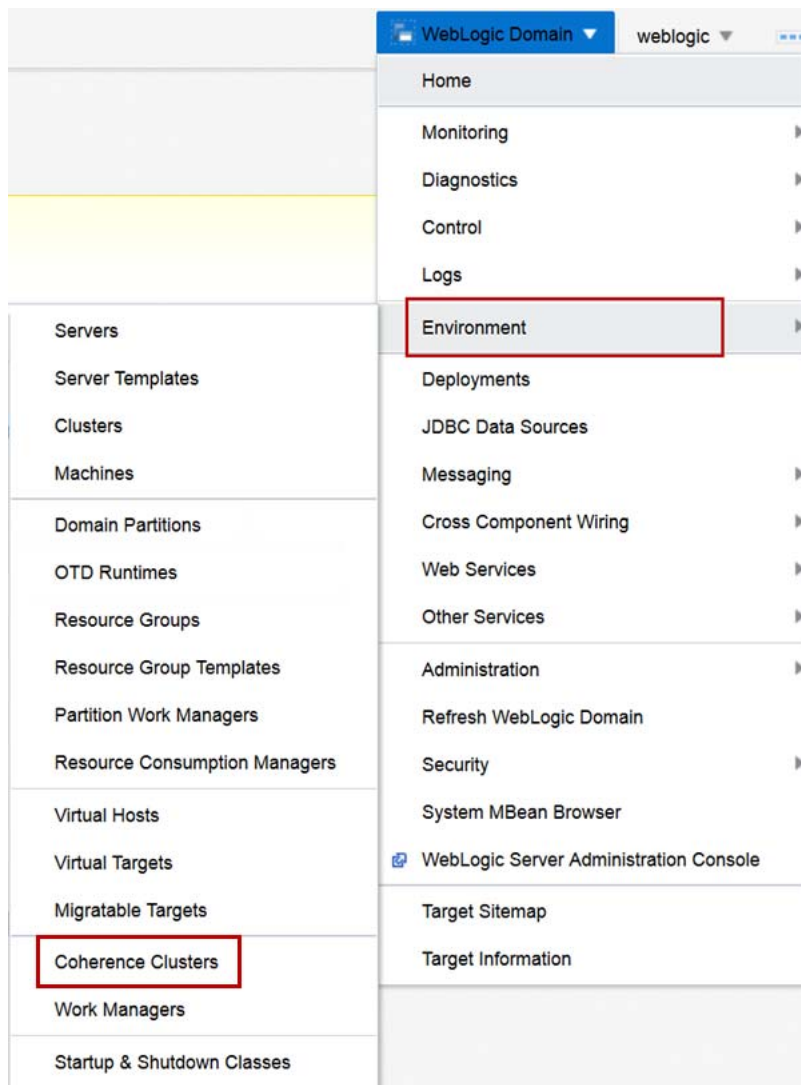
231.1.1.1

Members

Name	Type
appcluster	Oracle WebLogic Cluster
cachedcluster	Oracle WebLogic Cluster

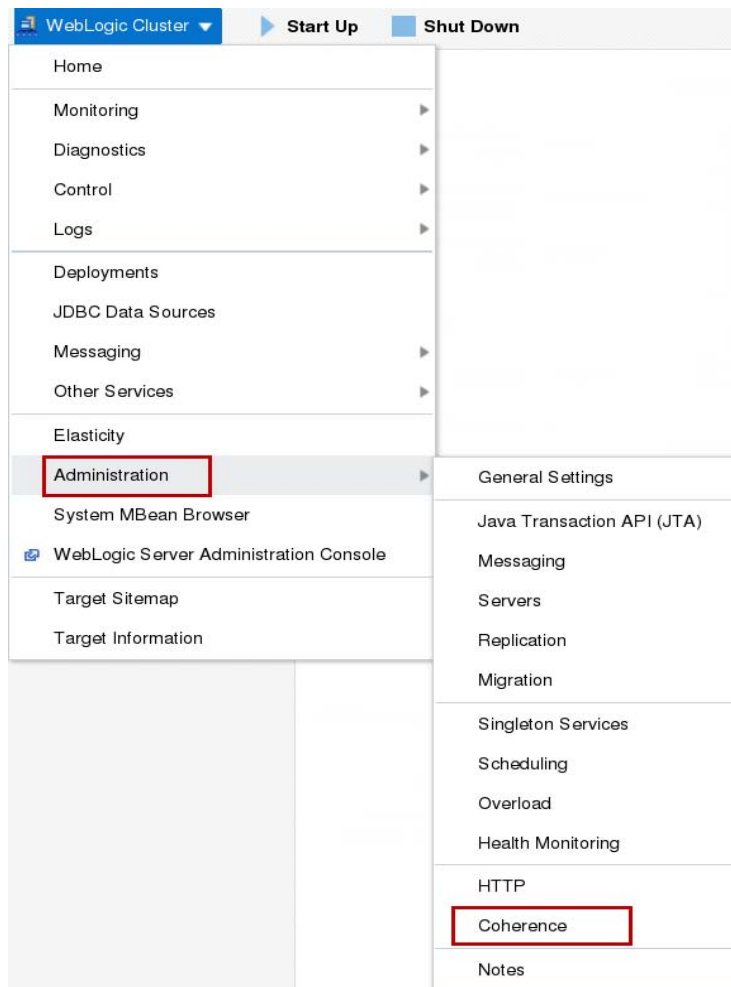
e. Click **Create**.

4. Ensure the appcluster is not storage enabled and the cachecluster is storage enabled.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters**.

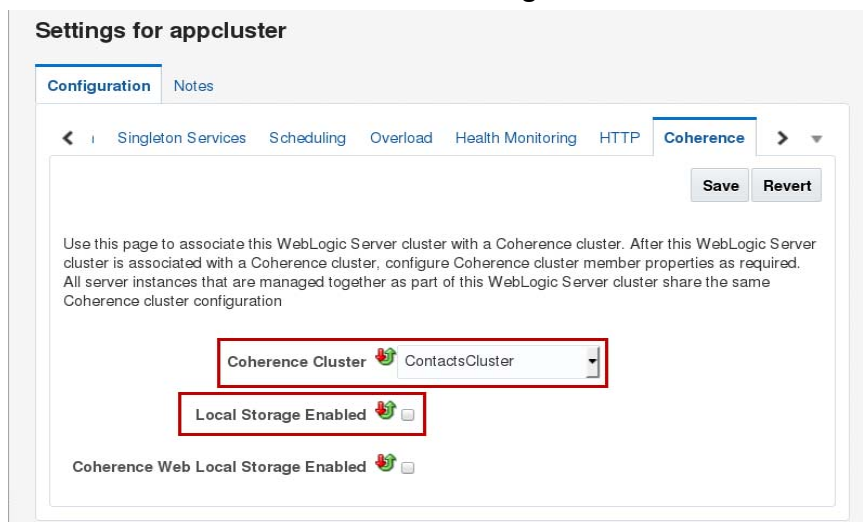


- b. Click the **appcluster** entry.

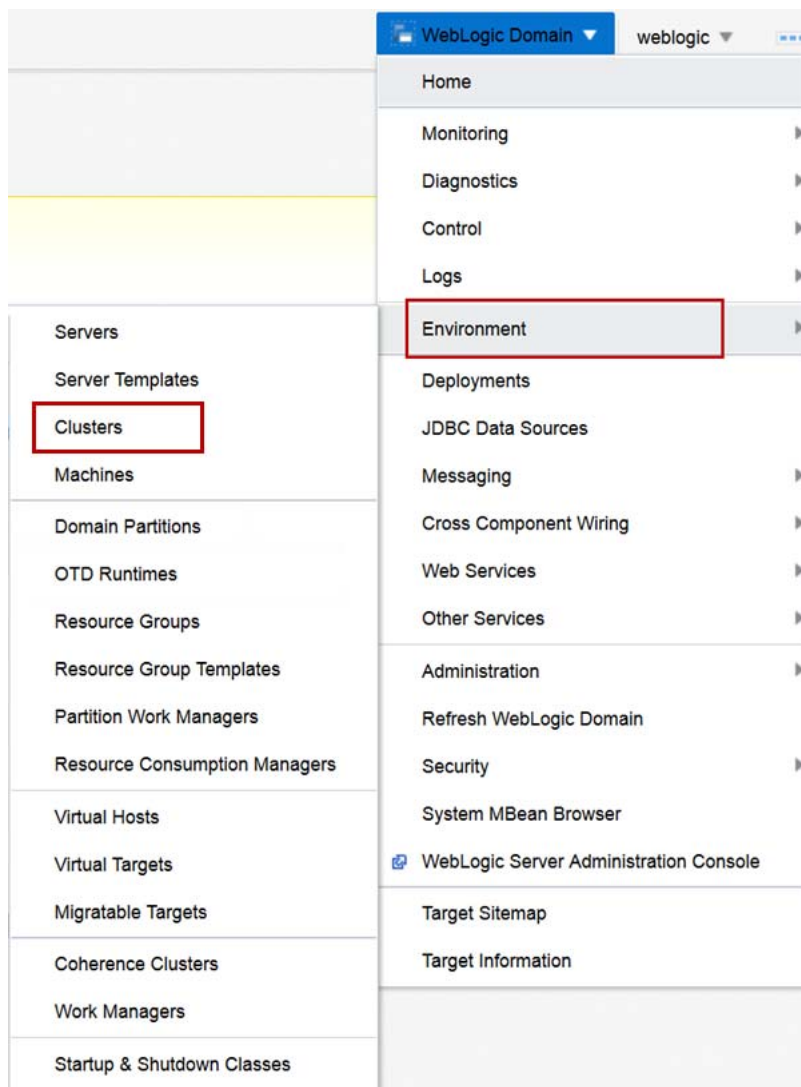
- c. From the **WebLogic Cluster** menu, select **Administration** and then select **Coherence**.



- d. For **Coherence Cluster**, select **ContactsCluster**.
e. Deselect **Local Storage Enabled**.
f. Deselect **Coherence Web Local Storage Enabled** and click **Save**.

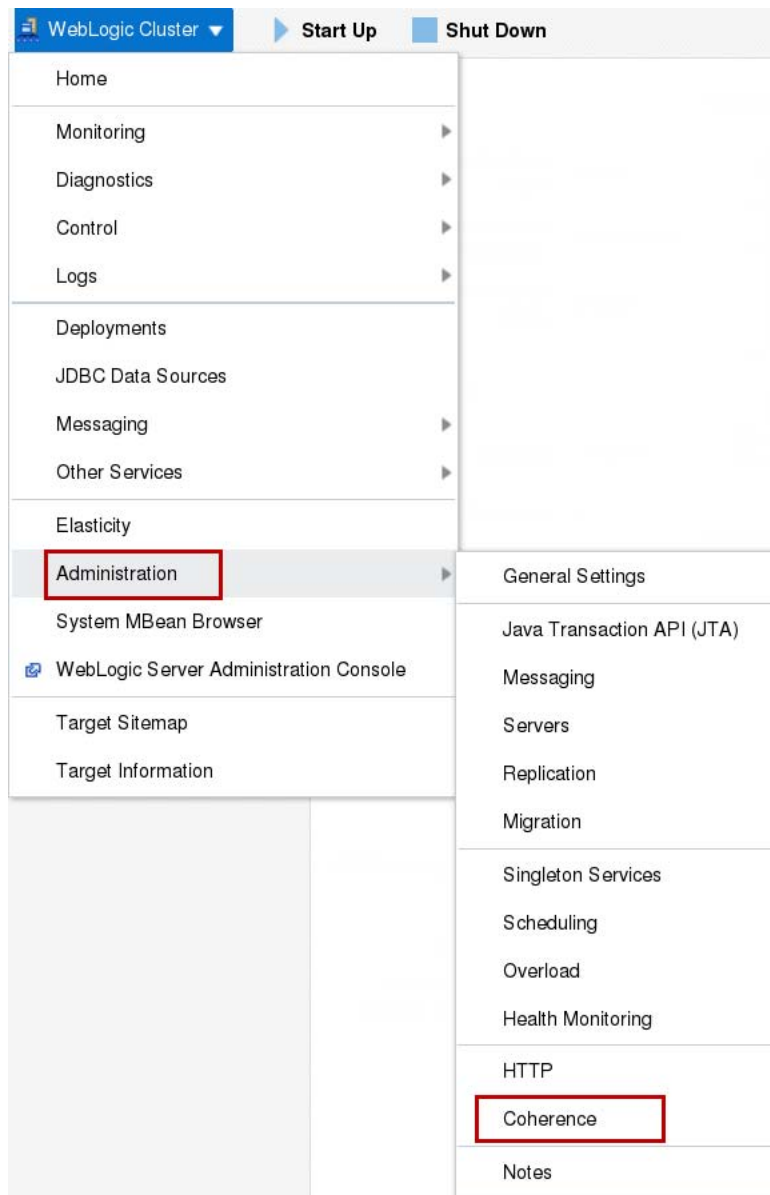


- g. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters**.



h. Click the **cachecluster** entry.

- i. From the **WebLogic Cluster** menu, select **Administration** and then select **Coherence**.



- j. For **Coherence Cluster**, select **ContactsCluster**.
k. Select **Local Storage Enabled**.
l. Deselect **Coherence Web Local Storage Enabled** and click **Save**.

Settings for cacheccluster

[Configuration](#)
[Notes](#)

[Singleton Services](#)
[Scheduling](#)
[Overload](#)
[Health Monitoring](#)
[HTTP](#)
[Coherence](#)

[Save](#)
[Revert](#)

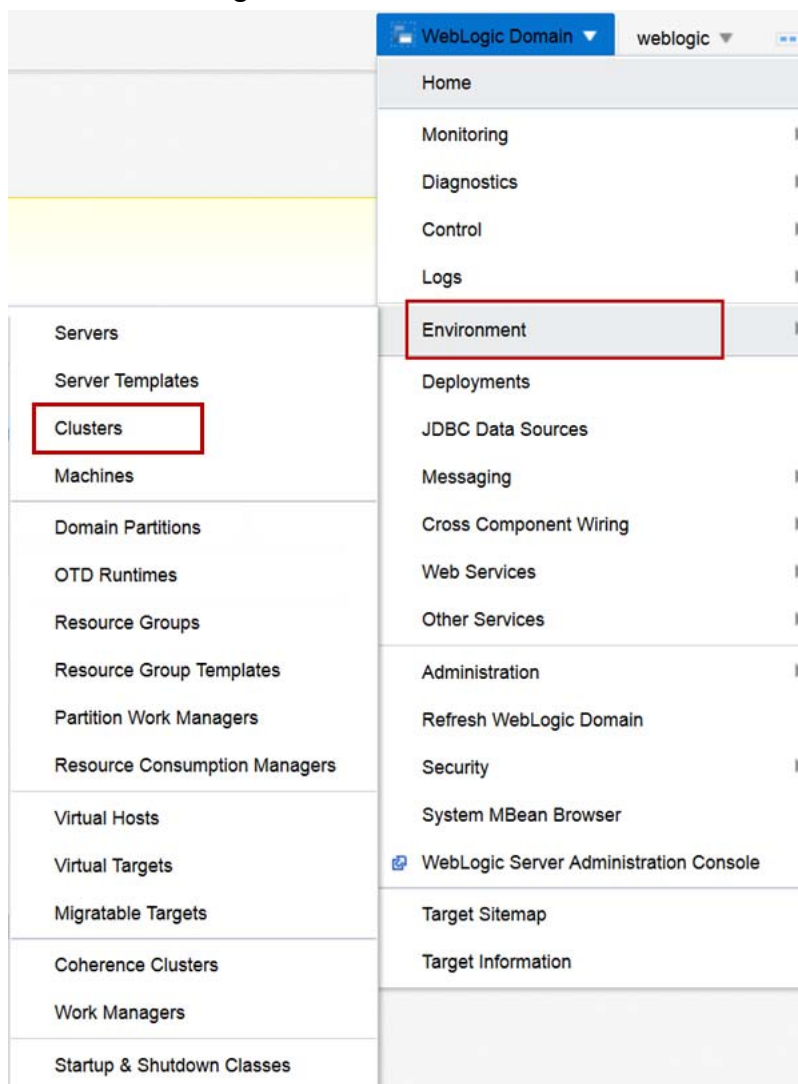
Use this page to associate this WebLogic Server cluster with a Coherence cluster. After this WebLogic Server cluster is associated with a Coherence cluster, configure Coherence cluster member properties as required. All server instances that are managed together as part of this WebLogic Server cluster share the same Coherence cluster configuration

Coherence Cluster

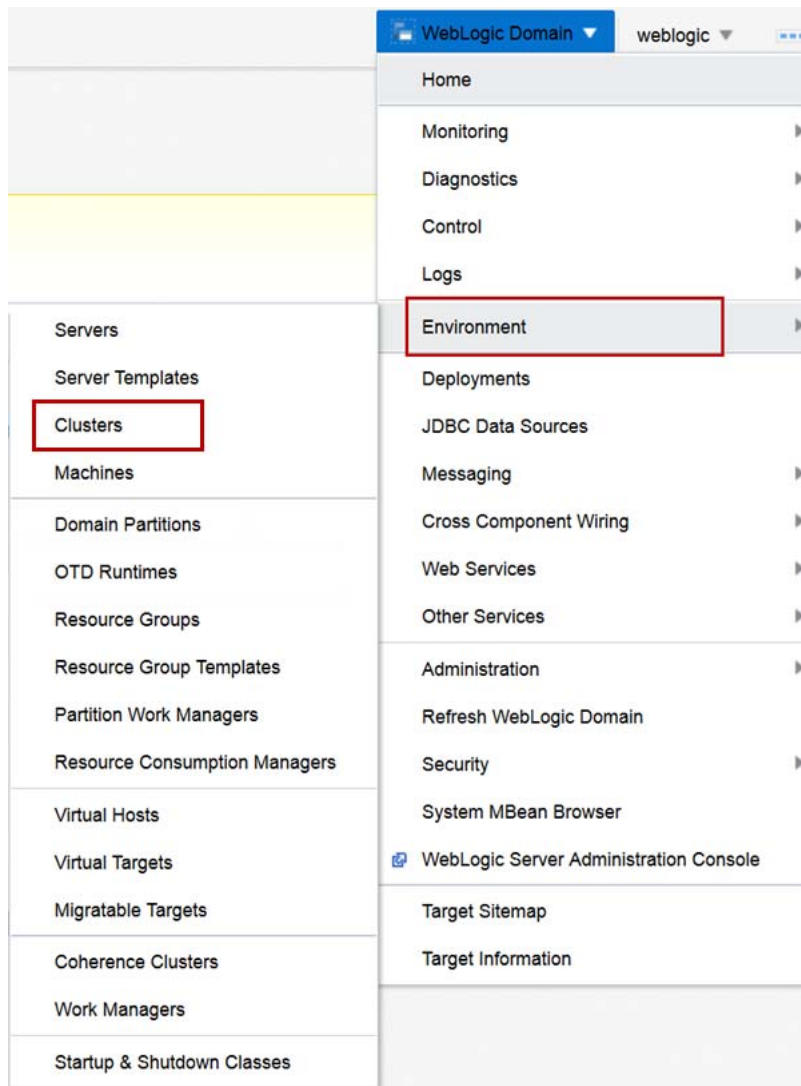
Local Storage Enabled
☒

Coherence Web Local Storage Enabled
☐

6. Restart the Dynamic Cluster appcluster.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters**.



- b. Click the table entry for **appcluster**.
 - c. Click **Control**, select **Shutdown**, and select **Force shutdown now**.
 - d. Click **Forcibly Shutdown Servers**.
 - e. Click the **Refresh** icon.
 - f. Click the table entry for **appcluster**.
 - g. Click **Control** and select **Start**.
 - h. Click **Start Servers**.
7. Start the Dynamic Cluster cachecluster.
- a. From the **WebLogic Domain** menu, select **Environment** and then click **Clusters**.



- b. Click the table entry for **cachecluster**.
- c. Click **Control** and select **Start**.
- d. Click **Start Servers**.

Practice Solution: Creating Coherence Clusters Managed by WebLogic Server

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating the dynamic cluster cachecluster or the coherence cluster, this script will not work. To run the script in that case, delete the partially created clusters before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
 - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice07-01
```
 - b. Run the solution script.

```
$ ./solution.sh
```
 - c. Enter any passwords as you are prompted.

Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_coherence_clusters.py`, which creates the new generic data source for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>> Solution for Practice 07-01 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

Practice 7-2: Creating Coherence Resource Group Templates

Overview

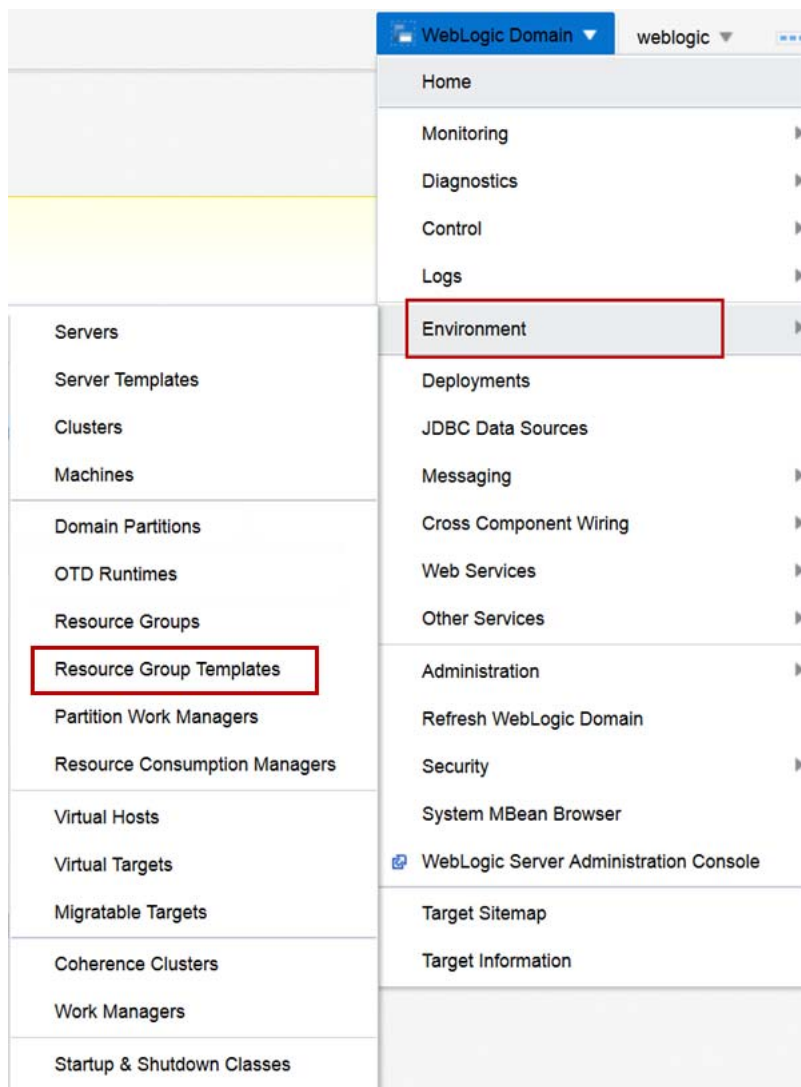
In this practice, you create the resource group templates required for the contact sample application—one resource group template for the client application and cluster and a second resource group template for the storage application and cluster.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster coherencecluster has been created and is running.

Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
 - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
 - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.
The Fusion Middleware Control home page for the **jrf_domain** is displayed.
2. Create the ContactClientResourceGroupTemplate.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates**.



- b. Click **Create**.
- c. For **Name**, enter `ContactClientResourceGroupTemplate` and click **Create**.
- d. Click the **ContactClientResourceGroupTemplate** table entry.
- e. Click the **Deployments** tab.
- f. Click **Deployment** and select **Deploy**.
- g. Select **Archive or exploded directory is on the server where Enterprise Manager is running**.
- h. Click **Browse**.
- i. Browse to `/u01/practices/part1/practice07-02/contact/ear/ExampleEAR.ear` and click **OK**.
- j. Click **Next**.
- k. In the **Distribution** section, select **Install only. Do not start.** and click **Next**.

Deploy Java EE Application: Deployment Settings
Back
Step 4 of 4
Next
Deploy
>>

Hide Deployment Summary

Archive Type

Java EE Application (EAR file)

Archive Location

/u01/practices/part1/practice07-02/contact/ear/ExampleEAR.ear

Deployment Plan

Create a new plan

Scope

Resource group template "ContactClientResourceGroupTemplate"

Deployment Type

Application

Application Name

ExampleEAR

Version

Not versioned

Context Root

example-web-app

Deployment Mode

Install only. Do not start.

Deployment Tasks

The table below lists common tasks that you may wish to do before deploying the application.

Name	Go To Task	Description
Configure Web Modules		Configure the Web modules in your application.
Configure Application Security		Configure application policy migration, credential migration and other security behavior.

Deployment Plan

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

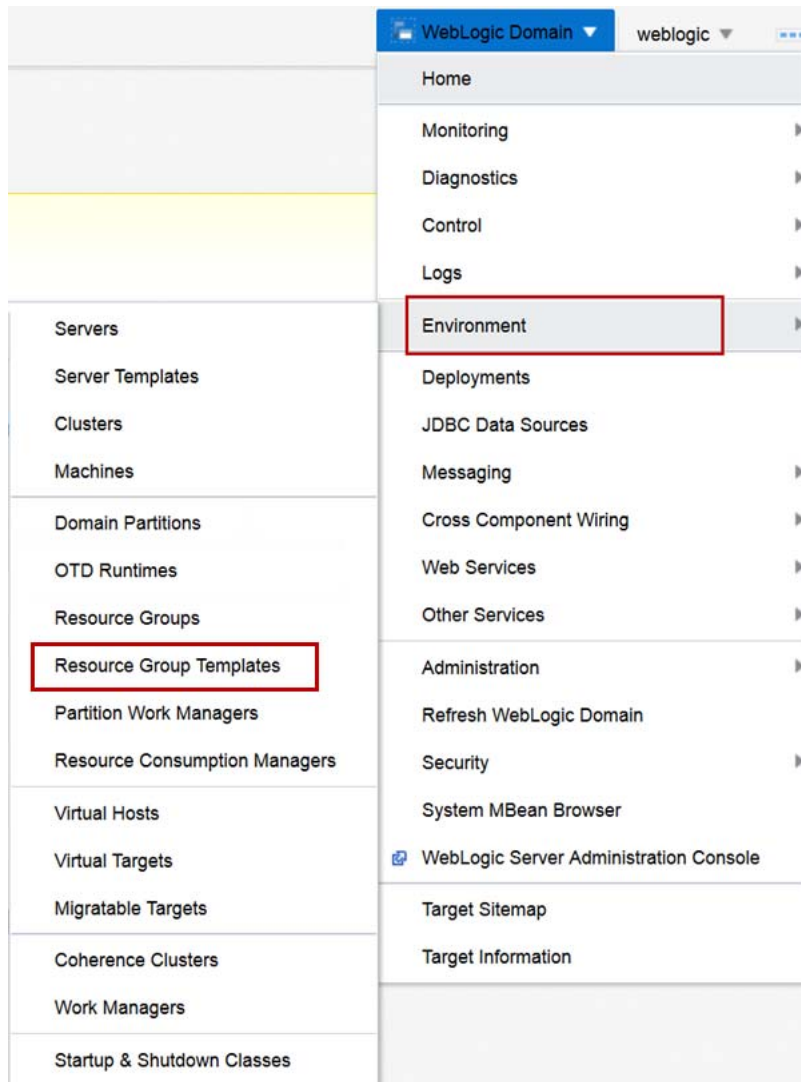
Edit Deployment Plan

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

Save Deployment Plan

- I. Review the Deployment Settings and click **Deploy**.
- m. Click **Close**.

3. Create the ContactStorageResourceGroupTemplate.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Resource Group Templates**.



- b. Click **Create**.
- c. For **Name**, enter ContactStorageResourceGroupTemplate and click **Create**.
- d. Click the **Deployments** tab.
- e. Click **Deployment** and select **Deploy**.
- f. Select **Archive or exploded directory is on the server where Enterprise Manager is running**.
- g. Click **Browse**.
- h. Browse to /u01/practices/part1/practice07-02/contact/ear/ExampleListenerEAR.ear and click **OK**.
- i. Click **Next**.
- j. In the **Distribution** section, select **Install only. Do not start.** and click **Next**.

Deploy Java EE Application: Deployment Settings

[Back](#)
Step 4 of 4
[Next](#)
[Deploy](#)
>>

Hide Deployment Summary

Archive Type

Java EE Application (EAR file)

Archive Location

/u01/practices/part1/practice07-02/contact/ear/ExampleListenerEAR.ear

Deployment Plan

Create a new plan

Scope

Resource group template "ContactStorageResourceGroupTemplate"

Deployment Type

Application

Application Name

ExampleListenerEAR

Version

Not versioned

Deployment Mode

Install only. Do not start.

Deployment Tasks

The table below lists common tasks that you may wish to do before deploying the application.

Name	Go To Task	Description
Configure EJBs		Configure the Enterprise Java Beans in your application.
Configure Application Security		Configure application policy migration, credential migration and other security behavior.

Deployment Plan

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

Edit Deployment Plan

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

Save Deployment Plan

- k. Review the Deployment Settings and click **Deploy**.
- l. Click **Close**.
- m. Click **Deployment** and select **Deploy**.
- n. Select **Archive or exploded directory is on the server where Enterprise Manager is running**.
- o. Click **Browse**.
- p. Browse to /u01/practices/part1/practice07-02/contact/gar/ExampleGAR.gar and click **OK**.
- q. Click **Next**.
- r. In the **Distribution** section, select **Install only. Do not start.** and click **Next**.

Deploy Java EE Application: Deployment Settings

[Back](#)
Step 4 of 4
[Next](#)
[Deploy](#)
>>

Hide Deployment Summary

Archive Type

Coherence Archive (GAR file)

Archive Location

/u01/practices/part1/practice07-02/contact/gar/ExampleGAR.gar

Deployment Plan

Create a new plan

Scope

Resource group template "ContactStorageResourceGroupTemplate"

Deployment Type

Application

Application Name

ExampleGAR

Version

Not versioned

Deployment Mode

Install only. Do not start.

Deployment Tasks

The table below lists common tasks that you may wish to do before deploying the application.

Name	Go To Task	Description
Configure Application Security		Configure application policy migration, credential migration and other security behavior.

Deployment Plan

You can optionally use the Edit Deployment Plan option to set more advanced deployment options which the deployment tasks above do not cover.

Edit Deployment Plan

You can optionally save the deployment plan to your local disk. You can redeploy this application later using your saved deployment plan and not have to edit the deployment plan.

Save Deployment Plan

- s. Review the Deployment Settings and click **Deploy**.
- t. Click **Close**.

Practice Solution: Creating Coherence Resource Group Templates

Perform the following tasks if you did not complete this practice and want to use the finished solution.

Note: If you started creating domain partitions, this script will not work. To run the script in that case, delete the partially created domain partitions before running this script.

Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster coherencecluster has been created and is running.

Solution Tasks

1. Connect to fmw1 as the `oracle` user.
2. Run the solution script.
 - a. Open a Terminal window as the `oracle` user and navigate to the practice directory.

```
$ cd /u01/practices/part1/practice07-02
```
 - b. Run the solution script.

```
$ ./solution.sh
```
 - c. Enter any passwords as you are prompted.

Note: This script starts the WebLogic Scripting Tool (WLST) passing it the WLST script `create_coherence_rg.py`, which creates the domain partitions for you. The WLST script prints out messages. Be patient and do not close the Terminal window until you see these messages:

```
...
>>> Solution for Practice 07-02 applied successfully!
```

```
Exiting WebLogic Scripting Tool.
```

3. Close the Terminal window.

Practice 7-3: Creating Domain Partitions Using Coherence Resources

Overview

In this practice, you create the domain partition for the contact sample application—one domain partition for bayland and a second domain partition for valley. Finally, you start the domain partitions and test the contact application and shared coherence cache.

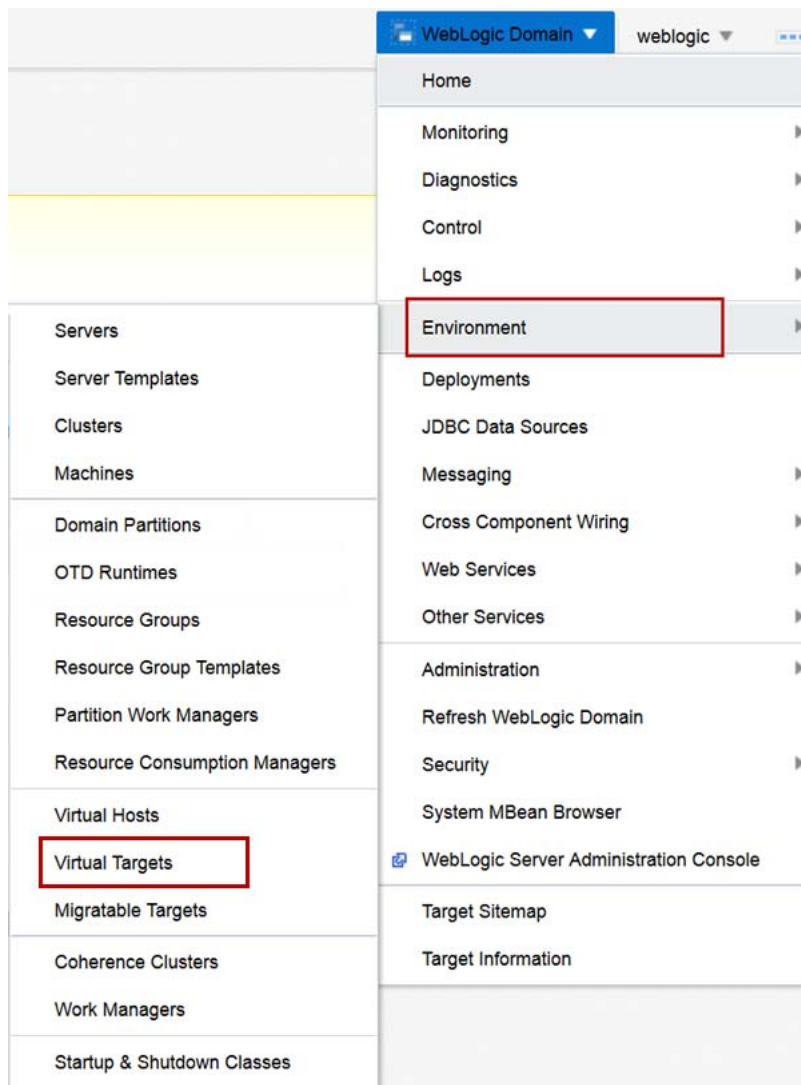
Assumptions

- WebLogic Server 12.2.1 Infrastructure is installed.
- The jrf_domain admin server is running.
- The nodemanager for host fmw1, nodemanager-fmw1 is running.
- The nodemanager for host fmw2, nodemanager-fmw2 is running.
- The machines fm1 and fmw2 have been created.
- The dynamic cluster appcluster has been created and is running.
- The dynamic cluster cachecluster has been created and is running.
- The resource group templates ContactClientResourceGroupTemplate and ContactStorageResourceGroupTemplate have been created.
- The Oracle Traffic Director configuration medrec has been created and registered.

Tasks

1. Log in to the jrf_domain WebLogic Server Domain using Fusion Middleware Control.
 - a. Open a Browser window and navigate to **http://fmw1:7001/em**.
 - b. For **User Name**, enter `weblogic` and for **Password**, enter the password from the Course Practice Environment: Security Credentials document for the weblogic user for the Oracle WebLogic Server product. Click **Login**.
The Fusion Middleware Control home page for the **jrf_domain** is displayed.

2. Create the baylandcontact-client virtual target.
 - a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets**.



- b. Click **Create**.
- c. For **Name**, enter baylandcontact-client.
- d. For **Uri Prefix**, enter /baylandcontact.

Create Virtual Target: General

Back

Step 1 of 2

Next

Cancel

Name *

baylandcontact-client

Uri Prefix

/baylandcontact

Hosts

Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

View

+ Add

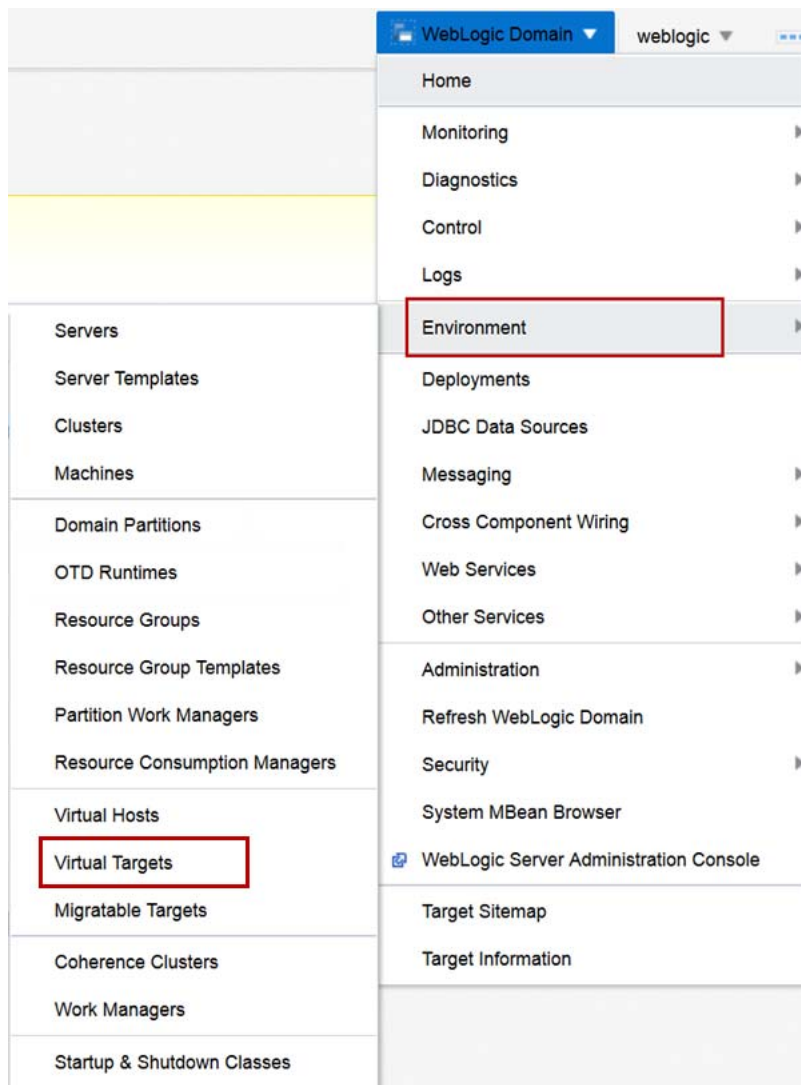
✕ Delete

Host Name
No host names specified for this virtual target.

► Advanced Options

- e. Click **Next**.
- f. Select **appcluster** from the **Cluster** menu.
- g. Click **Create**.

3. Create the baylandcontact-storage virtual target.
 - a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets**.



- b. Click **Create**.
- c. For **Name**, enter baylandcontact-storage.
- d. For **Uri Prefix**, enter /baylandcontact-storage.

Create Virtual Target: General

Back

Step 1 of 2

Next

Cancel

Name *

baylandcontact-storage

Uri Prefix

/baylandcontact-storage

Hosts

Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

View

+ Add

✕ Delete

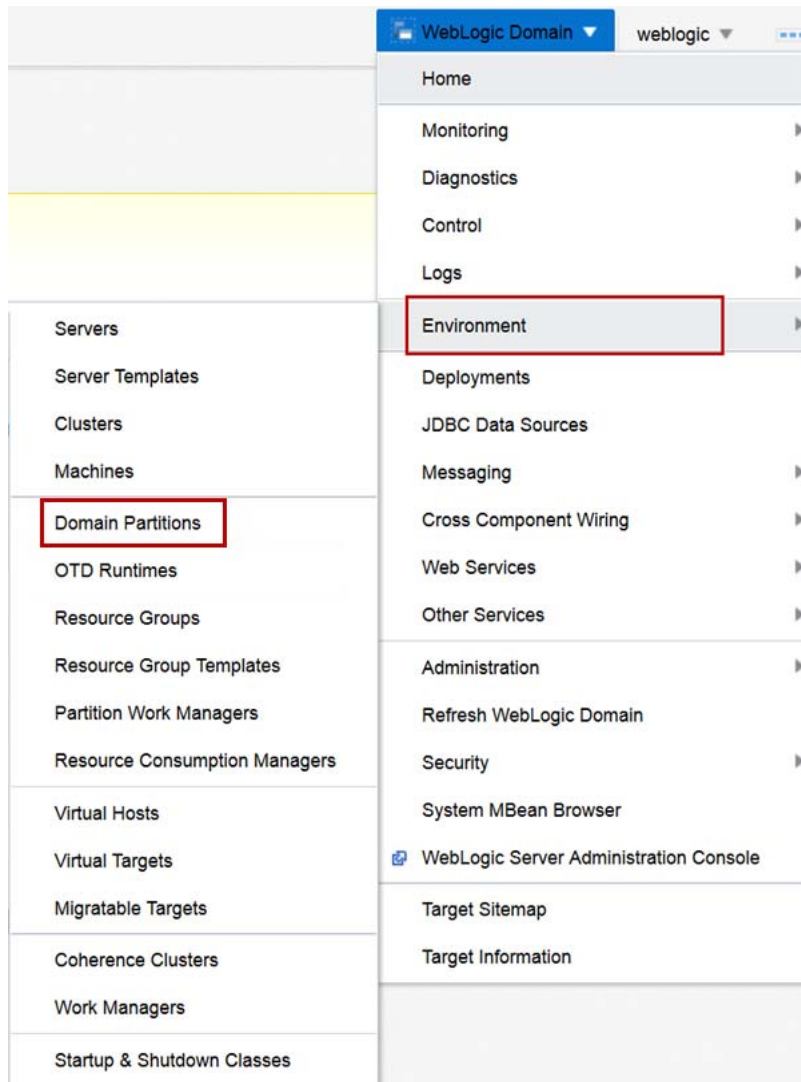
Host Name

No host names specified for this virtual target.

► Advanced Options

- e. Click **Next**.
- f. Select **cachecluster** from the **Cluster** menu.
- g. Click **Create**.

4. Create the baylandcontact domain partition.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter baylandcontact.
- d. For **Security Realm**, select bayland.
- e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing**.
- f. For **OTD Runtime**, select medrec.
- g. Click **Next**.

- h. For the Virtual Target **baylandcontact-client**, choose both **Select** and **Set as Default**.
- i. For the Virtual Target **baylandcontact-storage**, choose **Select** and click **Next**.

Create Domain Partition: Available Targets Back Step 2 of 4 Next Cancel

Select the virtual targets that will be available for this domain partition to use. Note that virtual targets can only be used by one partition; so, only available virtual targets are listed below.

Select	Virtual Target	Set as Default
☒	baylandcontact-client	☒
☒	baylandcontact-storage	☐

- j. For **Resource Group Name**, enter baylandcontact-client.
- k. For **Resource Group Template**, select **ContactClientResourceGroupTemplate**.
- l. Move the **baylandcontact-client** target to **Selected Targets**.
- m. Click **Next**.

Create Domain Partition: Summary Back Step 4 of 4 Next Create Cancel

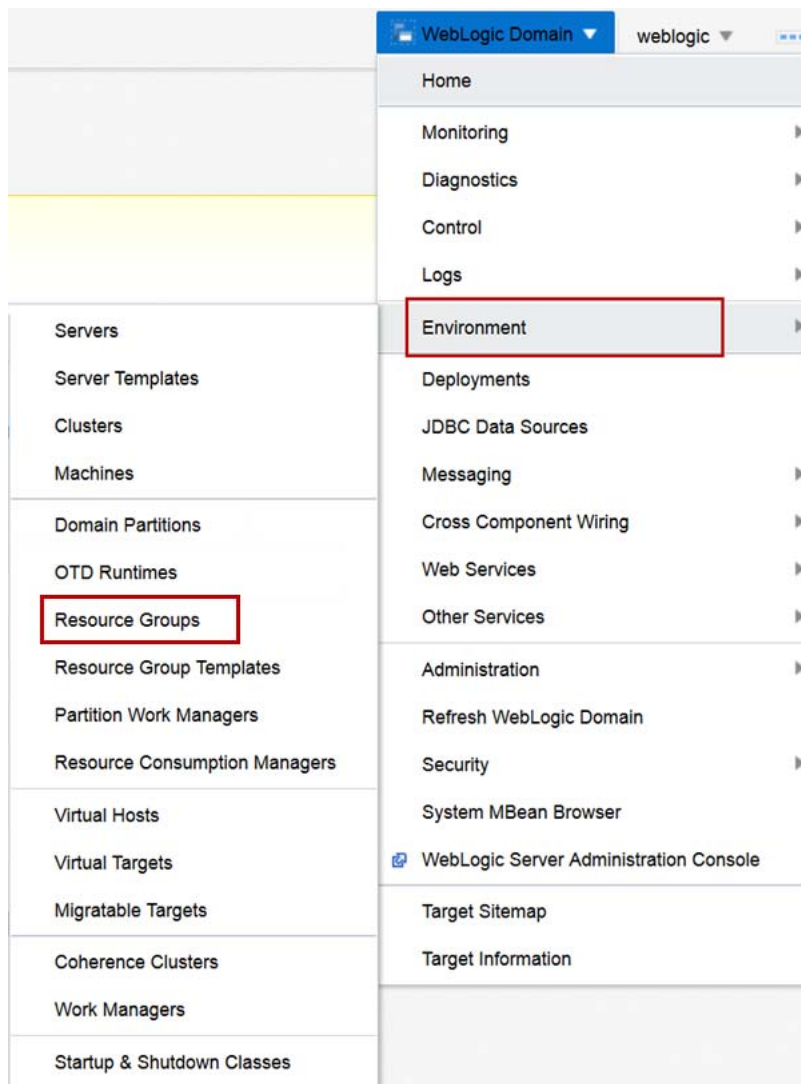
Information
This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes:

Name	baylandcontact
Realm	bayland
Primary Identity Domain	None
OTD Runtime	medrec
Available Targets	baylandcontact-client,baylandcontact-storage
Default Targets	baylandcontact-client
Resource Group	baylandcontact-client
Resource Group Template	ContactClientResourceGroupTemplate
Targets for the Resource Group	baylandcontact-client

- n. Click **Create**.
- o. Click the **baylandcontact** domain partition.

- p. From the **Domain Partition** menu, select **Administration** and select **Resource Groups**.



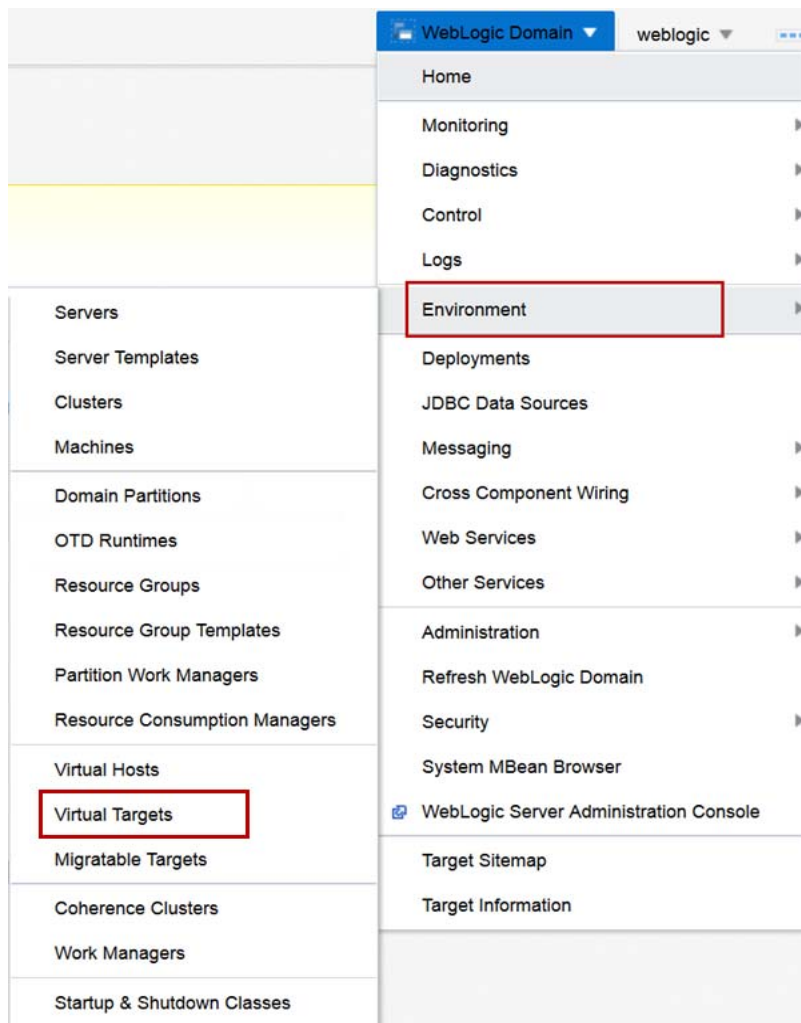
- q. Click **Create**.
- r. For **Name**, enter `baylandcontact-storage`.
- s. For **Resource Group Template**, select **ContactStorageResourceGroupTemplate**.

- t. For **Targets**, move **baylandcontact-storage** from **Available** to **Chosen** and click **Create**.

- u. Click the **Coherence Caches** tab.
- v. Click **Add**.
- w. For **Name**, enter `stores-shared-cache`.
- x. For **Coherence Cache Name**, enter `stores`.
- y. For **GAR (Grid Archive) Application Name**, enter `ExampleGAR`.
- z. Click **Shared** to allow this cache to be shared by multiple domain partitions.

- aa. Click **Save**.
The baylandcontact domain partition is created.

5. Start the baylandcontact domain partition.
 - a. In the **baylandcontact** domain partition details page, click **Start Up**.
 - b. When the Confirmation dialog box indicates the Start Operation is successful, click **Close**.
The baylandcontact domain partition successfully starts.
6. Test the baylandcontact domain partition.
 - a. In the Browser window, navigate to <http://fmw2:8080/baylandcontact/example-web-app>.
Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the baylandcontact deployment.
7. Create the valleycontact-client virtual target.
 - a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets**.



- b. Click **Create**.
- c. For **Name**, enter valleycontact-client.
- d. For **Uri Prefix**, enter /valleycontact.

Create Virtual Target: General Back Step 1 of 2 Next Cancel

Name *

Uri Prefix

Hosts

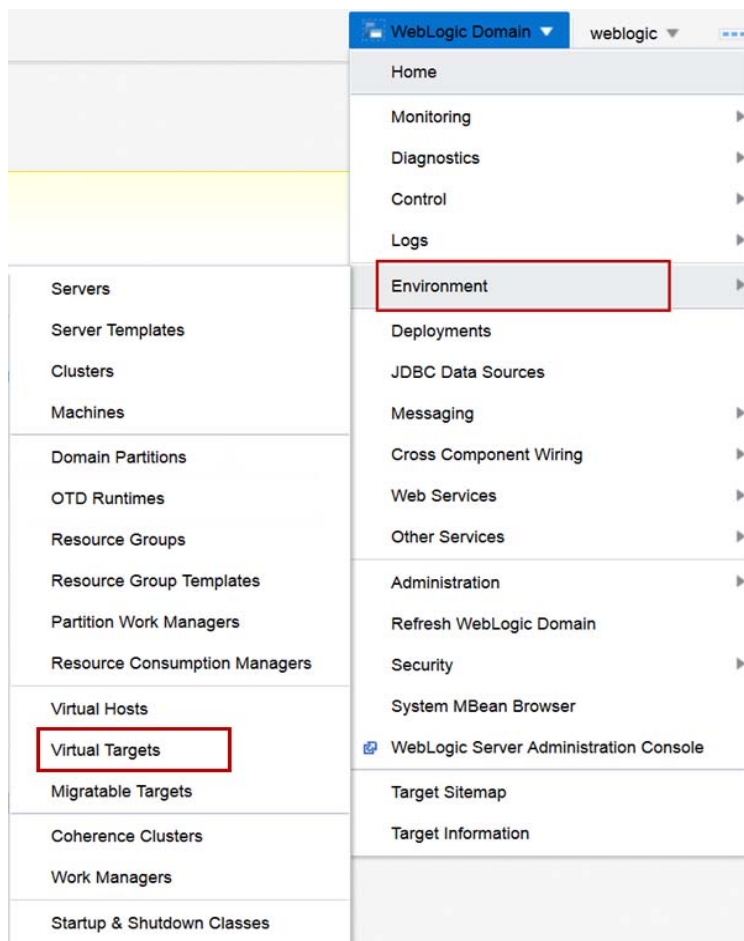
Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

View + Add × Delete

Host Name
No host names specified for this virtual target.

▶ Advanced Options

- e. Click **Next**.
- f. Select **appcluster** from the **Cluster** menu.
- g. Click **Create**.
8. Create the valleycontact-storage virtual target.
 - a. From the **WebLogic Domain** menu, select **Environment** and click **Virtual Targets**.



- b. Click **Create**.
- c. For **Name**, enter valleycontact-storage.
- d. For **Uri Prefix**, enter /valleycontact-storage .

Create Virtual Target: General

BackStep 1 of 2NextCancel

Name *

valleycontact-storage

Uri Prefix

/valleycontact-storage

Hosts

Specify the host name(s) that will be used for front-end access. If you are not using Oracle Traffic Director (OTD) to load balance connections for a domain partition, specify the actual host name(s) of the WebLogic server cluster or managed server. If you are using OTD to load balance connections for a domain partition, specify the host name(s) of the OTD administration server.

View ▾

+ Add

✕ Delete

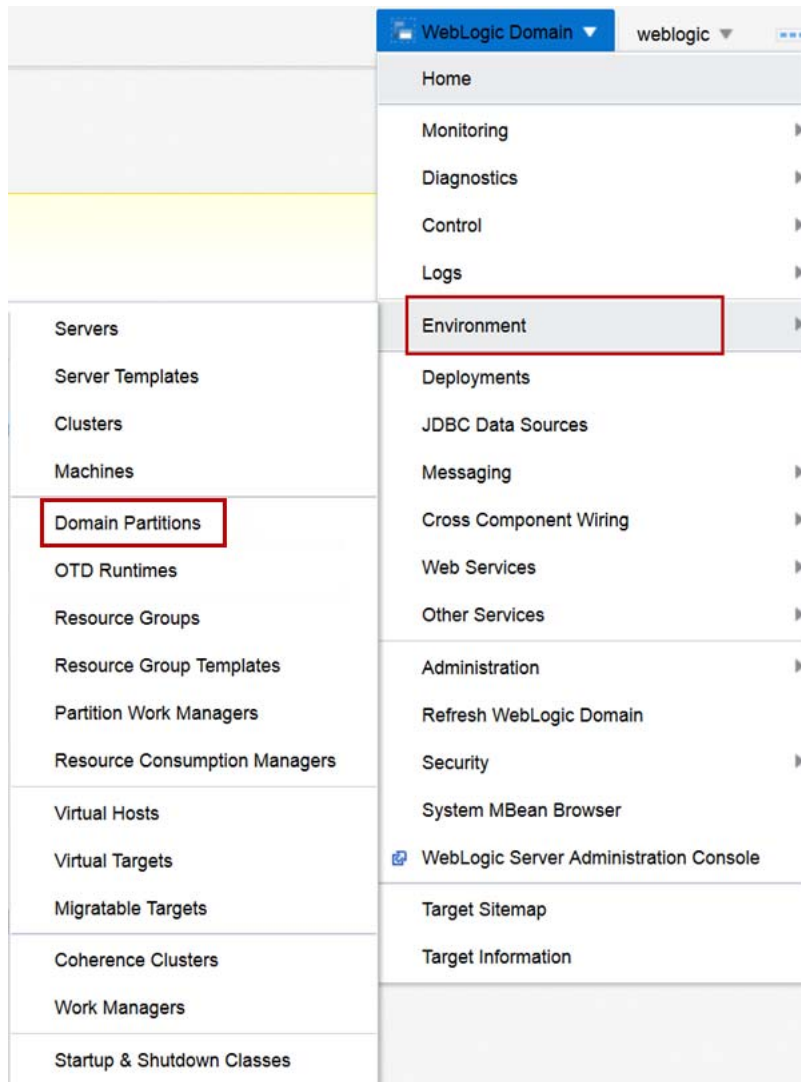
Host Name

No host names specified for this virtual target.

▶ Advanced Options

- e. Click **Next**.
- f. Select **cachecluster** from the **Cluster** menu.
- g. Click **Create**.

9. Create the valleycontact domain partition.
 - a. From the **WebLogic Domain** menu, select **Environment** and then click **Domain Partitions**.



- b. Click **Create**.
- c. For **Name**, enter valleycontact.
- d. For **Security Realm**, select valley.
- e. In the **Load Balancer Configuration** section of the page, select **Use OTD for load balancing**.
- f. For **OTD Runtime**, select medrec.
- g. Click **Next**.

- h. For the Virtual Target **valleycontact-client**, choose both **Select** and **Set as Default**.
- i. For the Virtual Target **valleycontact-storage**, choose **Select** and click **Next**.

Create Domain Partition: Available Targets Back Step 2 of 4 Next Cancel

Select the virtual targets that will be available for this domain partition to use. Note that virtual targets can only be used by one partition; so, only available virtual targets are listed below.

Select	Virtual Target	Set as Default
☒	valleycontact-client	☒
☒	valleycontact-storage	☐

- j. For **Resource Group Name**, enter `valleycontact-client`.
- k. For **Resource Group Template**, select **ContactClientResourceGroupTemplate**.
- l. Move the **valleycontact-client** target to **Selected Targets**.
- m. Click **Next**.

Create Domain Partition: Summary Back Step 4 of 4 Next Create Cancel

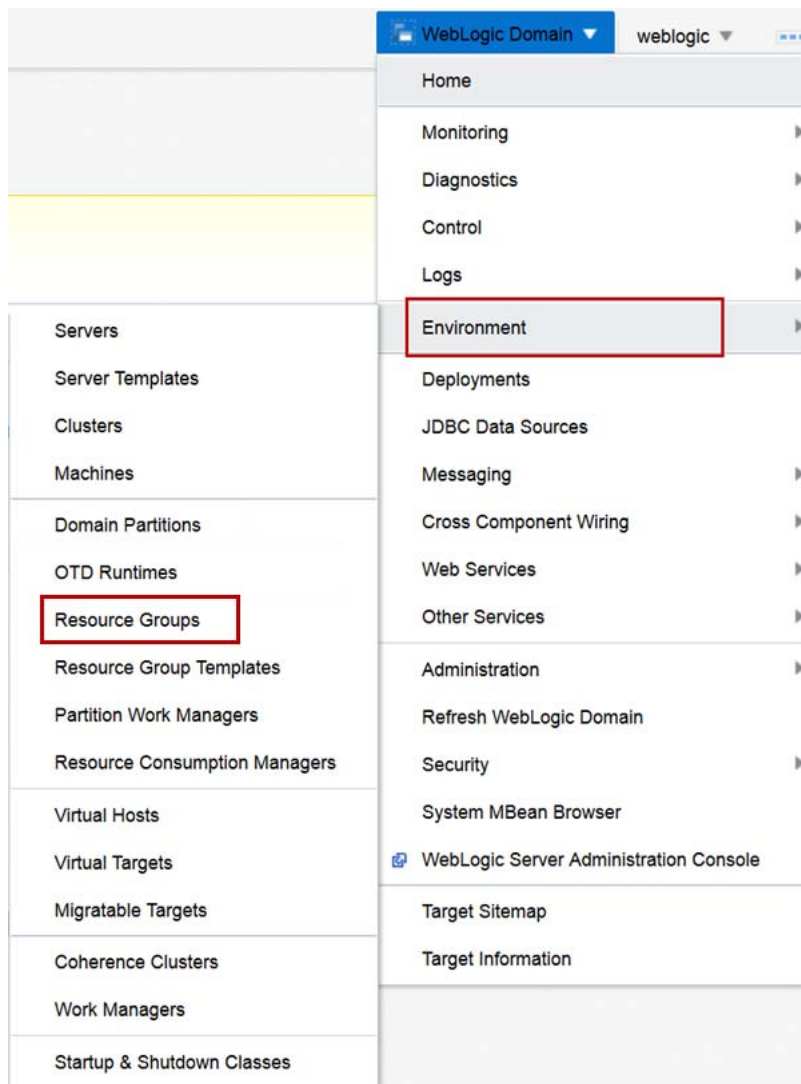
Information
This partition will be created and the changes activated immediately. You will not be able to roll back these changes.

A WebLogic domain partition will be created with the following attributes.

Name	valleycontact
Realm	valley
Primary Identity Domain	None
OTD Runtime	medrec
Available Targets	valleycontact-client, valleycontact-storage
Default Targets	valleycontact-client
Resource Group	valleycontact-client
Resource Group Template	ContactClientResourceGroupTemplate
Targets for the Resource Group	valleycontact-client

- n. Click **Create**.
- o. Click the **valleycontact** domain partition.

- p. From the **Domain Partition** menu, select **Administration** and select **Resource Groups**.



- q. Click **Create**.
- r. For **Name**, enter `valleycontact-storage`.
- s. For **Resource Group Template**, select **ContactStorageResourceGroupTemplate**.

- t. For **Targets**, move **valleycontact-storage** from **Available** to **Chosen** and click **Create**.

- u. Click the **Coherence Caches** tab.
- v. Click **Add**.
- w. For **Name**, enter `stores-shared-cache`.
- x. For **Coherence Cache Name**, enter `stores`.
- y. For **GAR (Grid Archive) Application Name**, enter `ExampleGAR`.
- z. Click **Shared** to allow this cache to be shared by multiple domain partitions.

- aa. Click **Save**.

The valleycontact domain partition is created with two resource groups – the valleycontact-client for client applications and the valleycontact-storage for coherence storage.

10. Start the valleycontact domain partition.
 - a. In the **valleycontact** domain partition details page, click **Start Up**.
 - b. When the Confirmation dialog box indicates that the Start Operation is successful, click **Close**.

The valleycontact domain partition successfully starts.
11. Test the valleycontact domain partition.
 - a. In the Browser window, navigate to <http://fmw2:8080/valleycontact/example-web-app>.

Note that when you created the domain partition and started it, the Lifecycle Manager performed the necessary configuration for OTD for the valleycontact deployment.
12. Test the stores shared coherence cache by viewing the contact sample application and creating more store cache entries, which should display in both application domain partitions.
 - a. In the browser window, navigate to <http://fmw2:8080/valleycontact/example-web-app/faces/StoreList.jsp>.

Note that the current cache size is 20.
 - b. In another browser window, navigate to <http://fmw2:8080/baylandcontact/example-web-app/faces/StoreList.jsp>.

Note that the current cache size is 20.
 - c. Click **Insert 20 Random Stores**.
 - d. Note that the current cache size is 40.
 - e. Refresh browser window that is displaying <http://fmw2:8080/baylandcontact/example-web-app/faces/StoreList.jsp>.

Note that the current cache size is 40. The applications in both baylandcontact and valleycontact are using the stores' shared coherence cache.